



UNIVERSITÀ DEGLI STUDI DI SALERNO
DIPARTIMENTO DI INFORMATICA

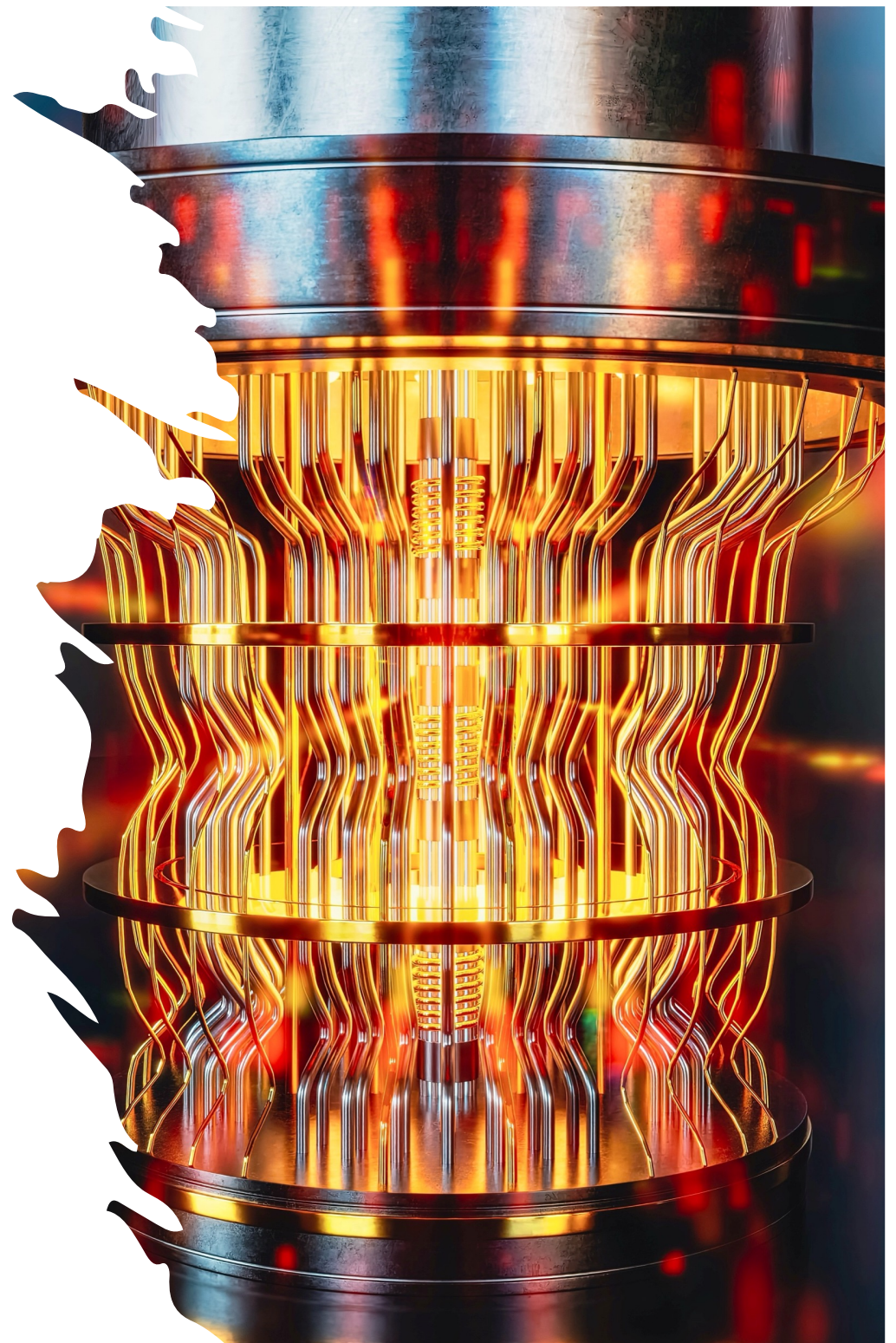
Course on Quantum Tech- nology for Security



Lecture 1 – Introduction to Quantum Computing

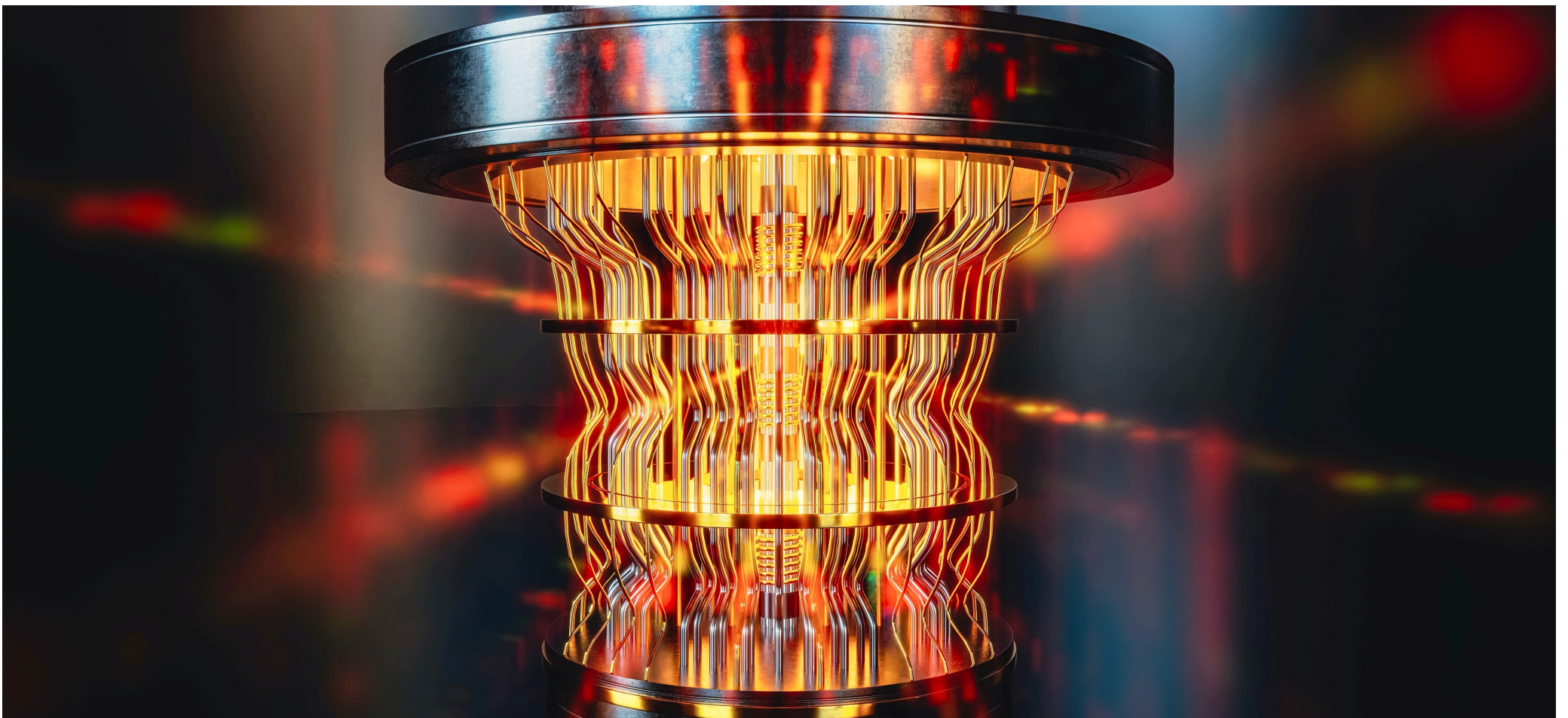


Prof. Esposito Christian



::.. Summary

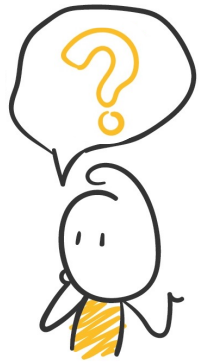
- Quantum Representation of Data
 - Qubits, Bloch sphere, Ket-Bra notation, superposition, basis in the Hilbert space.
 - No-Cloning theorem, quantum measurement. Quantum entanglement: Entanglement, Bell states, EPR paradox
- Basic principles of quantum computing:
 - Density matrix formalism
 - single- and n-qubit gates and quantum circuits.



Quantum Representation of Data

::... Binary Data Representation

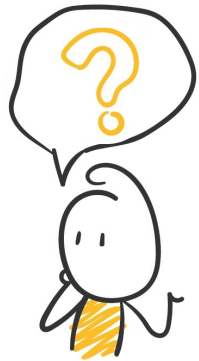
Informatics from "Informatique" refers to the study and use of machines for processing, storing, and managing information.



1. How can data be represented in machines for its storage and transmission?
2. How can processing operations be done on data?

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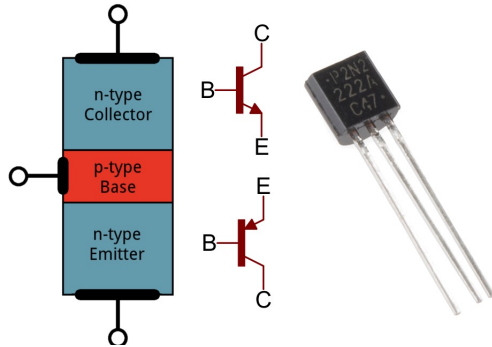


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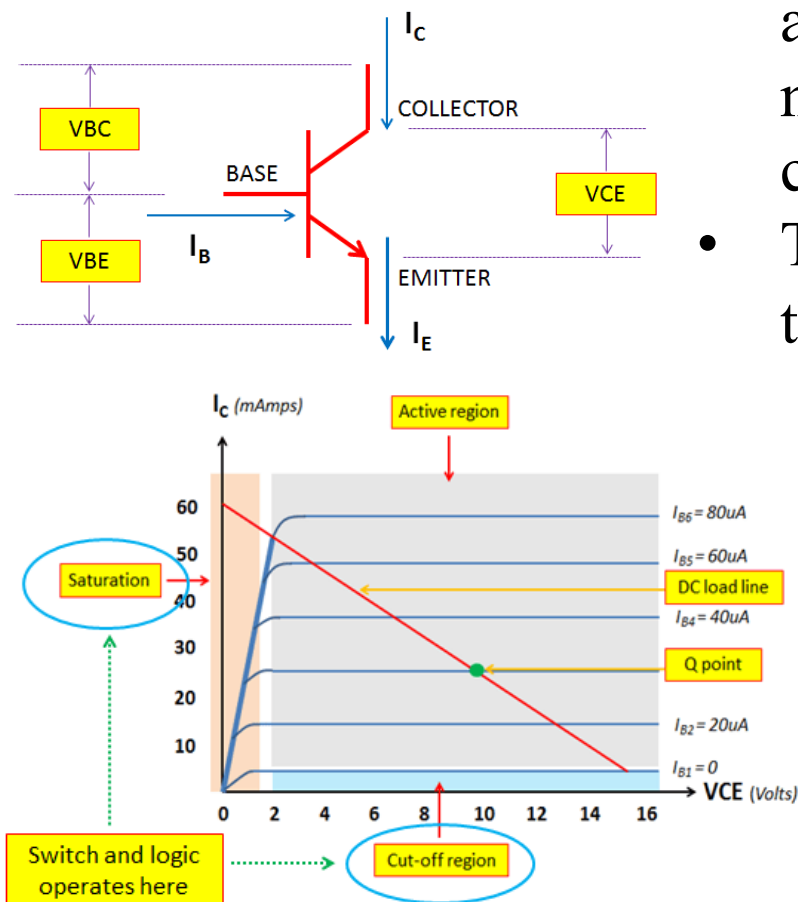
- A. Data is represented by a sequence of bits.
- B. Operations are made by means of combinatorial and sequential circuits.

... Binary Data Representation

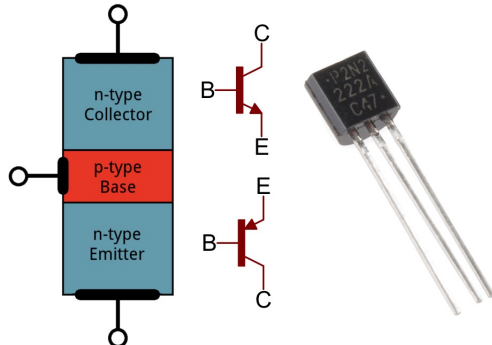


Modern computers are built from billions of transistors, which are basically tiny electronic switches:

- There are two N-type materials sandwiching the P-type material, or two diodes with their anodes connected. In the N-type material, negative charge is dominant, while positive charge is dominant in the P-type material.
- The current flowing into the base (I_B) drives the transistor into one of three possible states.

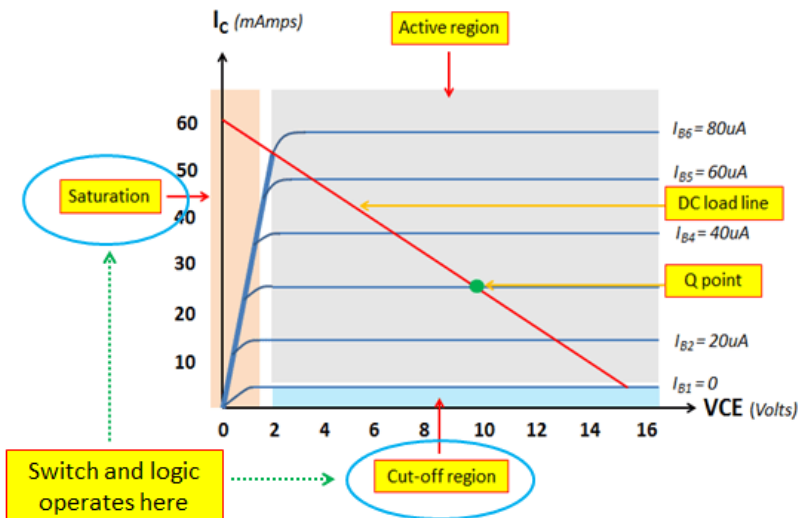
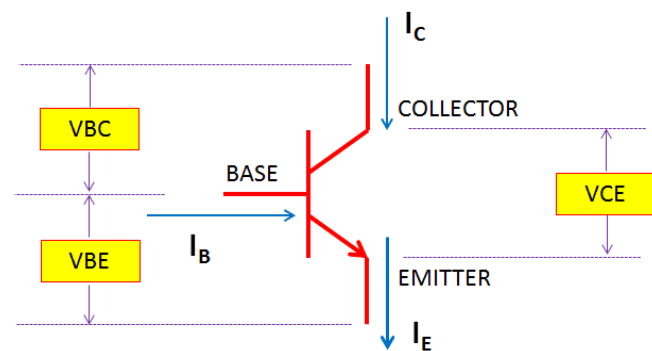


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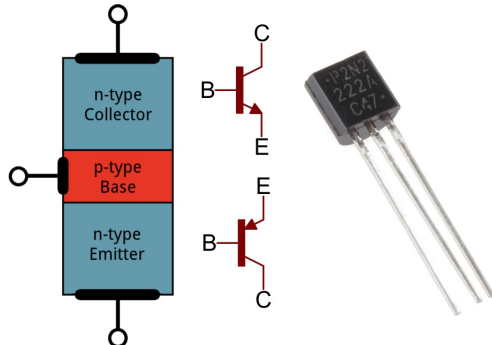
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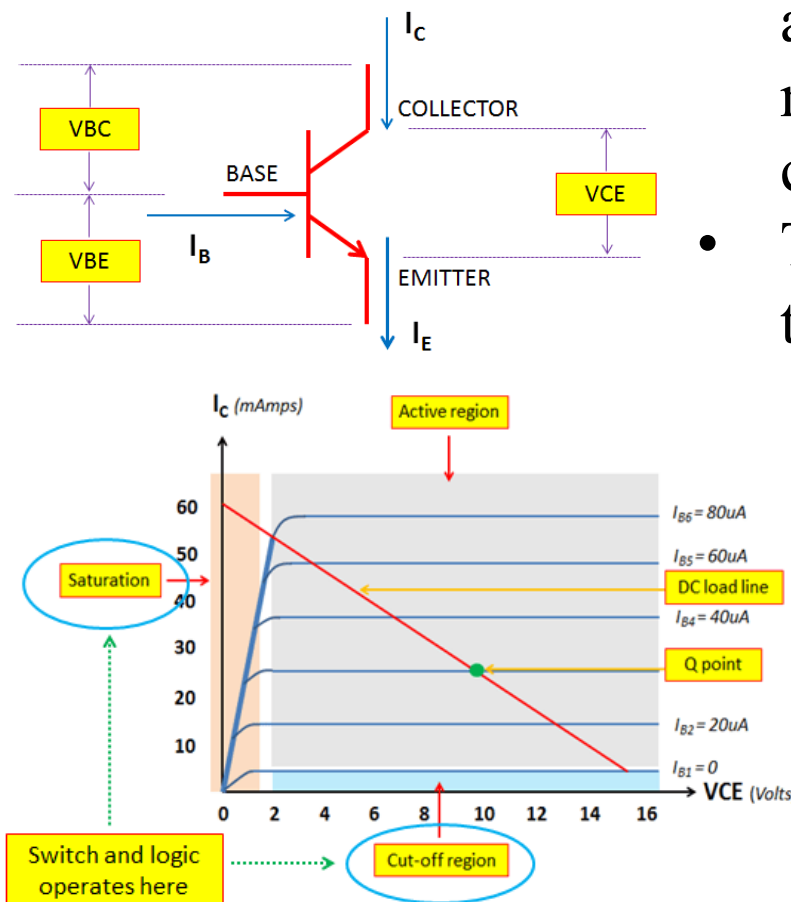
To turn on an NPN transistor, make the base about **0.6–0.7 V higher than the emitter** (forward-bias the base–emitter junction) and provide a base current so it can conduct from collector to emitter.

::... Binary Data Representation



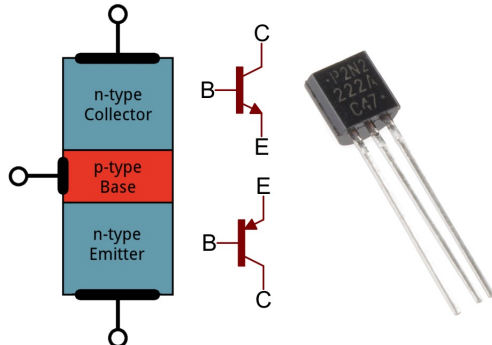
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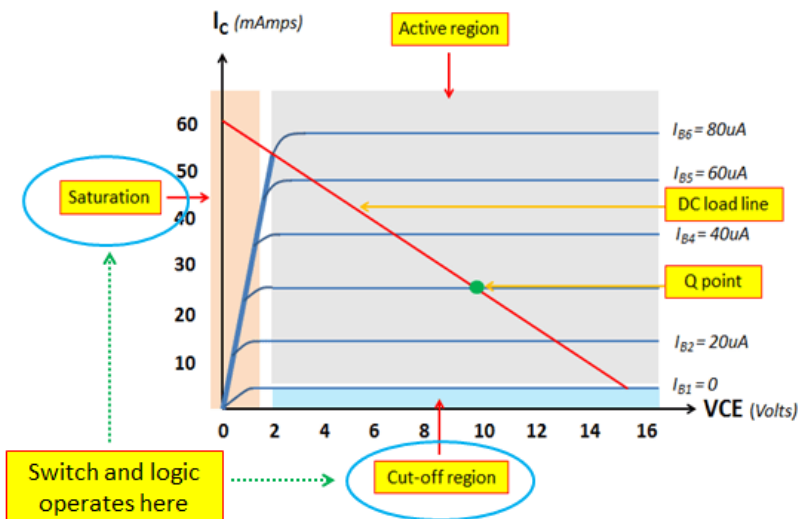
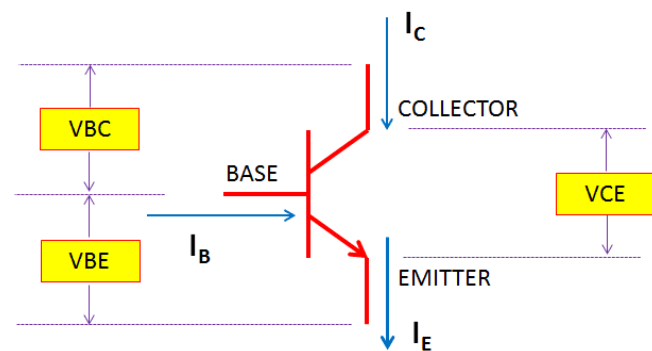
In the **cut-off** region, the base-emitter junction is not forward-biased, so $I_B \approx 0$ and $I_C \approx 0$, so the transistor is “off”.

::... Binary Data Representation



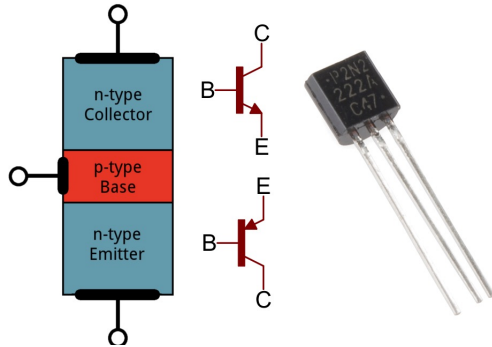
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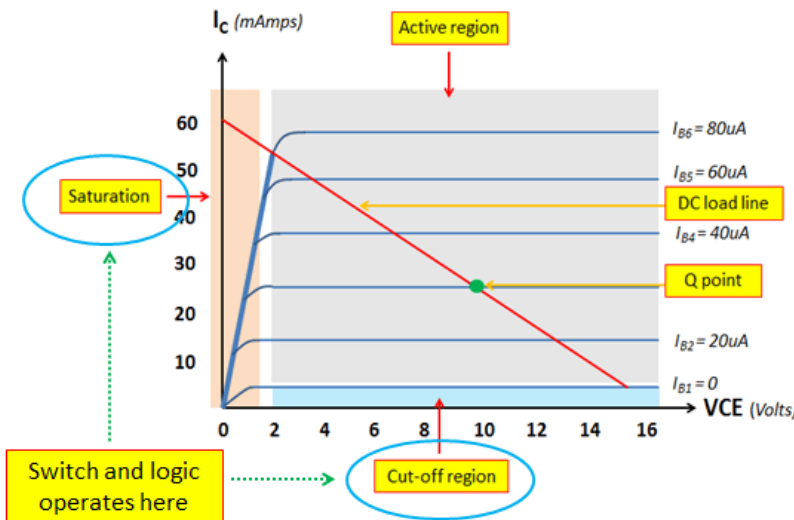
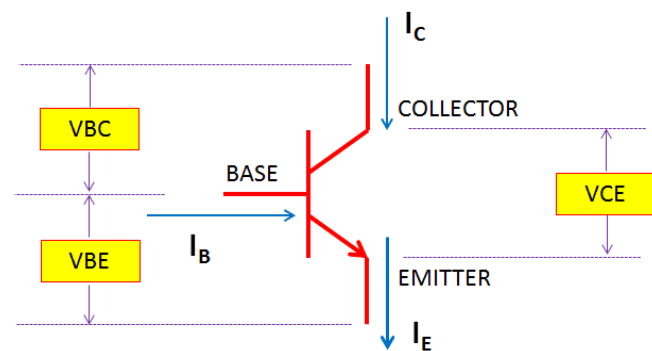
In the **active** region, $I_C \approx \beta I_B$ and the transistor is used for amplification, while in **saturation** both junctions are forward-biased, and the transistor is “fully on” with V_{CE} very small (typically $\sim 0.1\text{--}0.3$ V).

::... Binary Data Representation



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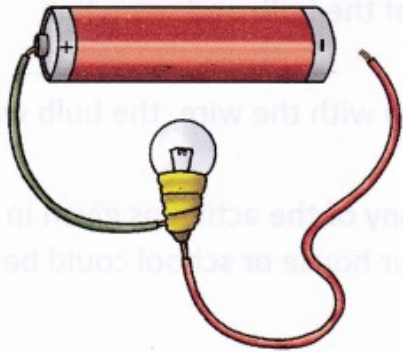


In digital circuits, designers force it to behave like a switch by driving it into two “extreme” regions:

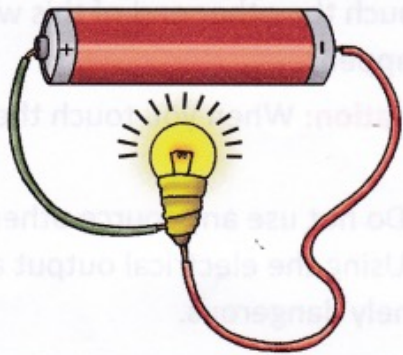
- **OFF** (logic 0) → cut-off;
- **ON** (logic 1) → saturation.

There is a 100% probability to represent logic 0 or 1 thanks to the two transistor's regions.

::... Binary Data Representation



(a) Open circuit—the bulb does not glow



(b) Closed circuit—the bulb lights up

In many circuits, you can map 0/1 to no current vs some current (or “small current vs larger current”) flowing within a wire. But in most digital electronics, bits are primarily represented by voltage levels, not by current, because:

- Current depends a lot on the load (what’s connected), so it’s less predictable,
- Voltage is easier to sense with clear thresholds (high/low), giving better noise tolerance.

Real electrical signals are messy: temperature changes, interference, manufacturing imperfections, etc.

If you try to represent data with many voltage levels (like 0–9), small noise can make a device misreads the value. With binary, wide “safe zones” can be defined like: anything near 0V counts as 0, anything near 1V counts as 1, with 100% probability. **That makes computers much more robust.**

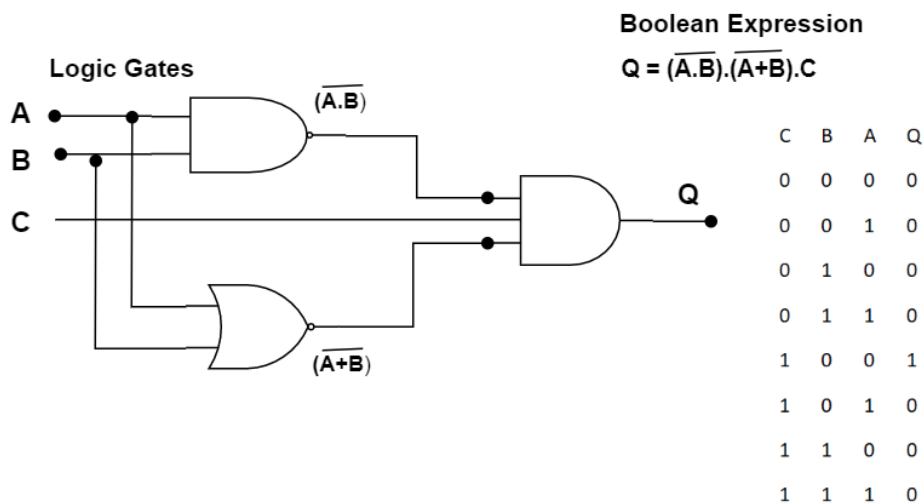
::... Binary Data Representation

Now that the binary representation of data has been agreed, we need to determine how operations on data can be done. “Doing operations” means building physical circuits whose outputs are predictable functions of their inputs (the output is 100% consistent with what is expected). The bridge from bits to operations is Boolean logic and the basic three operations of AND, OR and NOT or basic **logic gates**, being used to compute any other possible operation.

Any operation on bits can be described as a function like:

$$y = f(a, b, c)$$

where each variable is a bit. A computer implements these functions using logic gates.



::... Binary Data Representation

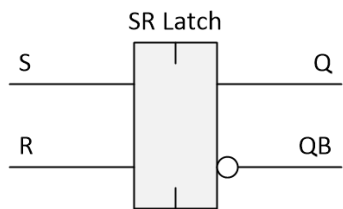
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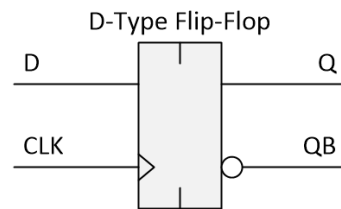
To compute step-by-step, computers need state, made with storage elements:

- Latches / flip-flops store 1 bit.
- Registers store many bits (e.g., 32/64).



R	S	$Q_{(new)}$	$QB_{(new)}$
0	0	$Q_{(old)}$	$QB_{(old)}$
0	1	1	0
1	0	0	1
1	1	0*	0*

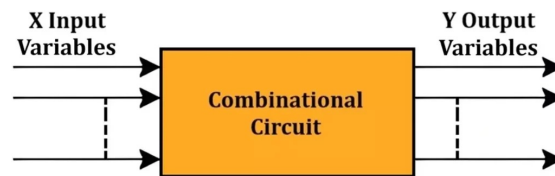
* Unstable configuration



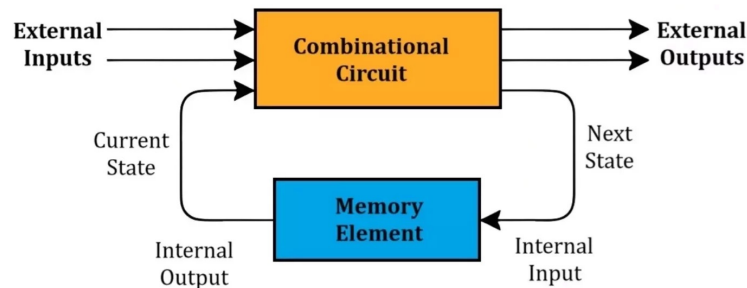
CLK	D	$Q_{(new)}$	$QB_{(new)}$
↑	0	0	1
↑	1	1	0
0	X	$Q_{(old)}$	$QB_{(old)}$
1	X	$Q_{(old)}$	$QB_{(old)}$
↓	X	$Q_{(old)}$	$QB_{(old)}$

::... Binary Data Representation

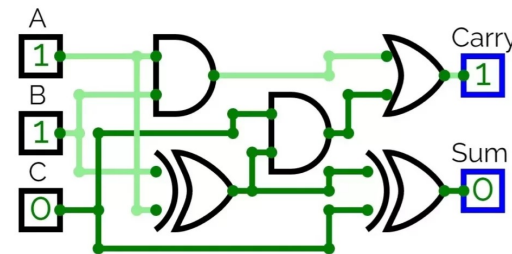
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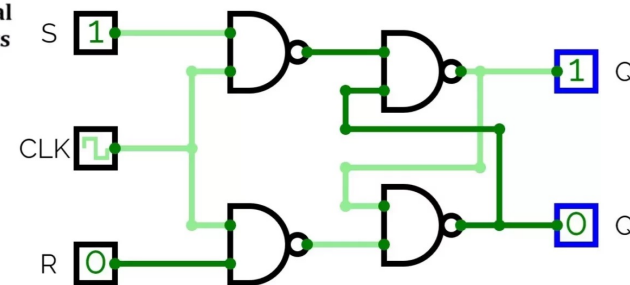
Combinational Circuit



Sequential Circuit



Full Adder



SR Flip Flop

::... Computing Boosting

Modern computing systems are becoming progressively more powerful through advances in semiconductor technology, computer architecture, and software engineering.

- At the **physical level**, improved manufacturing processes enable higher transistor density and faster switching, while also introducing new materials, device structures, and packaging techniques. In parallel, power and thermal constraints have become central design factors, so performance gains increasingly depend on energy efficiency, better cooling solutions, and more effective power-management strategies rather than raw frequency increases alone.
- At the **architectural level**, performance is enhanced by extracting more work from each clock cycle and by expanding parallel execution. In addition, heterogeneous designs have become critical: specialized accelerators deliver substantial speed-ups for targeted workloads by trading generality for high throughput and improved performance per watt.

::... Computing Boosting

- Another major driver of computational power is the **continued optimization** of the memory and data-movement subsystem. Because many applications are limited by memory bandwidth and latency rather than arithmetic speed, designers employ multi-level cache hierarchies, high-bandwidth memory technologies, faster interconnects, and increasingly advanced packaging methods to reduce communication costs between compute units and memory. Storage and networking improvements further enhance performance by minimizing bottlenecks in I/O and inter-node communication.
- At the **application level**, algorithmic improvements, better numerical methods, and domain-specific optimizations often yield gains that rival hardware upgrades.

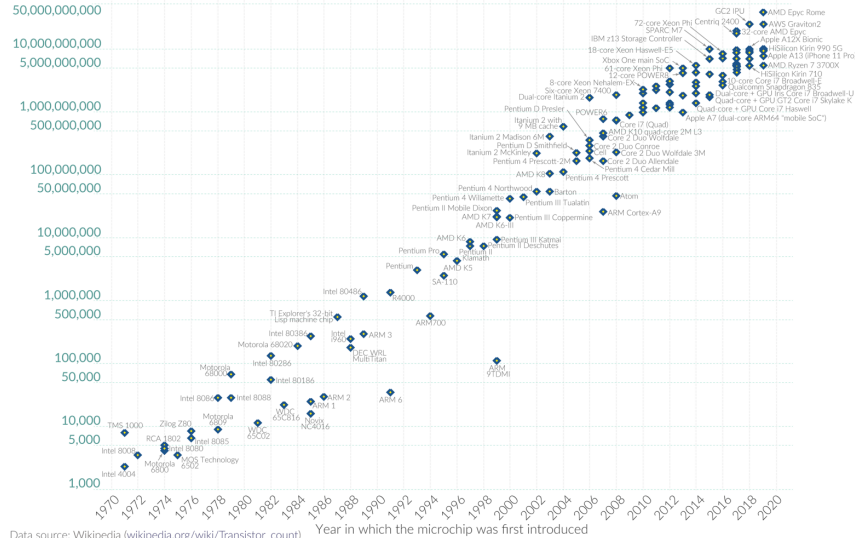
Together, these strategies form the foundation for sustained increases in computing performance.

::... Computing Boosting

Moore's Law: The number of transistors on microchips doubles every two years.

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

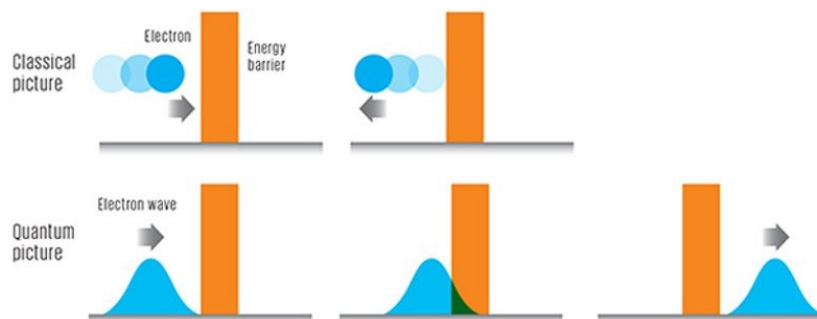
Transistor count



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)
OurWorldInData.org – Research and data to make progress against the world's largest problems. Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

We relied on **Moore's Law**, which predicted that the number of transistors on a microchip would double every two years. However, we are reaching the atomic scale, where classical physics starts to fail.

- **Size Constraints:** Transistors are now only a few nanometers wide. We simply cannot make them much smaller without hitting the size of an atom.
- **Quantum Tunneling:** When transistors get too small, electrons can "leak" through barriers (as the gate oxide). This creates heat and errors, making the processor unreliable.



- **Heat Dissipation:** Packing billions of transistors into a tiny space generates immense heat.

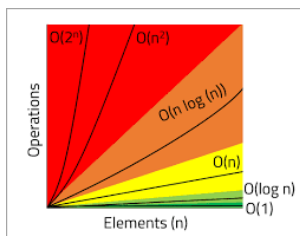
::... Computing Boosting

Classical computers are **linear**: they process information using bits and tackle problems one step at a time. While they are incredibly fast, some problems are "computationally expensive," meaning the time required to solve them grows exponentially with the size of the input.



- Caffeine Molecule Simulation: Simulating a simple molecule's energy states is nearly impossible for a classical supercomputer because of the sheer number of interactions.
- Cryptography: Breaking modern encryption would take a classical computer billions of years.

Data Structure	Time Complexity								Space Complexity	
	Average				Worst				Worst	
	Access	Search	Insertion	Deletion	Access	Search	Insertion	Deletion		
Array	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Stack	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Queue	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Singly-Linked List	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
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Binary Search Tree	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Cartesian Tree	$O(1)$	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
B-Tree	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Red-Black Tree	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Splay Tree	$O(1)$	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
AVL Tree	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
KD Tree	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(\log(n))$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$



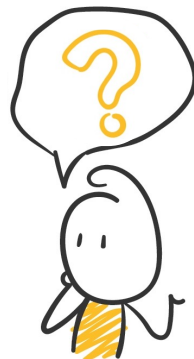
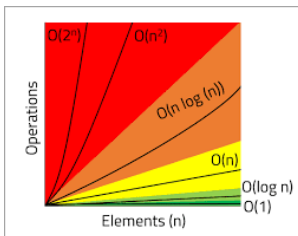
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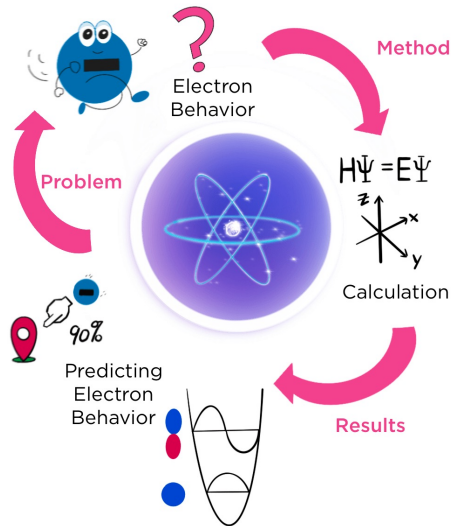
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How can we change this status quo to allow computing to go beyond physical limits?

::... Quantum Mechanics

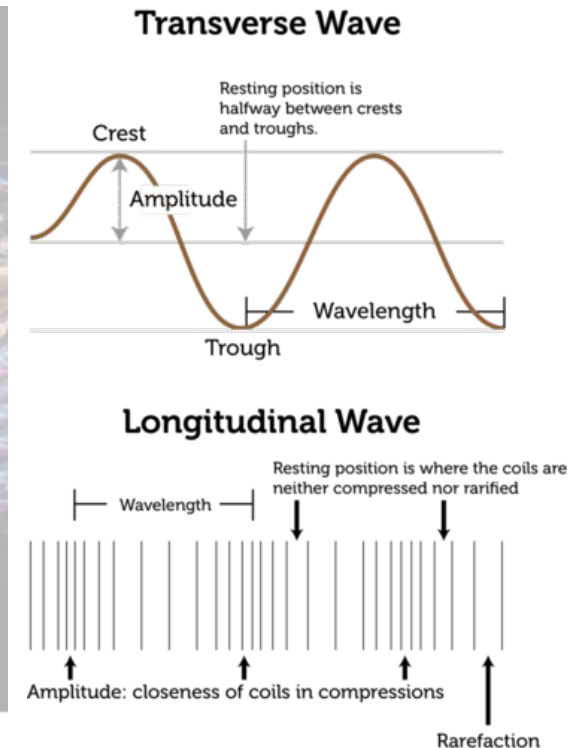
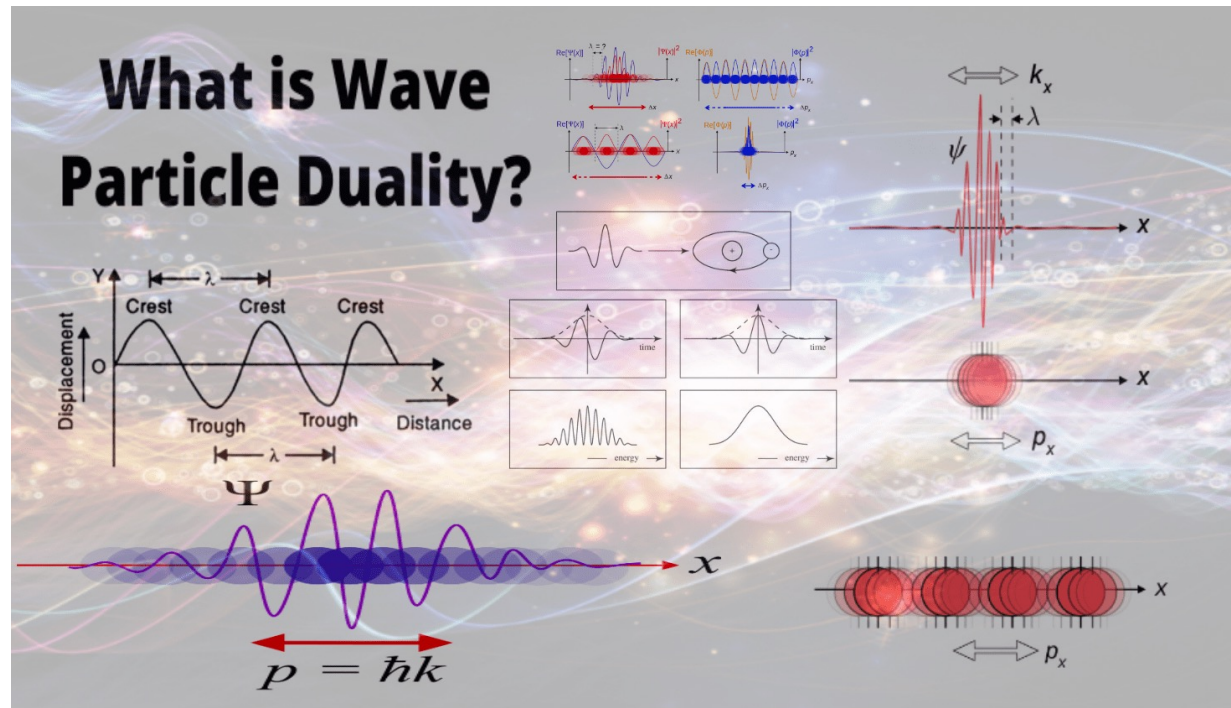


Quantum mechanics has attracted increasing attention from researchers seeking to boost computing. It is the branch of physics that describes how matter and energy behave at very small scales—atoms, electrons, photons, and other subatomic particles—where everyday “classical” intuition stops working.

Quantum originates from the Latin word *quantus*, meaning "how much" or "so much as», and denotes a specific "sum, amount, or portion". It was adopted in physics to describe discrete, measurable packets of energy, rather than a continuous stream.

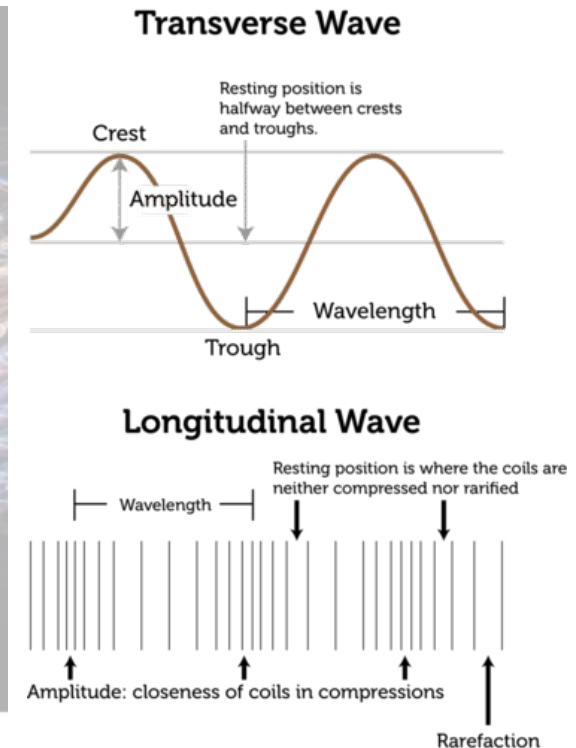
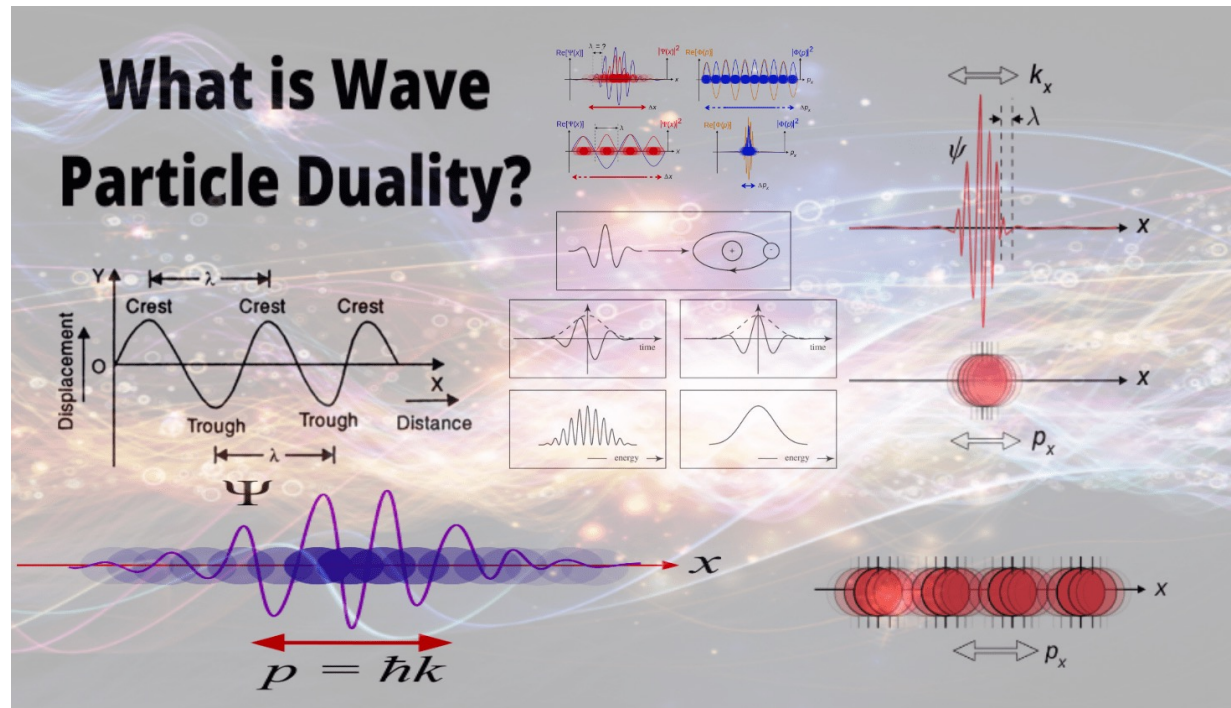
Quantum mechanics attracted so much interest for computing because it allows information to be processed in ways that classical bits and classical logic can't efficiently imitate.

::... Quantum Mechanics



Quantum entities don't act only like tiny billiard balls with a single definite path. Instead, they are described by a wavefunction, which spreads out in space and can produce interference and diffraction.

::: Quantum Mechanics



Quantum entities don't act only like tiny billiard balls with a single definite path. Instead, they are described by a wavefunction, which spreads out in space and can produce interference and diffraction

- Probability, not a physical “smear” of matter: The wavefunction isn't usually interpreted as a literal material wave. Let's consider its magnitude or amplitude as the maximum displacement moved by a point on a wave measured from its equilibrium position.

::... Quantum Mechanics

- The key role of the wave representation is that the square of its magnitude gives the probability density of finding the particle at a position:

$$P(x) \propto |\psi(x)|^2$$

De Broglie wavelength: Quantum particles have an associated wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$, where h is Planck's constant and p is momentum, given by the product of mass and speed.

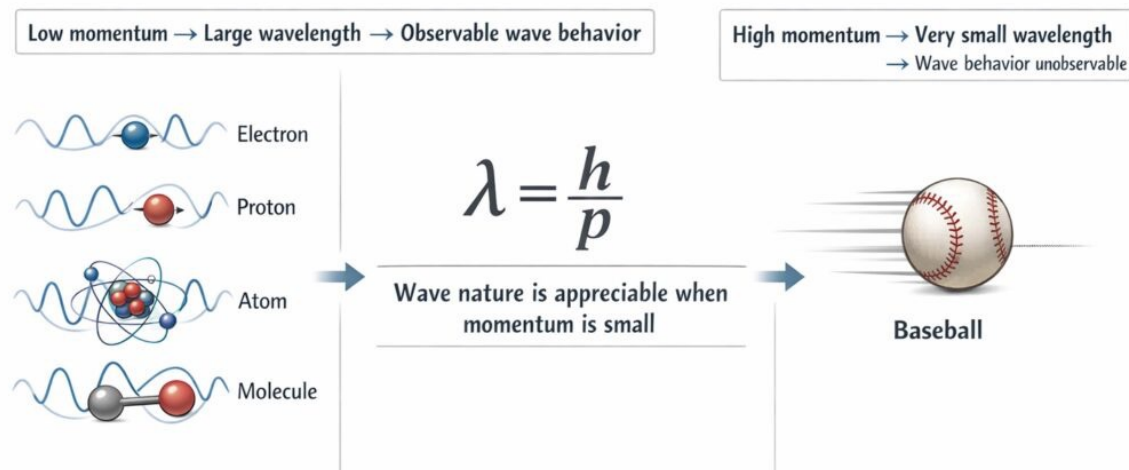
Smaller momentum implies a longer wavelength, where we achieve a more noticeable wave effects. For an electron, the wavelength is on the order of Angstroms ($10^{-10}m$), measurable. For macroscopic objects, the wavelength is so small that it is unobservable.

::... Quantum Mechanics

- The key role of the wave representation is that the square of its magnitude gives the probability density of finding the particle at a position:

$$P(x) \propto |\psi(x)|^2$$

De Broglie wavelength: Quantum particles have an associated wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$, where h is Planck's constant and p is momentum, given by the product of mass and speed.



::... Quantum Mechanics

The **Heisenberg uncertainty principle** states that there is a fundamental limit to how precisely certain pairs of physical properties can be known at the same time. The most famous pair is position x and momentum p : the more precisely you know a particle's position, the less precisely you can know its momentum, and vice versa.

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

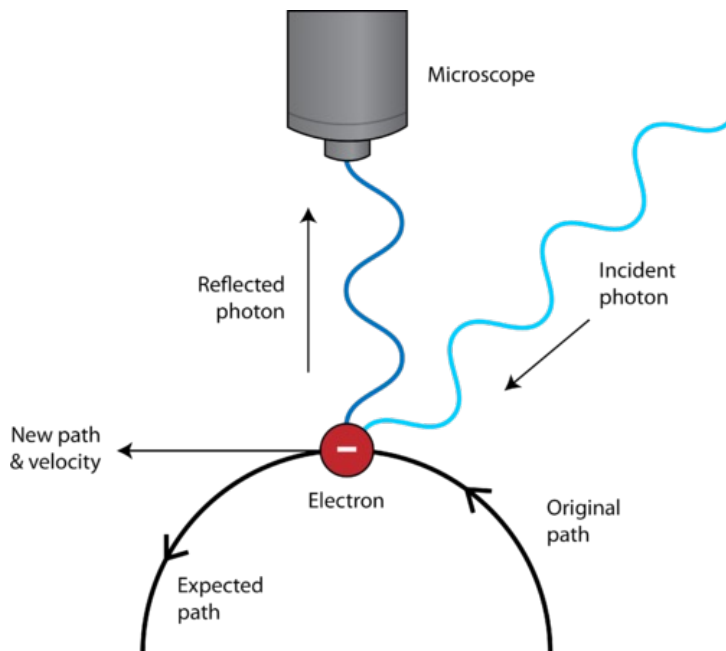
where Δx , is the uncertainty in position, Δp is the uncertainty in momentum, and $\hbar$ is the reduced Planck constant (equal to the Planck constant h divided by 2π $\hbar = h/2\pi$).

This is not just a limitation of measurement instruments: it is a property of nature that comes from the wave-like behavior of quantum particles and from the fact that position and momentum are represented by non-commuting operators in quantum mechanics. Similar uncertainty relations apply to other pairs, such as energy and time ($\Delta E \Delta t \gtrsim \hbar/2$).

::... Quantum Mechanics

The **Heisenberg uncertainty principle** can be better understood by the impossibility of determining simultaneously both the position and the velocity of a particle.

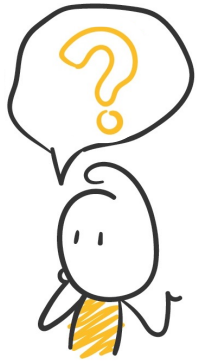
- The detection of an electron, for example, would be made by way of its interaction with photons of light. Since photons and electrons have nearly the same energy, any attempt to locate an electron with a photon will knock the electron off course, resulting in uncertainty about where the electron is located.



This uncertainty principle does not hold for large everyday objects because of their mass: if we are looking for something with a flashlight, the photons from the flashlight will not cause the object to move.

::... Quantum Mechanics

For a quantum particle, the Heisenberg uncertainty principle means it cannot have a perfectly “classical” state with both a sharply defined position and a sharply defined momentum at the same time



What are the consequences for a quantum particle?

::... Quantum Mechanics

For a quantum particle, the Heisenberg uncertainty principle means it cannot have a perfectly “classical” state with both a sharply defined position and a sharply defined momentum at the same time



- **No exact trajectory:** If you try to pin down the particle's position very precisely (small Δx), its momentum becomes intrinsically uncertain (large Δp). So, the idea of a single well-defined path like in classical mechanics breaks down.
- **Wavepacket spreading:** A particle localized in space is described by a wavepacket from many momentum components; because those components evolve differently, the packet typically spreads out over time.

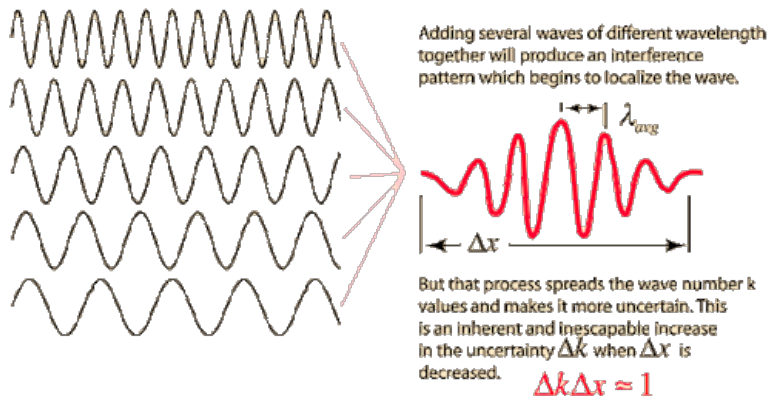
::... Quantum Mechanics

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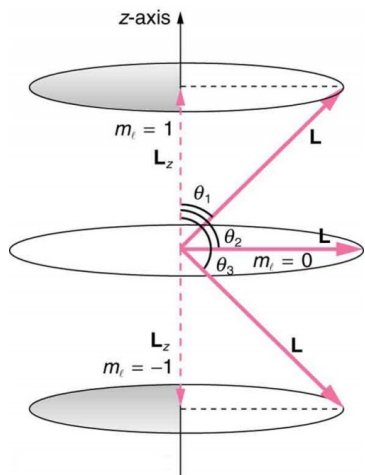


A wavepacket is a localized superposition of sinusoidal waves of different frequencies that travels as a unit, describing quantum particles in a limited region of space and representing wave-particle duality.

::... Quantum Mechanics

- “Quantum fluctuations” and zero-point energy: confining a particle (small Δx) forces a momentum spread, which implies nonzero kinetic energy. This is why systems like the harmonic oscillator have a nonzero ground-state energy (they can't sit perfectly still).

Some features of a quantum particle are described with binary pure states:



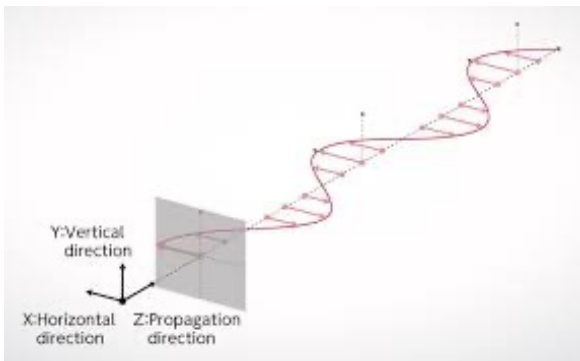
- Spin along z is the quantized component of a particle's intrinsic angular momentum, with values measured as multiples of $\frac{\hbar}{2}$. For spin-1/2 particles (electrons, protons), the values are $S_z = \pm \frac{\hbar}{2}$, corresponding to the "spin up" $\left(+\frac{\hbar}{2}\right)$ or "spin down" $\left(-\frac{\hbar}{2}\right)$ states. The measurement of spin along a direction is not continuous, but discrete (eigenvalues).

An electron in an orbital usually has spin (clockwise) or (counterclockwise). The Stern-Gerlach Experiment demonstrated the quantization of spin by showing the separation of a beam of atoms into two distinct directions (up/down): $|0\rangle = |\uparrow_z\rangle$, $|1\rangle = |\downarrow_z\rangle$.

::: Quantum Mechanics

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Some features of a quantum particle are described with binary pure states:



Photon polarization represents the direction of the electric-field oscillation of a light wave, which propagates forward; its electric field oscillates sideways (transverse to the direction of travel). The pattern of that oscillation is the polarization:

- **Linear polarization**: the field oscillates along a fixed direction
 $|H\rangle = \text{horizontal}, |V\rangle = \text{vertical}$
- **Circular polarization**: the field rotates as it propagates, tracing a circle
 $|R\rangle = \text{right-circular}, |L\rangle = \text{left-circular}$
- Elliptical polarization: the tip of the field traces an ellipse (linear and circular are special cases).

::... Qubit

Many important “quantum particle” features can be described with a two-state quantum system, as long as you focus on a *single binary degree of freedom*

1. Pick a binary property of the particle and call the two possibilities a basis:

Spin along z : $|0\rangle = |\uparrow_z\rangle$, $|1\rangle = |\downarrow_z\rangle$

Photon polarization: $|0\rangle = |H\rangle$, $|1\rangle = |V\rangle$

2. Describe the particle's state as a superposition

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

This captures the key quantum idea: the particle can be in a coherent combination of the two alternatives.

3. Measurements become simple probabilities, in the $\{|0\rangle, |1\rangle\}$ basis gives:

outcome 0 with probability $|\alpha|^2$

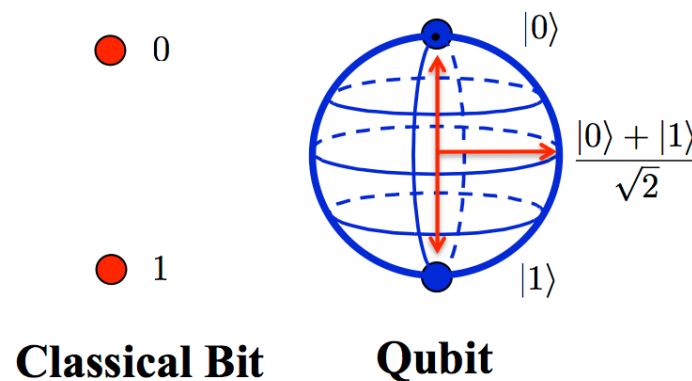
outcome 1 with probability $|\beta|^2$

::... Qubit

4. **Operations are rotations** (unitaries) - Any controlled evolution that stays within the two states is a 2×2 unitary matrix. A common form is

$$U = e^{-i\frac{\theta}{2}\mathbf{n} \cdot \boldsymbol{\sigma}}$$

which you can visualize as rotating the state on the Bloch sphere.



A **qubit** corresponds to the quantum state of a two-level quantum system $|\psi\rangle$, and is mathematically represented as a unit vector in a two-dimensional complex Hilbert space H , typically denoted as $\mathbb{C}^2$, defined by the two basis $|0\rangle, |1\rangle$. It is different to a classic bit that can be fully in one of the two logic states (0 or 1), as **the qubit can be anything in between these two states**.

::... Qubit

More precisely, one considers the Hilbert space $\mathcal{H} \cong \mathbb{C}^2$, endowed with the standard inner product, together with a fixed orthonormal basis usually denoted by $\{|0\rangle, |1\rangle\}$, called the computational basis. Any pure state of a qubit is represented by a unit vector $|\psi\rangle \in \mathcal{H}$, which can always be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where $\alpha, \beta \in \mathbb{C}$ satisfy the normalization condition

$$|\alpha|^2 + |\beta|^2 = 1.$$

The complex numbers α and β are called probability amplitudes. Their squared moduli determine the probabilities associated with measurement outcomes in the computational basis: according to the Born rule, a measurement of the qubit in the basis $\{|0\rangle, |1\rangle\}$ yields the outcome $|0\rangle$ with probability $|\alpha|^2$ and the outcome $|1\rangle$ with probability $|\beta|^2$.

A qubit generalizes the classical bit, since it is not restricted to the two discrete states 0 or 1, but may exist in any normalized superposition of them.

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$$\begin{bmatrix} a \\ b \end{bmatrix} \quad \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

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::... Qubit

$\mathcal{H}$ be a finite-dimensional Hilbert space over $\mathbb{C}$, endowed with an inner product $\langle \cdot, \cdot \rangle$. A family $\{\psi_i\}_{i=1}^n \subset \mathcal{H}$ is called an orthonormal basis if it satisfies

$$\langle \psi_i, \psi_j \rangle = \delta_{ij} \text{ for all } i, j,$$

where δ_{ij} is the Kronecker delta, equal to 1 if $i = j$ and 0 vice versa, and if it spans the whole space: every vector $|\psi\rangle \in \mathcal{H}$ admits a unique expansion

$$|\psi\rangle = \sum_{i=1}^n c_i |\psi_i\rangle, \quad c_i = \langle \psi_i, \psi \rangle.$$

The coefficients $c_i \in \mathbb{C}$ are the coordinates of the state with respect to the chosen basis, and Parseval's identity gives

$$\| \psi \|^2 = \langle \psi, \psi \rangle = \sum_{i=1}^n |c_i|^2.$$

Hence, if $|\psi\rangle$ is normalized, then $\sum_i |c_i|^2 = 1$, ensuring that the total probability of all measurement outcomes is equal to one.

::... Schrödinger's cat

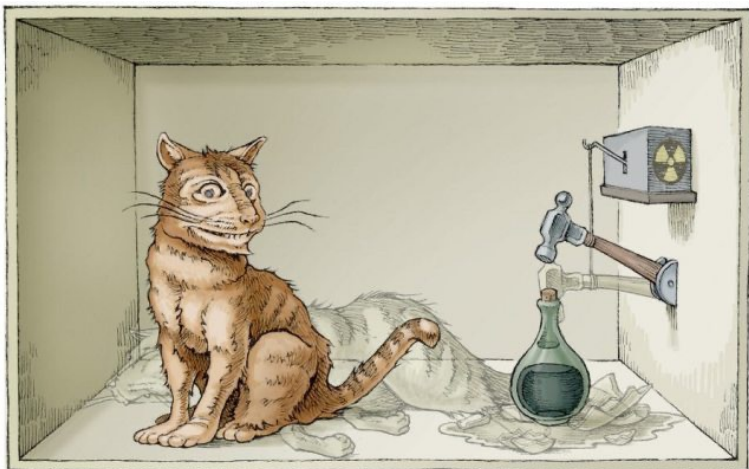
Schrödinger's cat is a thought experiment introduced by Erwin Schrödinger in 1935 to illustrate the conceptual strangeness of quantum mechanics when its principles are applied to macroscopic objects. The idea is not that real cats are ordinarily in quantum superpositions in any practical sense, but rather that the formalism of quantum theory seems to allow such a description if one extends it naively from microscopic systems to everyday-scale objects.



In the thought experiment, a cat is placed in a sealed box together with a radioactive atom, a detector, and a mechanism that releases poison if the atom decays. Quantum mechanically, before observation, the radioactive atom can be described as being in a superposition of two possibilities: decayed and not decayed.

::... Schrödinger's cat

Schrödinger's cat is a thought experiment introduced by Erwin Schrödinger in 1935 to illustrate the conceptual strangeness of quantum mechanics when its principles are applied to macroscopic objects. The idea is not that real cats are ordinarily in quantum superpositions in any practical sense, but rather that the formalism of quantum theory seems to allow such a description if one extends it naively from microscopic systems to everyday-scale objects.



Since the detector and poison mechanism are coupled to the atom, the entire system evolves into a joint state in which the microscopic superposition becomes correlated with macroscopic outcomes. In formal terms, the total state is an entangled superposition of “atom decayed, poison released, cat dead” and “atom not decayed, poison not released, cat alive”.

::... Schrödinger's cat

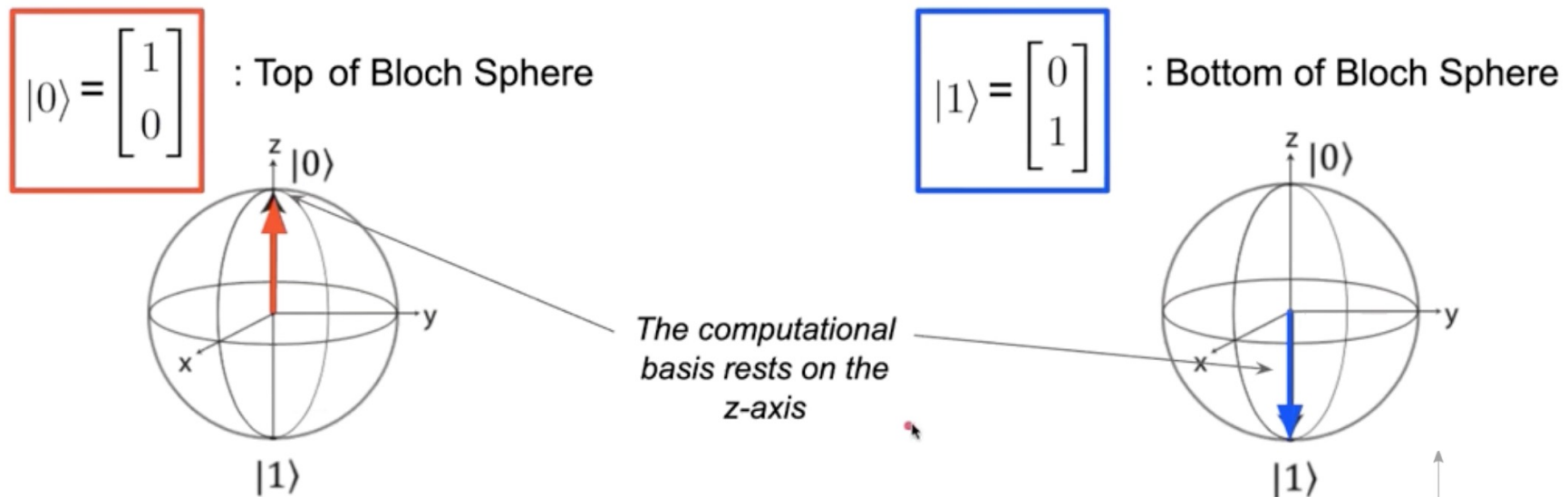
The paradox arises because quantum mechanics allows superpositions at the microscopic level, whereas our everyday experience suggests that the cat must be either alive or dead, not both. Schrödinger proposed this scenario to emphasize the tension between the mathematical formalism of quantum superposition and the apparent definiteness of classical reality.

The example is closely connected to the measurement problem. Before the box is opened, the system is described by a quantum state containing multiple possibilities. After observation, however, only one definite outcome is experienced. The question is how and why this transition from superposition to a single observed result occurs. Different interpretations of quantum mechanics answer this differently.

- Measurement causes the wavefunction to collapse to one outcome;
- No collapse occurs; instead, both outcomes persist;
- Interaction with the environment suppresses interference between the alternatives, making the system appear classical.

::... Basis in the Hilbert Space

Typically, qubits are expressed using the computational basis, corresponding to probability of measuring 0 or 1.

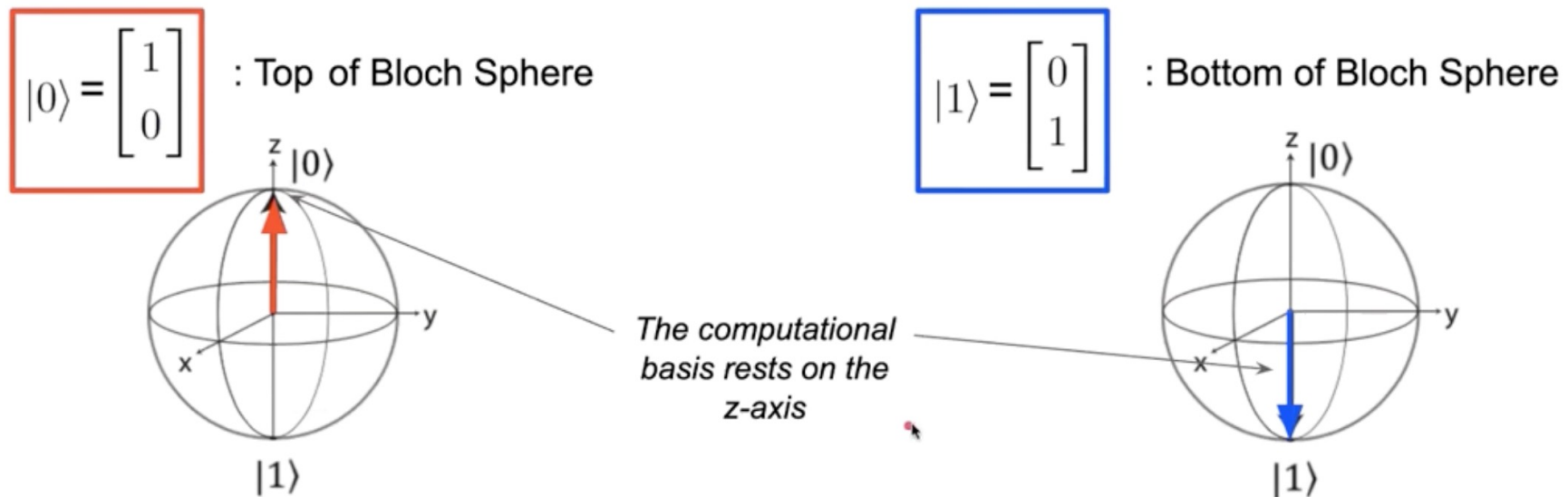


Another very useful representation of the quantum state is the Bloch sphere, which is the visual 3-dimensional intuitive representation of the state as points on the surface of the sphere, where the probability amplitudes are given by the two trigonometric functions.

The diagram shows a Bloch sphere with a point $|\psi\rangle$ on its surface. A blue vector connects the center of the sphere to the point $|\psi\rangle$. The angle between this vector and the positive z-axis is labeled θ . The angle between the projection of this vector onto the xy-plane and the positive x-axis is labeled ϕ . Below the sphere, the equation $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$ is written.

::... Basis in the Hilbert Space

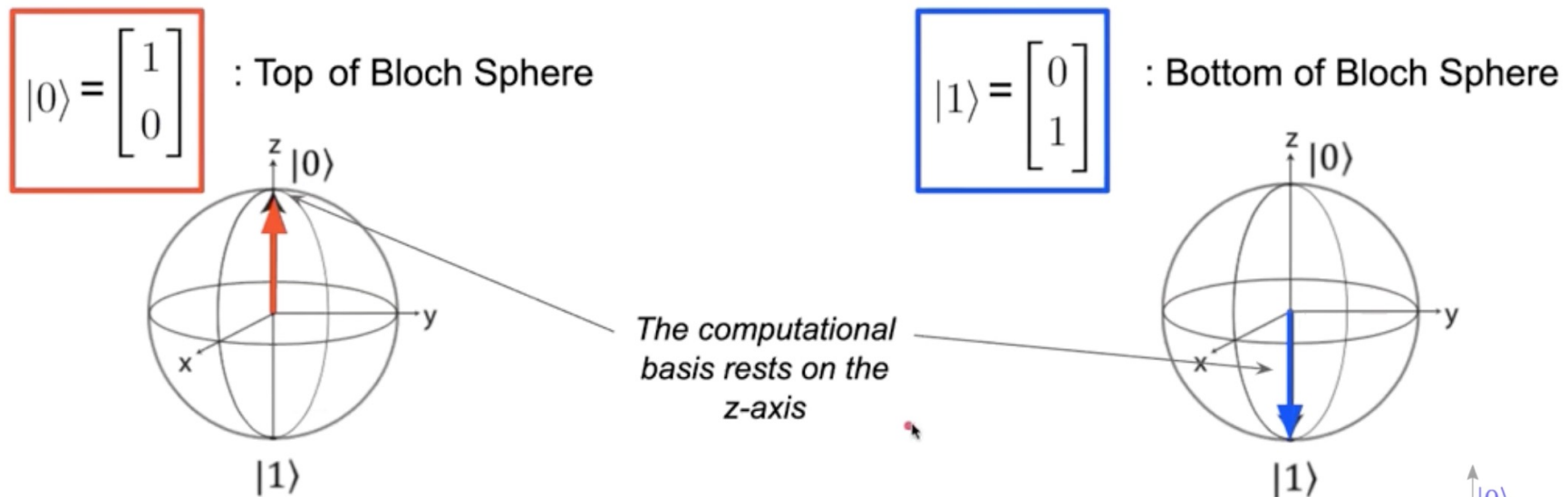
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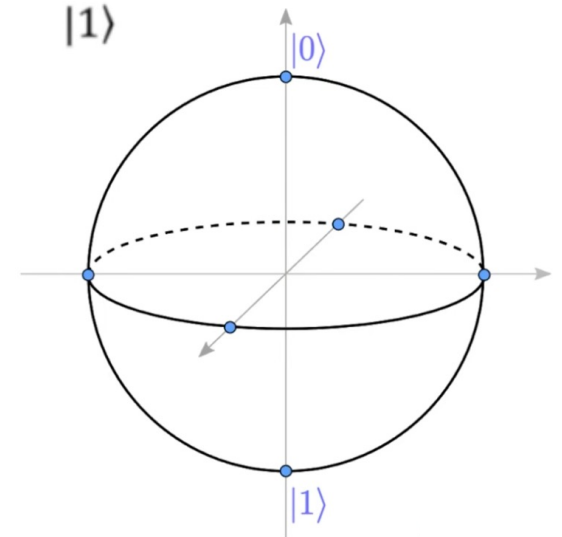
The computational basis $\{|0\rangle, |1\rangle\}$ is only one possible orthonormal basis. More generally, any pair of normalized and mutually orthogonal vectors $\{|\phi_0\rangle, |\phi_1\rangle\} \subset \mathbb{C}^2$ forms a valid qubit basis. In fact, every orthonormal basis of $\mathbb{C}^2$ can be obtained from the computational basis by a unitary transformation.

::... Basis in the Hilbert Space

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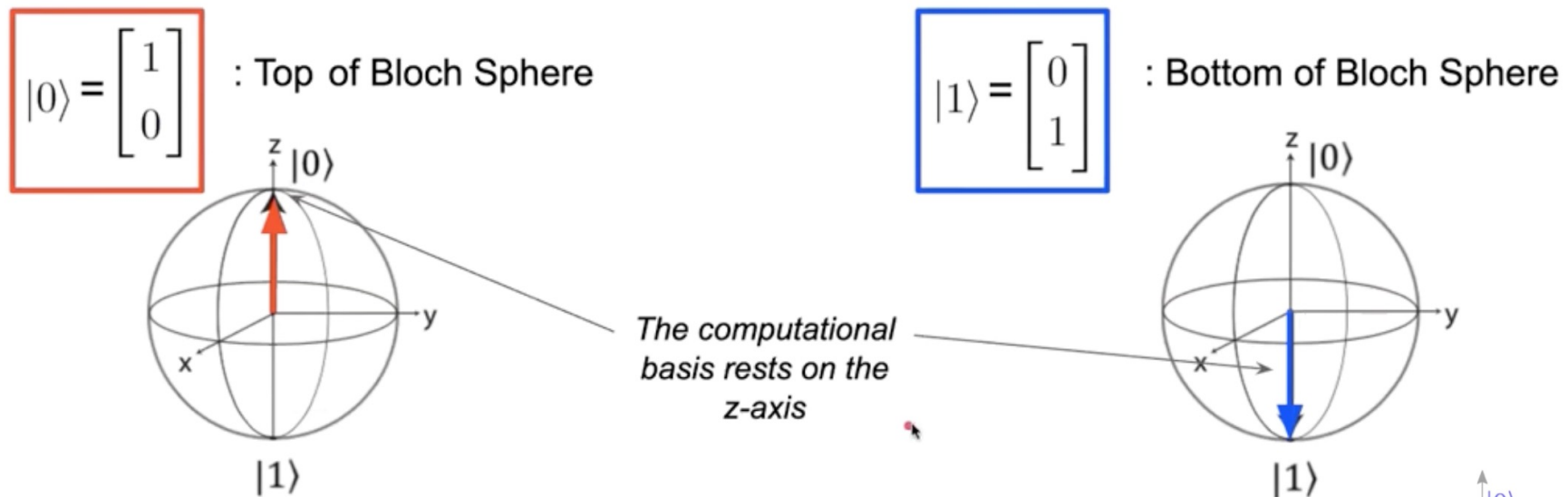


The $|0\rangle$, $|1\rangle$ states stand respectively at the north and south poles of the sphere.



::... Basis in the Hilbert Space

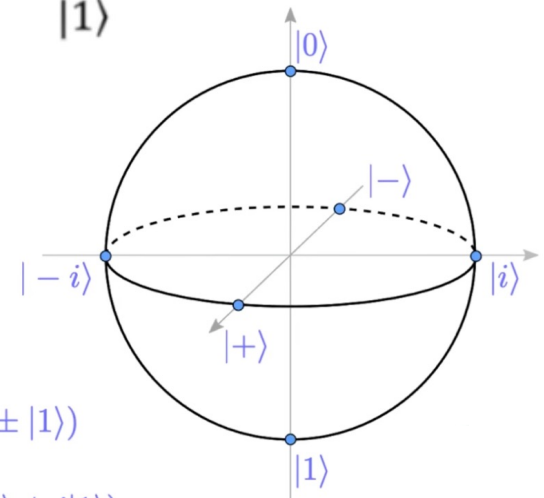
Typically, qubits are expressed using the computational basis, corresponding to probability of measuring 0 or 1.



The $|0\rangle, |1\rangle$ states stand respectively at the north and south poles of the sphere. There are other specific points that are arbitrary combinations of the states $|0\rangle, |1\rangle$, which can be alternative basis.

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$

$$|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$$



::... Basis in the Hilbert Space

Explicitly, if $U \in U(2)$, which denotes a unitary operator, and also is an element of the group $U(2)$ or the set of all such transformations, namely the set of all 2×2 unitary matrices satisfying

$$U^\dagger U = U U^\dagger = I, \text{ then}$$

$$|\phi_0\rangle = U |0\rangle, \quad |\phi_1\rangle = U |1\rangle$$

defines another orthonormal basis, because unitarity implies preservation of inner products:

$$\langle \phi_i, \phi_j \rangle = \langle i | U^\dagger U | j \rangle = \langle i | j \rangle = \delta_{ij}.$$

Besides the computational basis, two particularly important examples are the Hadamard basis and the circular, or Y -eigenbasis.

The Hadamard basis is given by

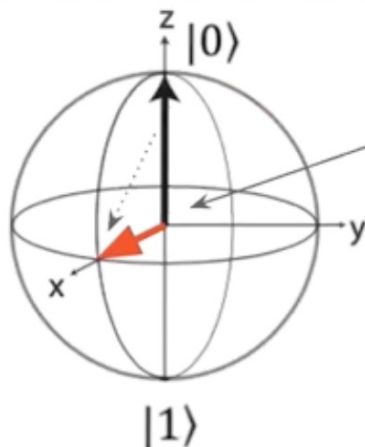
$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

which is the eigenbasis of the Pauli- X operator.

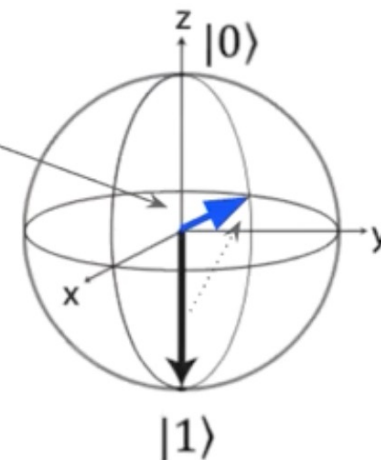
::... Basis in the Hilbert Space

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$



*The Hadamard,
or +/- basis,
rests on the
x-axis*



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::... Basis in the Hilbert Space

The circular basis is

$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}},$$

which is the eigenbasis of the Pauli- Y operator.

These are not merely alternative notations: they correspond to different measurement contexts, since measuring in different bases probes different observables.

More generally, every orthonormal basis of a qubit may be written, up to irrelevant global phases, in the form

$$\begin{aligned} |\phi_0\rangle &= \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle, \\ |\phi_1\rangle &= -e^{-i\varphi} \sin\left(\frac{\theta}{2}\right) |0\rangle + \cos\left(\frac{\theta}{2}\right) |1\rangle, \end{aligned}$$

for some $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$. One verifies directly that these vectors are normalized and orthogonal, and therefore they define a basis.

::... Basis in the Hilbert Space

The circular basis is

$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}},$$

which is the eigenbasis of the Pauli- Y operator.

These are not merely alternative notations: they correspond to different measurement contexts, since measuring in different bases probes different observables.

This parametrization shows that choosing a basis for a qubit is equivalent to choosing an orthonormal frame in $\mathbb{C}^2$, or geometrically, a measurement axis on the Bloch sphere.

::... Basis in the Hilbert Space

To move from one basis to another, one uses a change-of-basis matrix U_{ij} . Suppose $\{|\psi_i\rangle\}_{i=1}^n$ and $\{|\phi_i\rangle\}_{i=1}^n$ are two orthonormal bases of $\mathcal{H}$, then

$$U_{ij} = \langle \phi_i | \psi_j \rangle$$

is unitary, and it converts coordinates from the ψ -basis to the ϕ -basis. If

$$|\chi\rangle = \sum_{j=1}^n c_j |\psi_j\rangle = \sum_{i=1}^n d_i |\phi_i\rangle,$$

then the coordinate vectors $c = (c_1, \dots, c_n)^T$, and $d = (d_1, \dots, d_n)^T$, satisfy

$$d = Uc, \quad c = U^\dagger d.$$

Thus, “moving among bases” does not change the abstract state $|\chi\rangle$; it only changes its coordinate representation. In the qubit case, passing from the computational basis to the Hadamard basis is achieved by the Hadamard matrix

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

::... Basis in the Hilbert Space

If a qubit state has coordinates

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

then its coordinates in the basis $\{|+\rangle, |-\rangle\}$ are

$$\begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = H \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \alpha + \beta \\ \alpha - \beta \end{pmatrix},$$

so that

$$|\psi\rangle = \frac{\alpha + \beta}{\sqrt{2}} |+\rangle + \frac{\alpha - \beta}{\sqrt{2}} |-\rangle.$$

In this way, the formalism of Hilbert spaces makes precise both the existence of many admissible bases and the unitary transformations that relate them.

∴... Basis in the Hilbert Space

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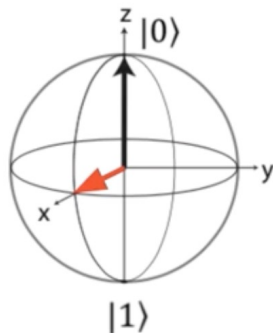
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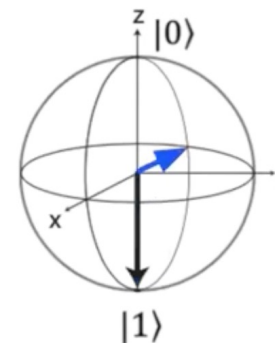
$$|0\rangle \Rightarrow |+\rangle$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



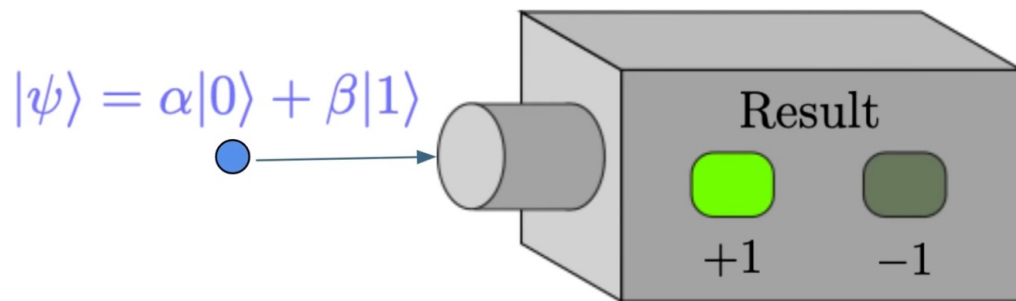
$$|1\rangle \Rightarrow |-\rangle$$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$



::... Measurements

When a qubit has been prepared, measurements try to extract information from it, answering the question: Is the qubit in state $|0\rangle$ or $|1\rangle$?



The measurement can be considered as a big box fed with the qubit in a prepared state considered as a big box fed with two states the qubit is.

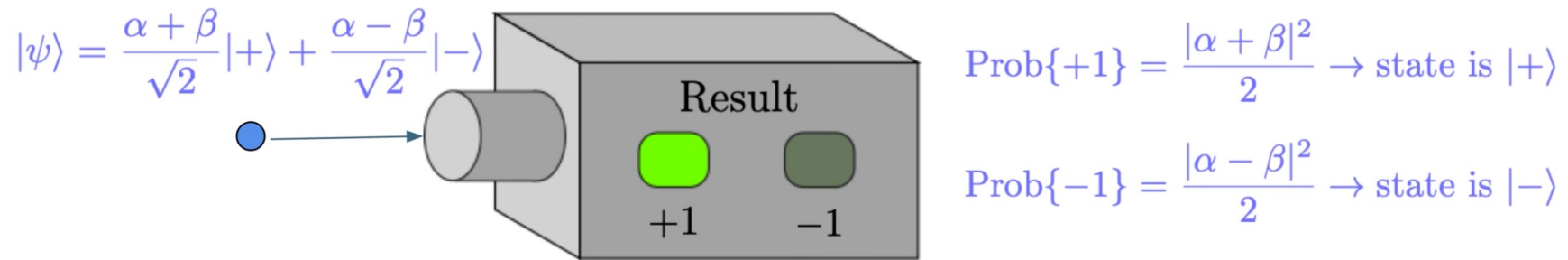
$\text{Prob}\{+1\} = |\alpha|^2 \rightarrow \text{state is } |0\rangle$

$\text{Prob}\{-1\} = |\beta|^2 \rightarrow \text{state is } |1\rangle$

The probabilities of being in one of the two states are given by the amplitudes in the superposition. After the measurements, the state collapses.

This is the Pauli-Z measurement, but it is not the only one we can do, as we have other two basis, so other two possible ways to make a measurements, by changing in the question the basis.

::... Measurements

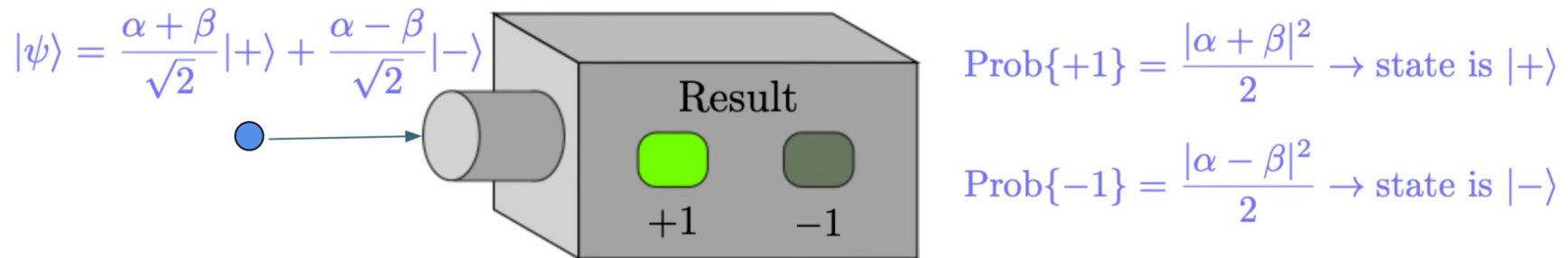


To know the probability of measure an outcome based an arbitrary base is given by the inner product of the basis elements with the qubit state:

$$\text{Prob}\{+1\} = |\langle b_0 | \psi \rangle|^2 \quad \text{Prob}\{-1\} = |\langle b_1 | \psi \rangle|^2$$

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::... Measurements



To know the probability of measure an outcome based an arbitrary base is given by the inner product of the basis elements with the qubit state:

Pauli Z basis $\{|0\rangle, |1\rangle\}$

$$\begin{aligned}\text{Prob}\{+1\} &= |\langle 0|\psi\rangle|^2 \\ &= |\alpha|^2\end{aligned}$$

$$\begin{aligned}\text{Prob}\{-1\} &= |\langle 1|\psi\rangle|^2 \\ &= |\beta|^2\end{aligned}$$

Pauli Y basis $\{|i\rangle, |-i\rangle\}$

$$\begin{aligned}\text{Prob}\{+1\} &= |\langle i|\psi\rangle|^2 \\ &= \frac{|\alpha - i\beta|^2}{2}\end{aligned}$$

$$\begin{aligned}\text{Prob}\{-1\} &= |\langle -i|\psi\rangle|^2 \\ &= \frac{|\alpha + i\beta|^2}{2}\end{aligned}$$

::... Measurements

A measurement is not defined solely by the state being measured, but also by the **observable** chosen by the experimenter. Mathematically, an observable is represented by a self-adjoint operator A on the Hilbert space, and its possible measurement outcomes are the eigenvalues of A .

The corresponding eigenvectors form an orthonormal basis (or, more generally, an orthogonal decomposition into eigenspaces). Measuring the system “in a given basis” therefore means performing the projective measurement associated with the observable whose eigenvectors are precisely that basis. For this reason, different bases correspond to different physical questions asked of the same qubit.

For a single qubit, the computational basis $\{|0\rangle, |1\rangle\}$ is the eigenbasis of the Pauli- Z operator,

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

since

$$Z|0\rangle = +|0\rangle, \quad Z|1\rangle = -|1\rangle.$$

::... Measurements

Thus, measuring in the computational basis is equivalent to measuring the observable Z . In this measurement, the two outcomes correspond to the eigenvalues $+1$ and -1 , and the state collapses onto $|0\rangle$ or $|1\rangle$.

If the qubit is in the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then the probabilities of obtaining the two outcomes are $|\alpha|^2$ and $|\beta|^2$. By contrast, the Hadamard basis $\{|+\rangle, |-\rangle\}$ is the eigenbasis of the Pauli- X operator,

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

because

$$X|+\rangle = +|+\rangle, \quad X|-\rangle = -|-\rangle.$$

Measuring in the basis $\{|+\rangle, |-\rangle\}$ is therefore a measurement of the observable X , not of Z . Although the same qubit state is being measured, the probabilities are now computed from its expansion in the X -basis.

Rewriting $|\psi\rangle$ in $\{|+\rangle, |-\rangle\}$ gives

$$|\psi\rangle = \frac{\alpha + \beta}{\sqrt{2}}|+\rangle + \frac{\alpha - \beta}{\sqrt{2}}|-\rangle,$$

::... Measurements

so the probabilities of the two outcomes are

$$\left| \frac{\alpha + \beta}{\sqrt{2}} \right|^2 \text{ and } \left| \frac{\alpha - \beta}{\sqrt{2}} \right|^2 .$$

These are generally different from the probabilities obtained in the computational basis, which shows explicitly that the measurement statistics depend on the chosen basis, hence on the chosen observable.

A simple example makes this clear. Suppose the qubit is prepared in the state $|0\rangle$. If one measures Z , the outcome is certain: one obtains the eigenvalue $+1$ with probability 1, and the post-measurement state remains $|0\rangle$. However, if one measures X , then

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}} ,$$

so the outcomes $+1$ and -1 occur with equal probability $1/2$. Thus, the same physical state $|0\rangle$ is perfectly definite with respect to the observable Z , but maximally indefinite with respect to the observable X .

::... Measurements

This is the precise sense in which different bases probe different properties of the qubit.

The converse example is equally instructive. If the qubit is prepared in the state $|+\rangle$, then an X -measurement gives the outcome $+1$ with certainty, because $|+\rangle$ is an eigenstate of X . But in the computational basis,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

so a Z -measurement yields $|0\rangle$ or $|1\rangle$ each with probability $1/2$. Again, the state is definite for one observable and indeterminate for another.

This reflects the fact that X and Z are distinct, non-commuting observables:

$$XZ \neq ZX.$$

Non-commutativity is the mathematical expression of measurement incompatibility: a state cannot, in general, have simultaneously sharp values for both observables.

::... Measurements

The same idea extends to the basis

$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, | -i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}},$$

which is the eigenbasis of the Pauli- Y operator,

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

A measurement in this basis corresponds to probing the observable Y .

Geometrically, the three Pauli observables X , Y , and Z correspond to three orthogonal axes on the Bloch sphere. Measuring in the Z -basis asks whether the qubit is aligned with the north-south axis, measuring in the X -basis asks whether it is aligned with the left-right axis, and measuring in the Y -basis asks whether it is aligned with the front-back axis. The different bases therefore represent different experimental orientations, each revealing a different aspect of the same quantum state.

The exact **physical meaning** of observables depends on how the abstract qubit is encoded.

::... Measurements

If the qubit is a spin- $\frac{1}{2}$ particle and $|0\rangle, |1\rangle$ are chosen as spin-up and spin-down along the z -axis, then the observable Z corresponds to spin measured along the z -axis, while the other observable are related to the other axis. In that case, measuring X means asking whether the spin points along $+x$ or $-x$.

If the qubit is encoded in photon polarization and $|0\rangle, |1\rangle$ represent horizontal and vertical polarization, then X corresponds to measuring in the **diagonal/anti-diagonal polarization basis**, then measuring X means asking whether the photon is polarized along one diagonal direction or the orthogonal one, while Y corresponds to measurement in the circular polarization basis.

Thus, the Pauli operators Z, Y, X always represents the same abstract observable in qubit theory: **X detects the real part of the coherence, Y detects the imaginary part of the coherence, and Z detects the difference in occupation between the computational basis states,** but its concrete laboratory meaning depends on the physical realization of the qubit.

::... Pauli Operators

The **Pauli operators** are the three fundamental linear operators acting on the two-dimensional complex Hilbert space $\mathbb{C}^2$ of a single qubit. They are represented by the matrices

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are usually considered together with the identity operator

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

satisfying $X^2 = Y^2 = Z^2 = I$.

Their action on the computational basis $\{|0\rangle, |1\rangle\}$ is given by

$$\begin{aligned} X|0\rangle &= |1\rangle, & X|1\rangle &= |0\rangle, \\ Y|0\rangle &= i|1\rangle, & Y|1\rangle &= -i|0\rangle, \\ Z|0\rangle &= |0\rangle, & Z|1\rangle &= -|1\rangle. \end{aligned}$$

Thus, X exchanges the basis states, Z changes the phase of $|1\rangle$, and Y combines a bit flip with a phase factor.

::... Pauli Operators

They also satisfy the fundamental commutation and anticommutation relations

$$[X, Y] = 2iZ, [Y, Z] = 2iX, [Z, X] = 2iY,$$

where $[A, B] = AB - BA$

$$\{X, Y\} = \{Y, Z\} = \{Z, X\} = 0,$$

where $\{A, B\} = AB + BA$.

These relations are what make the Pauli operators the algebraic foundation of single-qubit theory, spin- $\frac{1}{2}$ systems, and Bloch-sphere rotations.

::... The ket-bra notation

The **ket–bra notation** (or **Dirac notation**) is the standard symbolic language used to represent quantum states, dual states, inner products, and operators in a compact and mathematically precise way. It is especially convenient because it separates the abstract quantum state from any particular matrix representation, while still allowing one to pass immediately to coordinates once a basis is chosen.

A **ket**, written $|\psi\rangle$, denotes a vector in the Hilbert space of the system, and the ket notation is the abstract version of a column vector.

○

$|0\rangle$

Probability of measuring

$\rightarrow 0: 100\%$
 $\rightarrow 1: 0\%$

Phase : Positive (+)

Quantum State : $1|0\rangle + 0|1\rangle$

●

$|1\rangle$

Probability of measuring

$\rightarrow 0: 0\%$
 $\rightarrow 1: 100\%$

Phase : Positive (+)

Quantum State : $0|0\rangle + 1|1\rangle$

○

●

Probability of measuring

$\rightarrow 0: 50\%$
 $\rightarrow 1: 50\%$

Phase : Positive (+)

Quantum State : $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

○

●

Probability of measuring

$\rightarrow 0: 50\%$
 $\rightarrow 1: 50\%$

Phase : Negative (-)

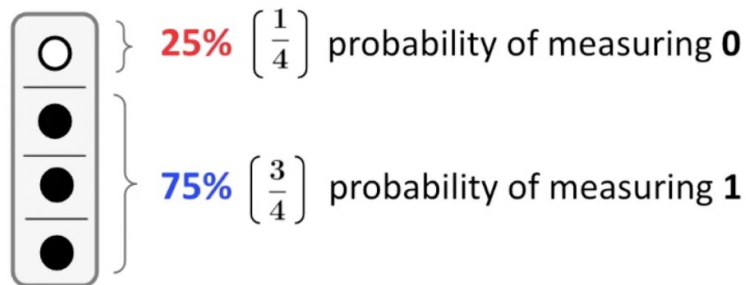
Quantum State : $\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$

State of Qubit
 $|\psi\rangle = a|0\rangle + b|1\rangle$
 $|a|^2$: Probability measuring 0
 $|b|^2$: Probability measuring 1
 $+/-$: Indicates phase
 Constrained by: $|a|^2 + |b|^2 = 1$

::... The ket-bra notation

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State of Qubit
 $|\psi\rangle = a|0\rangle + b|1\rangle$
 $|a|^2$: Probability measuring 0
 $|b|^2$: Probability measuring 1
+/- : Indicates phase
Constrained by: $|a|^2 + |b|^2 = 1$

$$|\psi\rangle = \frac{1}{\sqrt{4}}|0\rangle + \frac{\sqrt{3}}{\sqrt{4}}|1\rangle$$
$$|a|^2 = \left(\frac{1}{\sqrt{4}}\right)^2 \quad |b|^2 = \left(\frac{\sqrt{3}}{\sqrt{4}}\right)^2$$

::... The ket-bra notation

The **ket–bra notation** (or **Dirac notation**) is the standard symbolic language used to represent quantum states, dual states, inner products, and operators in a compact and mathematically precise way. It is especially convenient because it separates the abstract quantum state from any particular matrix representation, while still allowing one to pass immediately to coordinates once a basis is chosen.

A **bra**, written $\langle\psi|$, denotes the dual vector associated with the ket $|\psi\rangle$, or its adjoint. Formally, if $|\psi\rangle \in \mathcal{H}$, then $\langle\psi| \in \mathcal{H}^*$, the dual space of continuous linear functionals on $\mathcal{H}$. In finite-dimensional quantum mechanics, the bra is obtained from the ket by taking the conjugate transpose:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \Rightarrow \langle\psi| = (\alpha^* \quad \beta^*).$$

Accordingly, if $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then

$$\langle\psi| = \alpha^*\langle 0| + \beta^*\langle 1|.$$

::... The ket-bra notation

The expression formed by a bra acting on a ket,

$$\langle \phi | \psi \rangle = \langle \phi || \psi \rangle,$$

is the **inner product** between the states $|\phi\rangle$ and $|\psi\rangle$. This is a complex number. If $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle$, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then

$$\langle \phi | \psi \rangle = \gamma^* \alpha + \delta^* \beta.$$

$$\langle \psi | \psi \rangle = |\alpha|^2 + |\beta|^2,$$

which equals 1 for a normalized qubit state. The orthonormality of the computational basis is expressed by

$$\langle 0 | 0 \rangle = 1, \langle 1 | 1 \rangle = 1, \langle 0 | 1 \rangle = 0, \langle 1 | 0 \rangle = 0.$$

In qubit theory, the inner product has direct physical meaning because it determines amplitudes and probabilities. For example, if $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then the amplitude of finding the qubit in the state $|0\rangle$ is

$$\langle 0 | \psi \rangle = \alpha,$$

and the probability of obtaining that outcome in a measurement in the computational basis is

$$|\langle 0 | \psi \rangle|^2 = |\alpha|^2.$$

::... The ket-bra notation

$$|\langle 1 | \psi \rangle|^2 = |\beta|^2.$$

This is the Born rule expressed naturally in ket–bra notation.

The reverse combination,

$$|\phi\rangle\langle\psi|,$$

is called an **outer product**. Unlike an inner product, it is not a scalar but a **linear operator** on the Hilbert space. In matrix language, $|\phi\rangle\langle\psi|$ is the product of a column vector and a row vector.

For qubits, outer products are fundamental because they describe projectors, basis operators, and quantum gates. For example,

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

are the projectors onto the basis states $|0\rangle$ and $|1\rangle$. Their sum gives the identity operator:

$$I = |0\rangle\langle 0| + |1\rangle\langle 1|.$$

::... The ket-bra notation

This is the completeness relation for the computational basis. It means that every qubit state can be decomposed in that basis, and it is also the algebraic expression of the fact that $\{|0\rangle, |1\rangle\}$ spans $\mathbb{C}^2$.

$$\begin{aligned}|0\rangle\langle 0|\psi\rangle &= |0\rangle\langle 0|(\alpha|0\rangle + \beta|1\rangle) \\ &= \alpha|0\rangle\langle 0|0\rangle + \beta|0\rangle\langle 0|1\rangle \\ &= \alpha|0\rangle\end{aligned}$$

$$\begin{aligned}|1\rangle\langle 1|\psi\rangle &= |1\rangle\langle 1|(\alpha|0\rangle + \beta|1\rangle) \\ &= \alpha|1\rangle\langle 1|0\rangle + \beta|1\rangle\langle 1|1\rangle \\ &= \beta|1\rangle\end{aligned}$$

$|0\rangle\langle 0|$ “**projects**” onto $|0\rangle$

$|1\rangle\langle 1|$ “**projects**” onto $|1\rangle$

This is a way to formally express the operation of measurements with respect to the observable Z of the computational basis.

::... The ket-bra notation

This is the completeness relation for the computational basis. It means that every qubit state can be decomposed in that basis, and it is also the algebraic expression of the fact that $\{|0\rangle, |1\rangle\}$ spans $\mathbb{C}^2$.

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$|0\rangle\langle 0|$ “**projects**” onto $|0\rangle$

$|1\rangle\langle 1|$ “**projects**” onto $|1\rangle$

The same can be done w.r.t. the other two basis:

$$|+\rangle\langle +|\psi\rangle = \frac{\alpha + \beta}{\sqrt{2}}|+\rangle$$

$$|-\rangle\langle -|\psi\rangle = \frac{\alpha - \beta}{\sqrt{2}}|-\rangle$$

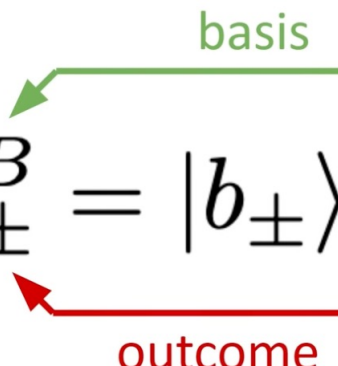
$$|i\rangle\langle i|\psi\rangle = \frac{\alpha - i\beta}{\sqrt{2}}|i\rangle$$

$$|-i\rangle\langle -i|\psi\rangle = \frac{\alpha + i\beta}{\sqrt{2}}|-i\rangle$$

::... The ket-bra notation

This is the completeness relation for the computational basis. It means that every qubit state can be decomposed in that basis, and it is also the algebraic expression of the fact that $\{|0\rangle, |1\rangle\}$ spans $\mathbb{C}^2$.

We can generalise this result by expressing projections in any generic basis of the Hilbert space as follows:

$$\Pi_{\pm}^B = |b_{\pm}\rangle\langle b_{\pm}|$$


The diagram shows the equation $\Pi_{\pm}^B = |b_{\pm}\rangle\langle b_{\pm}|$. A green arrow points from the word "basis" to the ket $|b_{\pm}\rangle$. A red arrow points from the word "outcome" to the bra $\langle b_{\pm}|$.

Pauli X basis

$$\Pi_{+}^X = |+\rangle\langle +|$$

$$\Pi_{-}^X = |-\rangle\langle -|$$

Pauli Y basis

$$\Pi_{+}^Y = |i\rangle\langle i|$$

$$\Pi_{-}^Y = |-i\rangle\langle -i|$$

Pauli Z basis

$$\Pi_{+}^Z = |0\rangle\langle 0|$$

$$\Pi_{-}^Z = |1\rangle\langle 1|$$

::... The ket-bra notation

Using outer products, many important qubit operators can be written in a compact and transparent way. For instance, the Pauli operators may be expressed as

$$\begin{aligned}X &= |0\rangle\langle 1| + |1\rangle\langle 0|, \\Y &= -i|0\rangle\langle 1| + i|1\rangle\langle 0|, \\Z &= |0\rangle\langle 0| - |1\rangle\langle 1|.\end{aligned}$$

These formulas make their action immediately clear. For example,

$$X|0\rangle = |1\rangle, X|1\rangle = |0\rangle,$$

so X flips the computational basis states.

$$\begin{aligned}|0\rangle\langle 0| &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & |\psi\rangle\langle\psi| &= \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix} \\|1\rangle\langle 1| &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\end{aligned}$$

Outer product can be used to also describe the quantum state, written in the matrix form rather than the column one of the ket.

::... The ket-bra notation

Using outer products, many important qubit operators can be written in a compact and transparent way. For instance, the Pauli operators may be expressed as

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These formulas make their action immediately clear. For example,

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so X flips the computational basis states.

Dirac notation is also the natural language for **expectation values**. If A is an observable or, more generally, a linear operator on the qubit, then the expectation value of A in the state $|\psi\rangle$ is $\langle\psi|A|\psi\rangle$.

This is a scalar. For example, the expectation value of Z in the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

is

$$\langle\psi|Z|\psi\rangle = |\alpha|^2 - |\beta|^2.$$

::... The ket-bra notation

Using outer products, many important qubit operators can be written in a compact and transparent way. For instance, the Pauli operators may be expressed as

$$\begin{aligned}X &= |0\rangle\langle 1| + |1\rangle\langle 0|, \\Y &= -i|0\rangle\langle 1| + i|1\rangle\langle 0|, \\Z &= |0\rangle\langle 0| - |1\rangle\langle 1|.\end{aligned}$$

These formulas make their action immediately clear. For example,

$$X|0\rangle = |1\rangle, X|1\rangle = |0\rangle,$$

so X flips the computational basis states.

$$\langle Z \rangle = \langle \psi | Z | \psi \rangle$$

$$\begin{aligned}\langle Z \rangle &= (\alpha^* \quad \beta^*) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\&= (\alpha^* \quad \beta^*) \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\&= |\alpha|^2 - |\beta|^2\end{aligned}$$

::... The ket-bra notation

Hence, ket–bra notation not only represents states, but also encodes the probabilistic predictions of measurements.

The notation extends naturally to **alternative bases**. For instance, using the Hadamard basis the same qubit state may also be written as

$$| \psi \rangle = \langle + | \psi \rangle | + \rangle + \langle - | \psi \rangle | - \rangle.$$

This shows that the ket notation is independent of the chosen basis: the abstract state $| \psi \rangle$ does not change, only its expansion coefficients do. The coefficients are always obtained by taking inner products with the corresponding bra vectors.

For systems of more than one qubit, Dirac notation uses the tensor product. If $| a \rangle$ and $| b \rangle$ are single-qubit states, then the joint state is written

$$| a \rangle \otimes | b \rangle,$$

often abbreviated as

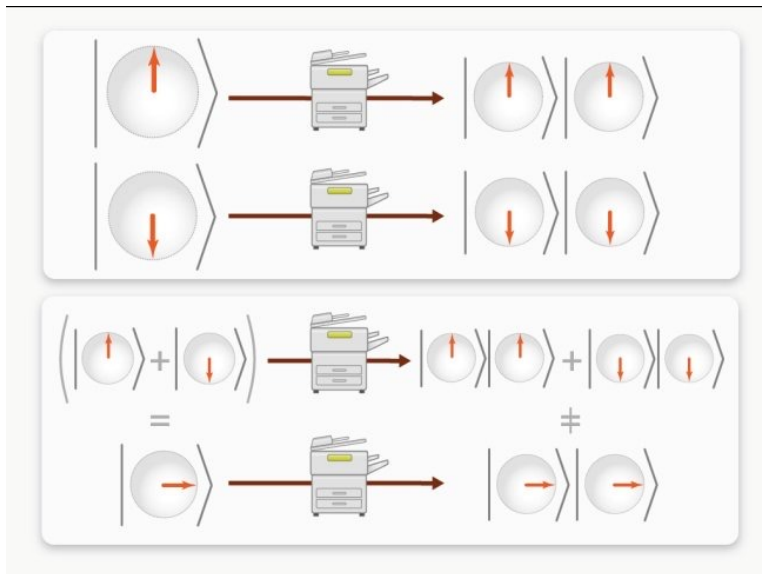
$$| ab \rangle.$$

::... The ket-bra notation

Conceptually, the main advantage of ket–bra notation is that it distinguishes clearly between vectors, dual vectors, scalars, and operators. The ket $|\psi\rangle$ is a state vector, the bra $\langle\psi|$ is its dual, the bracket $\langle\phi|\psi\rangle$ is a scalar overlap, and the outer product $|\phi\rangle\langle\psi|$ is an operator.

In qubit theory, this notation provides a unified way to express state preparation, basis expansion, measurement amplitudes, projectors, quantum gates, and multipartite states with both mathematical precision and physical transparency.

::... No-cloning Theorem



The no-cloning theorem: there exists no physically admissible operation capable of producing a perfect copy of an arbitrary unknown quantum state, which is the superposition of the two possible states.

Formally, there is no universal unitary transformation that maps $|\psi\rangle|0\rangle$ to $|\psi\rangle|\psi\rangle$ for every possible input state $|\psi\rangle$.

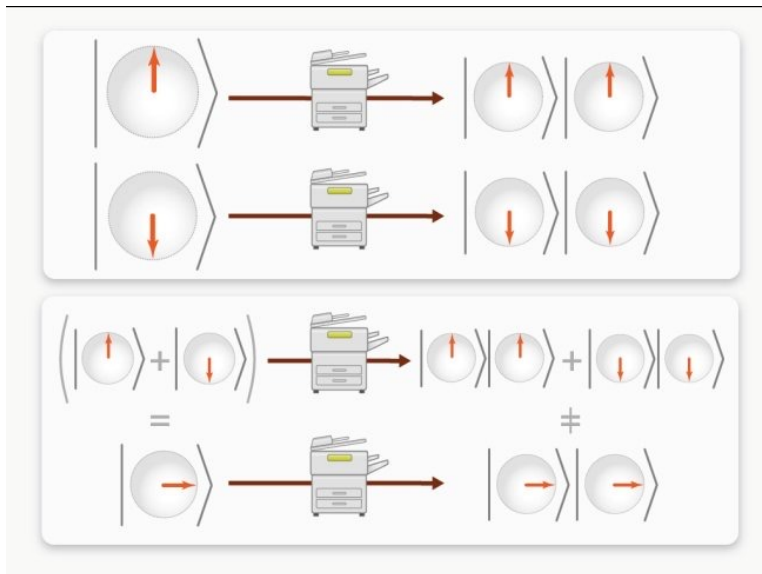
Proof (by contradiction): Suppose that there exists a unitary C (that copies an arbitrary unknown qforte), then:

$$C(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

Let is consider two quantum states $|\psi_1\rangle$ and $|\psi_2\rangle$, so that

$C(|\psi_1\rangle \otimes |0\rangle) = |\psi_1\rangle \otimes |\psi_1\rangle$ and $C(|\psi_2\rangle \otimes |0\rangle) = |\psi_2\rangle \otimes |\psi_2\rangle$, but C is unitary so $C^\dagger C = I$.

::... No-cloning Theorem



The **no-cloning theorem**: there exists no physically admissible operation capable of producing a perfect copy of an arbitrary unknown quantum state, which is the superposition of the two possible states. Formally, there is no universal unitary transformation that maps $|\psi\rangle|0\rangle$ to $|\psi\rangle|\psi\rangle$ for every possible input state $|\psi\rangle$.

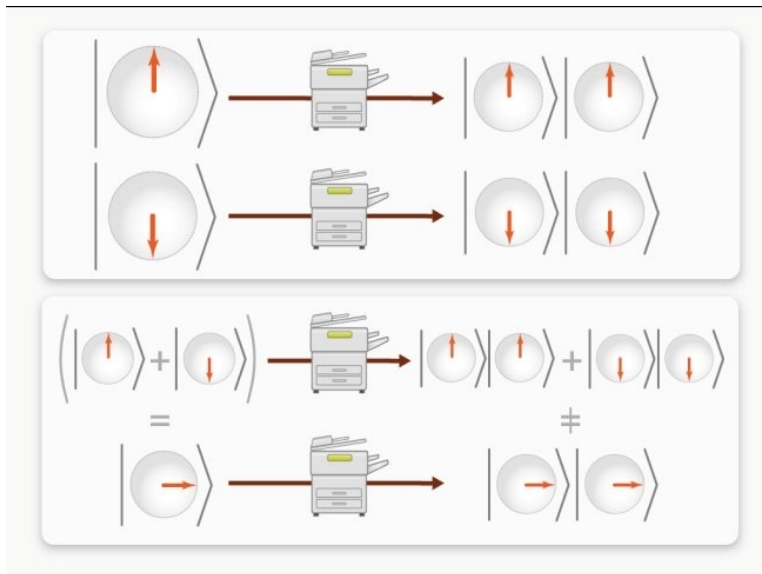
Proof (by contradiction): Suppose that there exists a unitary C (that copies an arbitrary unknown qforte), then:

$$C(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

$$\begin{aligned} \langle\psi_1|\psi_2\rangle &= \langle\psi_1|\psi_2\rangle \langle 0|0\rangle = \\ &= (\langle\psi_1| \otimes \langle 0|) C^\dagger C(|\psi_2\rangle \otimes |0\rangle) = \\ &= (\langle\psi_1| \otimes \langle\psi_1|)(|\psi_2\rangle \otimes |\psi_2\rangle) = (\langle\psi_1|\psi_2\rangle)^2 \end{aligned}$$

There is a contradiction, and the theorem is proved.

::... No-cloning Theorem

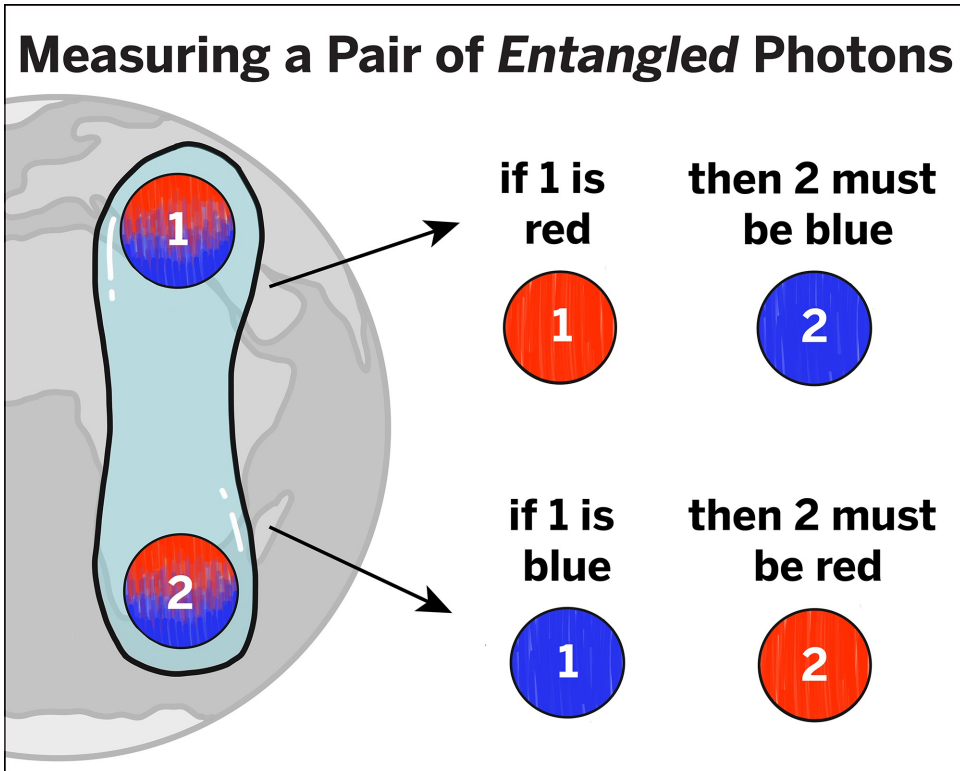


The **no-cloning theorem**: there exists no physically admissible operation capable of producing a perfect copy of an arbitrary unknown quantum state, which is the superposition of the two possible states. Formally, there is no universal unitary transformation that maps $|\psi\rangle|0\rangle$ to $|\psi\rangle|\psi\rangle$ for every possible input state $|\psi\rangle$.

This impossibility is a consequence of the linearity of quantum mechanics, and it implies that quantum information cannot be duplicated in the same unrestricted way as classical information. As a result, many classical strategies based on copy-ing data, such as straightforward redundancy-based error correction, cannot be transferred directly to the quantum setting. Quantum measurement is intrinsically disturbing: when a qubit is measured, the outcome reveals only partial information by the measurement basis, and the pre-measurement state is generally altered by the observation.

::... Entanglement

Another defining feature of qubits is **entanglement**, which arises in composite quantum systems: when two or more qubits are entangled, the global state of the system cannot be decomposed into a tensor product of independent single-qubit states. In such cases, the subsystem states are not individually sufficient to characterize the whole system, because the essential information resides in the correlations between the qubits.



A canonical example is provided by the Bell states, which are maximally entangled two-qubit states. In these states, measurement outcomes obtained on one qubit are strongly correlated with those obtained on the other, in a way that cannot be reproduced by any classical model based solely on local hidden variables.

::... Entanglement

A **Bell state** is any one of the four canonical **maximally entangled** pure states of a two-qubit system. In the computational basis $\{|00\rangle, |11\rangle, |01\rangle, |10\rangle\}$, they are

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle).$$

These four states form an orthonormal basis of the two-qubit Hilbert space $\mathbb{C}^2 \otimes \mathbb{C}^2$, usually called the **Bell basis**. None of them is a product state; each represents genuine entanglement between the two qubits.

Their most important structural feature is that they are **maximally entangled**. This means that although the joint two-qubit state is pure, each individual qubit, considered by itself, is completely indefinite. For example, if the pair is in the state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle),$$

then neither qubit alone has a definite state.

::... Entanglement

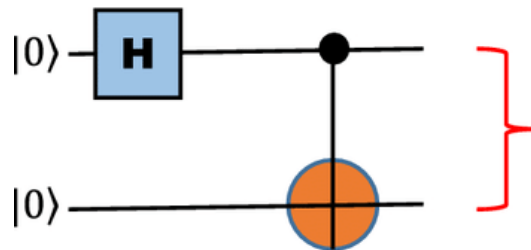
So each single qubit looks maximally mixed, but the pair as a whole has perfect correlations. If both qubits are measured in the computational basis, the outcomes are individually random, yet they always agree: one obtains either 00 or 11, each with probability $1/2$. By contrast, in $|\Psi^+\rangle$ the outcomes are always opposite, namely 01 or 10.

A useful operational fact is that Bell states can be created from simple product states by a short quantum circuit. Starting from $|00\rangle$, one applies a Hadamard gate to the first qubit and then a CNOT gate, obtaining

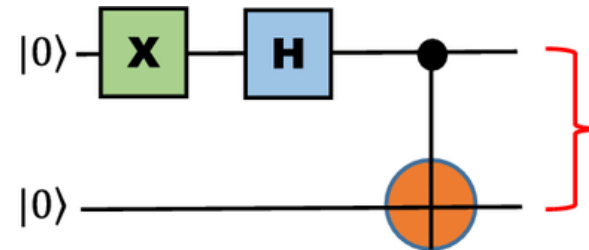
$$|00\rangle \xrightarrow{H \otimes I} \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = |\Phi^+\rangle.$$

::... Entanglement

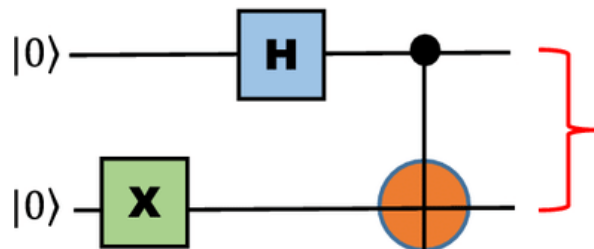
Specific quantum circuits allows to prepare a pair of entangled qubits:



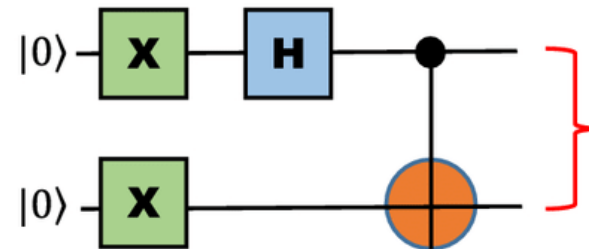
$$|\psi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



$$|\psi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$



$$|\phi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$



$$|\phi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

... Entanglement

$C = 2.99792458 \times 10^8$
 $\hbar = 6.6260699 \times 10^{-34}$
 $g = 6.6740719 \times 10^{-11}$
 $M = 2.17647051 \times 10^{-8}$
 $l = 1.61612928 \times 10^{-35}$
 $P = 8.987954 \times 10^{16}$
 $P_m = 1.6726219 \times 10^{-27}$
 $A = 5.2917721 \times 10^{-11}$
 $M_e = 9.10938358 \times 10^{-31}$
 $K = \frac{1}{4\pi\epsilon_0} = 8.987 \times 10^9$
 $E_e = 8.187102 \times 10^{-14}$
 $R = \frac{c}{4\pi A} = 1.0973731$

$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} = \frac{8\pi g(\frac{mc^2}{2})^2}{C^4} = 2\pi g m^2 = Ch = 2\pi |m c^2 = \frac{2\pi P l P_m C^2}{(4\pi/3)}$

$10^{29} = \frac{g 10^{85} P_m \alpha}{C^2}$

$g(P) = g \frac{m^2}{P_m^2}$

$l = \frac{g m}{C^2}$

$P l = \frac{g(P)(\frac{4\pi}{3} P_m)}{C^2}$

$\frac{P_m^2}{m_e^2} = \frac{\pi \alpha}{128.5 A}$

$A^2 = \frac{g(P) \pi m_e}{128.5 C^2}$

$Ch = 2\pi A m_e C^2 \alpha$

$137.036 = \frac{g m^2}{K e^2} = \frac{Ch}{K e^2} = \frac{1}{\alpha}$

2^h qubit

$n^{\frac{1}{2\pi}} = e^{2\pi i}$
 $Ch = EL$
 e^+

$i\hbar \frac{\partial \psi}{\partial t} = \frac{\hbar c}{i} \left(\sum a_i \frac{\partial \psi}{\partial x_i} \right) + a_4 m c^2 \psi$

$2ChR2A = 2Ke^2 = Ch\alpha + m_e C^2 \alpha A$

ER

The **Einstein-Podolsky-Rosen (EPR) paradox** states that by measuring one particle, one could predict with certainty the corresponding property of the distant particle, without directly disturbing it. EPR took this to suggest that the distant particle must already possess a definite “element of reality” for that property.

The **Bohm spin version** of the EPR setup: if two observers measure spin along the same axis on the two particles, their results are perfectly anti-correlated.

::... Entanglement

Bell's theorem - If one assumes local hidden variables of the sort EPR hoped might complete quantum mechanics, then the correlations among measurements must satisfy certain inequalities. Quantum mechanics predicts violations of those inequalities for suitable measurements on entangled states, and Bell's theorem is precisely that no theory satisfying those locality-type constraints can reproduce all quantum predictions. In this sense, the EPR paradox raised the conceptual problem, while Bell's theorem turned it into a mathematically testable statement.

::... Noisy World - Mixed states

Up to now, only noiseless quantum states have been considered, but this is not sufficient for real-world applications.

Desired
state
 $|\psi\rangle$

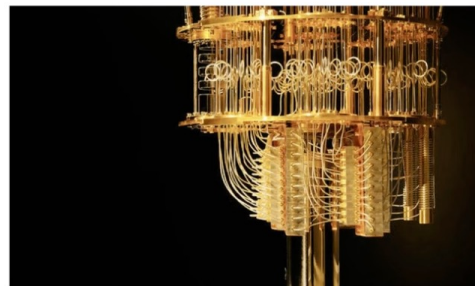


Quantum lab

Prepared state $|\psi'\rangle \neq |\psi\rangle$

In reality, the preparation phase return $|\psi_1\rangle$ with probability p_1 , $|\psi_2\rangle$ with probability p_2 , and so on. This also applies to the information processing. Even if we were able to have a preparation to an idea state $|\psi\rangle$, then the application of a unitary operator is not ideal.

Desired
unitary
 U



Quantum computer

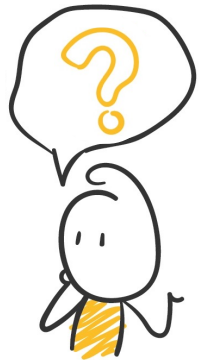
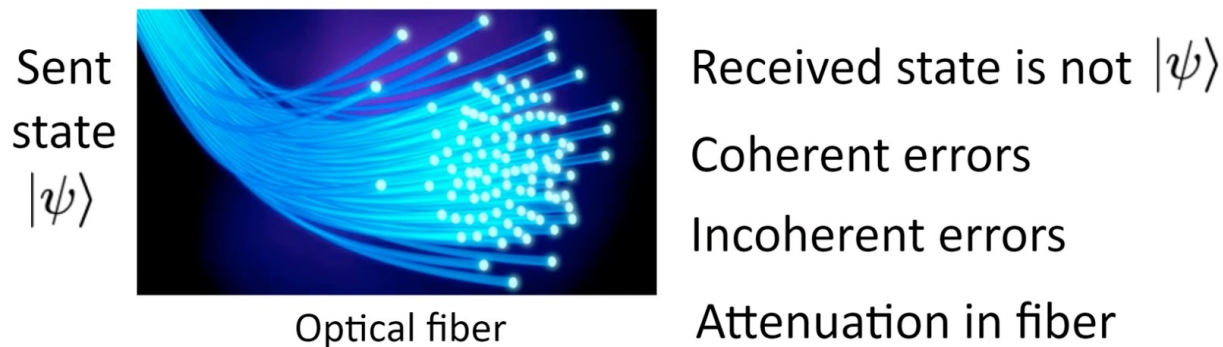
Output is not $U|\psi\rangle$

Coherent error: $U' \neq U$

Incoherent error: not
even a unitary.

::... Noisy World - Mixed states

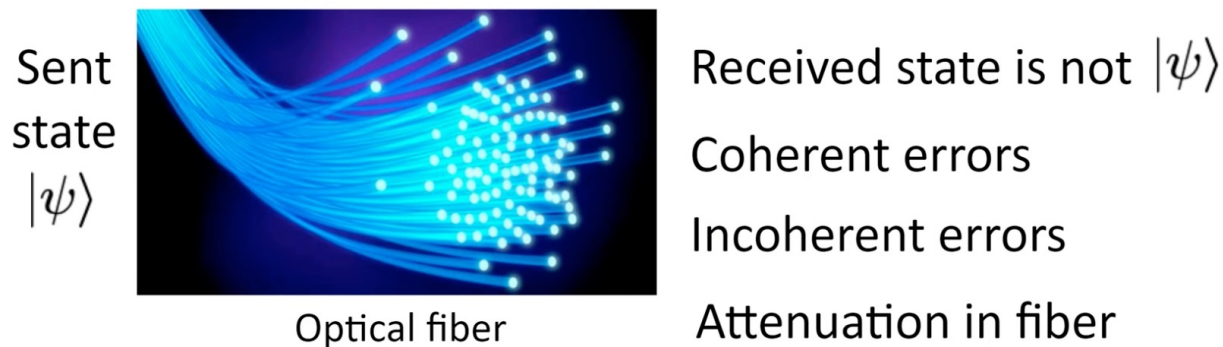
The same can happen when we try to communicate a prepared idea state $|\psi\rangle$ over a very long optical fiber.



How can we describe these effects?
How can we deal with them?

::... Noisy World - Mixed states

The same can happen when we try to communicate a prepared idea state $|\psi\rangle$ over a very long optical fiber.



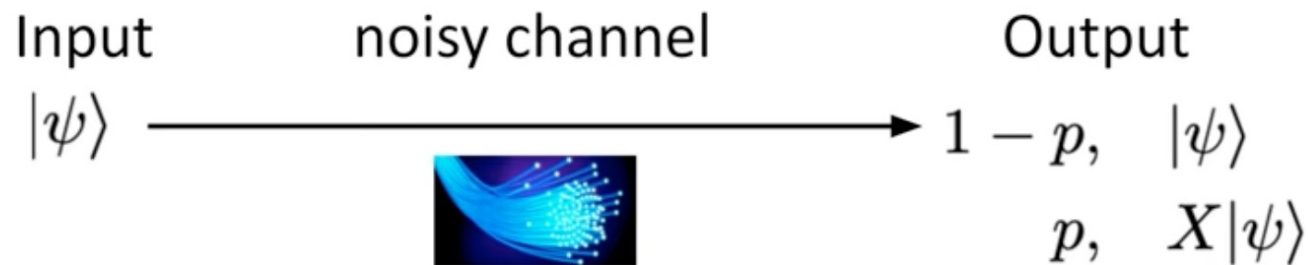
Superpositions of kets represent the perfect knowledge of the quantum states, and are not able to describe the probabilistic nature of a set of possible quantum states $|\psi_i\rangle$ each with probability p_i . The solution is using a mixture of outer products:

$$p_1|\psi_1\rangle\langle\psi_1| + p_2|\psi_2\rangle\langle\psi_2| + \cdots + p_n|\psi_n\rangle\langle\psi_n|$$

This is the description of noisy **mixed states**.

::... Noisy World - Mixed states

Let us consider a flip channel:



Output: $\rho = (1 - p)|\psi\rangle\langle\psi| + pX|\psi\rangle\langle\psi|X$

indicating that the output is the probability p that the state did not change and the probability $(1-p)$ that the state has been flipped.

$$|\psi\rangle = 0 \quad \rho = (1 - p)|0\rangle\langle 0| + p|1\rangle\langle 1| = \begin{pmatrix} 1 - p & 0 \\ 0 & p \end{pmatrix}$$

$$|\psi\rangle = 1 \quad \rho = (1 - p)|1\rangle\langle 1| + p|0\rangle\langle 0| = \begin{pmatrix} p & 0 \\ 0 & 1 - p \end{pmatrix}$$

::... Noisy World - Mixed states

A density matrix is the most general description of a quantum state.

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

This formalism can describe pure states with probabilities $p_i = 0$ with the only exception of one related to the pure state $|\psi_j\rangle$, as well as mixed states with unitary and unitary noise affecting real quantum computations.

Pure states has to be normalised, i.e., $|\alpha|^2 + |\beta|^2 = 1$; the normalization has to be imposed also in the density matrix formalisms. To this aim, let us introduce the trace of a matrix as follows:

Diagram illustrating the trace of a matrix. The trace is defined as the sum of the diagonal elements of a square matrix A :

$$\text{Tr}\{A\} = \sum_i A_{ii}$$

Labels in the diagram: "trace" (green), "square matrix" (red), "sum of diagonal terms" (blue).

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix}$$

$$\text{Tr}\{A\} = A_{11} + A_{22} + A_{33}$$

::... Noisy World - Mixed states

A density matrix is the most general description of a quantum state.

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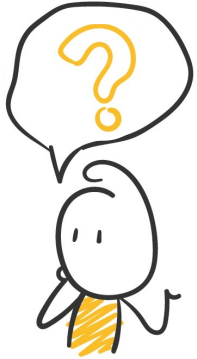
Pure states has to be normalised, i.e., $|\alpha|^2 + |\beta|^2 = 1$; the normalization has to be imposed also in the density matrix formalisms. The normalization is imposed as follows:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad \text{Tr}\{\rho\} = 1$$

$$|\psi\rangle\langle\psi| = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} (\alpha^* \quad \beta^*) = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix} \quad \text{Tr}\{|\psi\rangle\langle\psi|\} = |\alpha|^2 + |\beta|^2 = 1$$

::... Noisy World - Mixed states

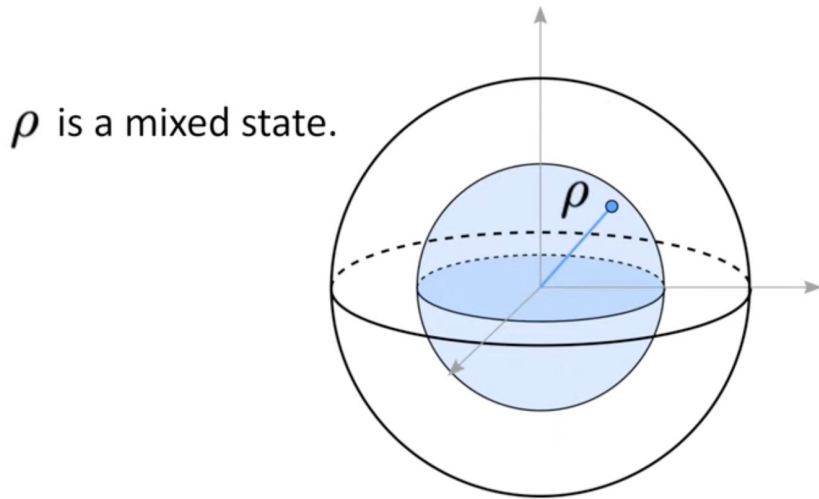
A pure state is a point on the Bloch sphere.



what about a mixed state? Can we have a visual representation on the sphere?

::... Noisy World - Mixed states

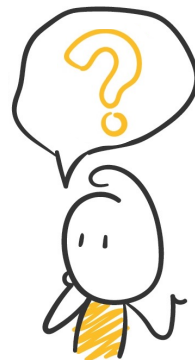
A pure state is a point on the Bloch sphere, while a mixed state in a density matrix representation is a point ρ living inside of the sphere, and the radius connecting the center of the sphere and point representing the mixed state has length less than 1.



Equal superposition

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

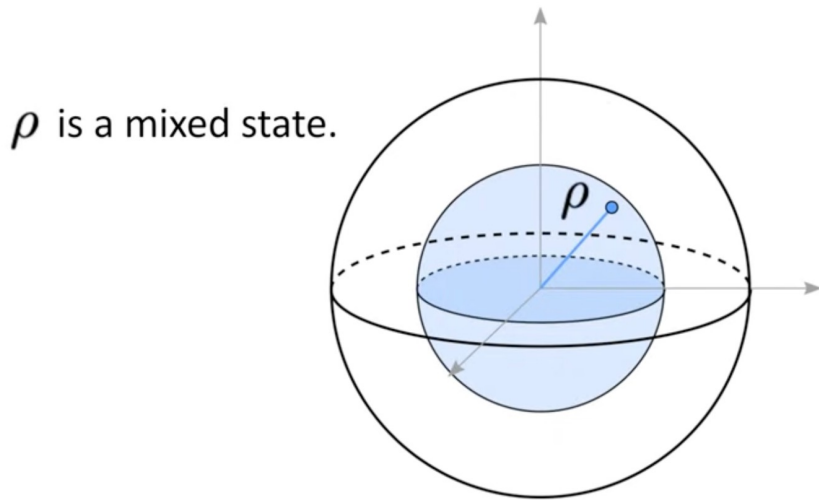
When the radius is 0 then the mixed state is equal to the center, which is the equal superposition of the two pure states $|0\rangle, |1\rangle$.



Is this correct?

::... Noisy World - Mixed states

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Equal superposition

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Maximally mixed state

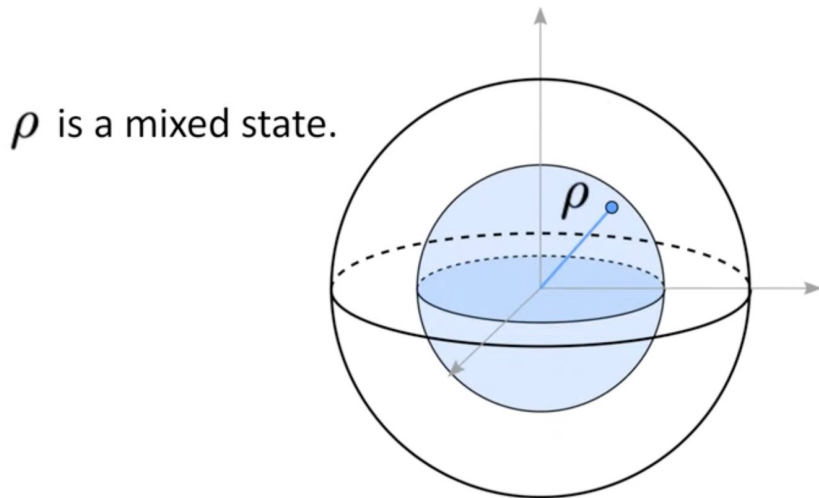
$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

When the radius is 0 then the mixed state is equal to the center, which is the equal superposition of the two pure states $|0\rangle, |1\rangle$.

The correct one is the maximally mixed state, as in the previous case we have perfect knowledge, while now we have uncertainty.

::... Noisy World - Mixed states

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Equal superposition

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Maximally mixed state

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

Measure both in the Pauli Z basis.

$$\begin{aligned}\text{Prob}\{+1\} &= \frac{1}{2} \text{ (state } |0\rangle) \\ \text{Prob}\{-1\} &= \frac{1}{2} \text{ (state } |1\rangle)\end{aligned}$$

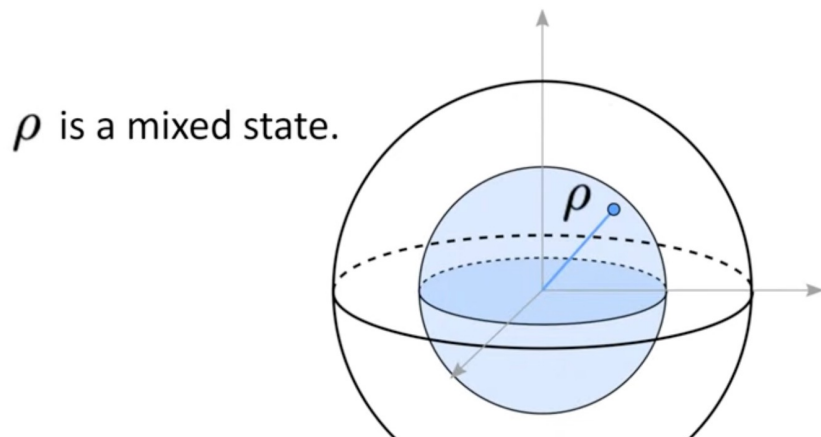
$$\begin{aligned}\text{Prob}\{+1\} &= \frac{1}{2} \text{ (state } |0\rangle) \\ \text{Prob}\{-1\} &= \frac{1}{2} \text{ (state } |1\rangle)\end{aligned}$$



What is the difference among them?
When I measure them, I get the same results.

::... Noisy World - Mixed states

A pure state is a point on the Bloch sphere, while a mixed state in a density matrix representation is a point ρ living inside of the sphere, and the radius connecting the center of the sphere and point representing the mixed state has length less than 1.



Equal superposition

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Measure in the Pauli X basis.

$$= |+\rangle$$

$$\text{Prob}\{+1\} = 1 \text{ (state } |+\rangle)$$

$$\text{Prob}\{-1\} = 0 \text{ (state } |-\rangle)$$

Maximally mixed state

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

$$= \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -|$$

$$\text{Prob}\{+1\} = \frac{1}{2} \text{ (state } |+\rangle)$$

$$\text{Prob}\{-1\} = \frac{1}{2} \text{ (state } |-\rangle)$$

The two states are very different, and to notice this, measurements need to be done in a different basis.

::... Noisy World - Mixed states

Let's summarize the difference between pure and mixed states:

	Pure states	Mixed states
Notation	$ \psi\rangle$	ρ
Represented by	Vectors and matrices	Matrices only
Normalization	$ \langle\psi \psi\rangle ^2 = 1$	$\text{Tr}\{\rho\} = 1$
Knowledge of the state	Perfect	Imperfect

::... Noisy World - Mixed states

Noise occurring during state preparation, processing, and communication transforms pure states into mixed states. How can we measure with a single number the deviation between the expected state and the one we obtain?

::... Noisy World - Mixed states

Noise occurring during state preparation, processing, and communication transforms pure states into mixed states. How can we measure with a single number the deviation between the expected state and the one we obtain?

Fidelity is such a number and capture how different is the output state from the expected one, and can be formally expressed as

$$F(\rho, |\psi\rangle) = \langle\psi|\rho|\psi\rangle$$

where ρ is the obtained output and $|\psi\rangle$ is the desired output. Fidelity gives the quality of the output and varies between 0 and 1:

$$0 \leq F(\rho, |\psi\rangle) \leq 1$$

when fidelity is equal to 0 then the obtained output is orthogonal to the desired one, and we can distinguish the obtained output from the desired one. If fidelity is 1, the obtained output is equal to the desired one.

::... Noisy World - Mixed states

Let us consider that the obtained output is the expected one:

$$\rho = |\psi\rangle\langle\psi|, \quad F(\rho, |\psi\rangle) = \langle\psi|\psi\rangle^2 = 1$$

Consider now that ρ is the maximally mixed state due to a very noise operation or channel:

$$\begin{aligned} \rho &= \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|, \quad |\psi\rangle = |0\rangle \rightarrow F(\rho, |\psi\rangle) = \langle 0|\rho|0\rangle \\ &= \frac{1}{2} \langle 0|0\rangle^2 + \frac{1}{2} \langle 0|1\rangle^2 \\ &= \frac{1}{2} \end{aligned}$$

Let us consider we wanted to initialize a two-qubit system to the pure state $|00\rangle$ but we obtain the mixed state $\rho = \frac{1}{4} (|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + |11\rangle\langle 11|)$, the the fidelity is equal to

$$F(\rho, |\psi\rangle) = \langle 00|\rho|00\rangle = \frac{1}{4}$$

::... Noisy World - Mixed states

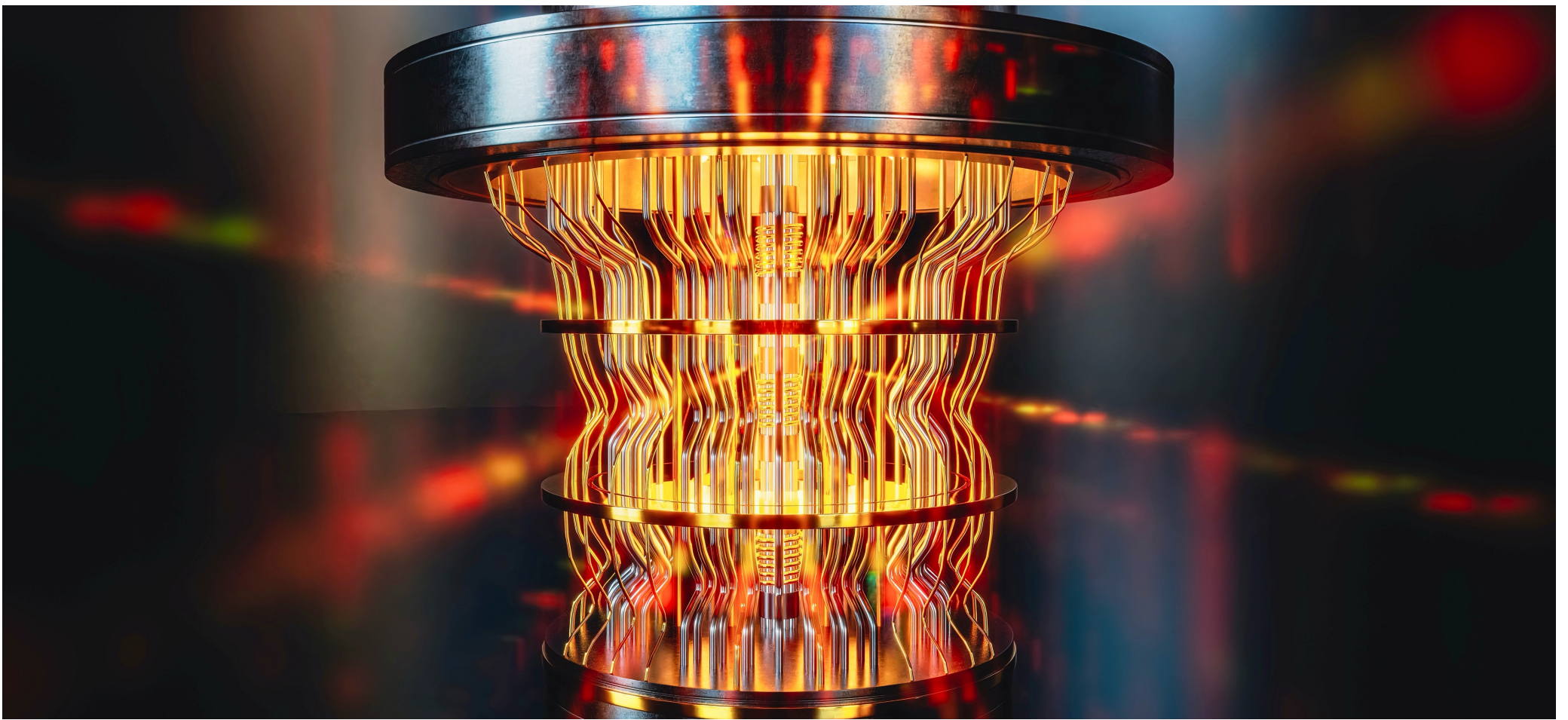
This result for a generic system of N qubit to be initialised to the 0 state can be generalised as

$$F(\rho, |\psi\rangle) = \frac{1}{2^N}$$

Let us consider the flip channel, the obtained output is:

$$\begin{aligned}\rho &= (1 - p) |\psi\rangle\langle\psi| + pX|\psi\rangle\langle\psi|X, \quad F(\rho, |0\rangle) = \langle 0|\rho|0\rangle \\ &= (1 - p)\langle 0|0\rangle^2 + p|\langle 0|1\rangle|^2 = 1 - p\end{aligned}$$

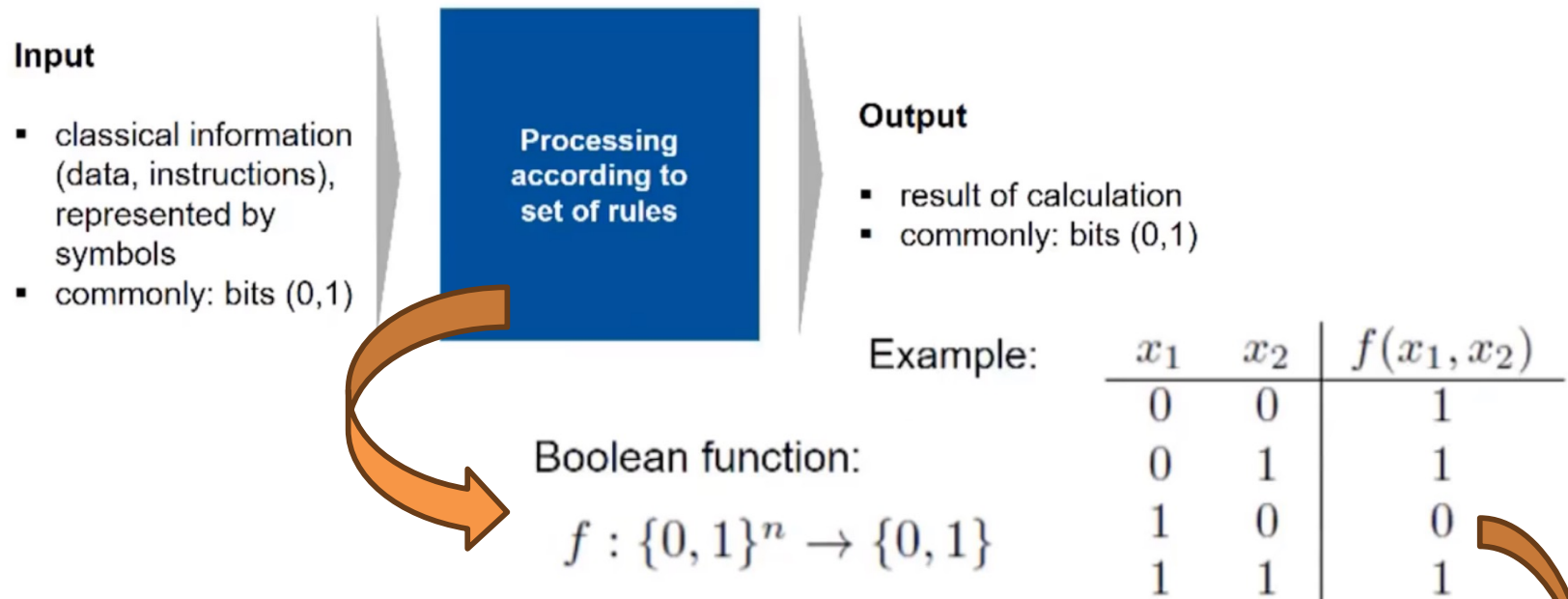
Fidelity is the probability that the obtained output state passing the test to be in the same state as the desired one. Certain protocol have the requirement in terms of fidelity.



Introduction to Quantum Computations

::... Quantum Computations

In the classic computations, the following scheme is



where the rules can be deterministic (produce the same output always and pass through the same intermediate states) or probabilistic (the state evolution and the relative output generation are not always the same but have a random nature of some sort).

Boolean functions can be built from elementary logic gates:

$$f(x_1, x_2) = (\neg x_1 \wedge \neg x_2) \vee (\neg x_1 \wedge x_2) \vee (x_1 \wedge x_2)$$

::... Quantum Computations

In the classic computations, the universal set is composed of three possible operations:

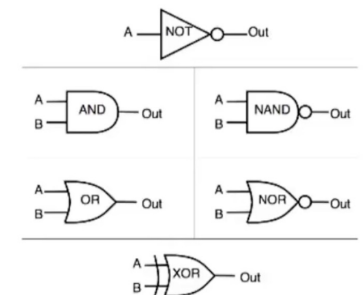
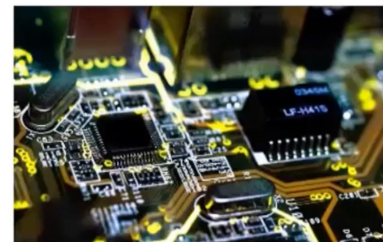
- $\wedge$ - and;
- $\vee$ - or;
- $\neg$ - not;

Such that any other function is a mere combination of these element operations: $\text{NAND}(x_1, x_2) := \neg(x_1 \wedge x_2)$.

Vice versa, NAND can be used to realize all the other possible boolean functions, even the elementary ones:

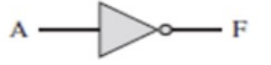
- $x_1 \wedge x_2 = \text{NAND}(\text{NAND}(x_1, x_2), \text{NAND}(x_1, x_2))$
- $x_1 \vee x_2 = \text{NAND}(\text{NAND}(x_1, x_1), \text{NAND}(x_2, x_2))$
- $\neg x = \text{NAND}(x, x)$

The inside of a computer is make of a series of these tiny gates.



::... Quantum Computations

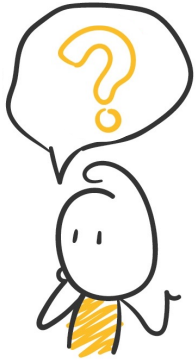
A gate is reversible if the output allows for reconstructing the input, and this is true for the not gate, but for those with two inputs are not reversible. If we know the output of an AND gate is 0, the combination of inputs that leads to those outputs is not only one (a row in the truth table), so we cannot determine for sure which input determined that output.

Name	Graphical Symbol	Algebraic Function	Truth Table															
AND		$F = A \cdot B$ or $F = AB$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	F	0	0	0	0	1	0	1	0	0	1	1	1
A	B	F																
0	0	0																
0	1	0																
1	0	0																
1	1	1																
OR		$F = A + B$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	F	0	0	0	0	1	1	1	0	1	1	1	1
A	B	F																
0	0	0																
0	1	1																
1	0	1																
1	1	1																
NOT		$F = \overline{A}$ or $F = A'$	<table><tr><th>A</th><th>F</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	F	0	1	1	0									
A	F																	
0	1																	
1	0																	
NAND		$F = \overline{AB}$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	F	0	0	1	0	1	1	1	0	1	1	1	0
A	B	F																
0	0	1																
0	1	1																
1	0	1																
1	1	0																
NOR		$F = \overline{A + B}$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	F	0	0	1	0	1	0	1	0	0	1	1	0
A	B	F																
0	0	1																
0	1	0																
1	0	0																
1	1	0																
XOR		$F = A \oplus B$	<table><tr><th>A</th><th>B</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	F	0	0	0	0	1	1	1	0	1	1	1	0
A	B	F																
0	0	0																
0	1	1																
1	0	1																
1	1	0																

Some gates are irreversible, so we lose information when we apply them.

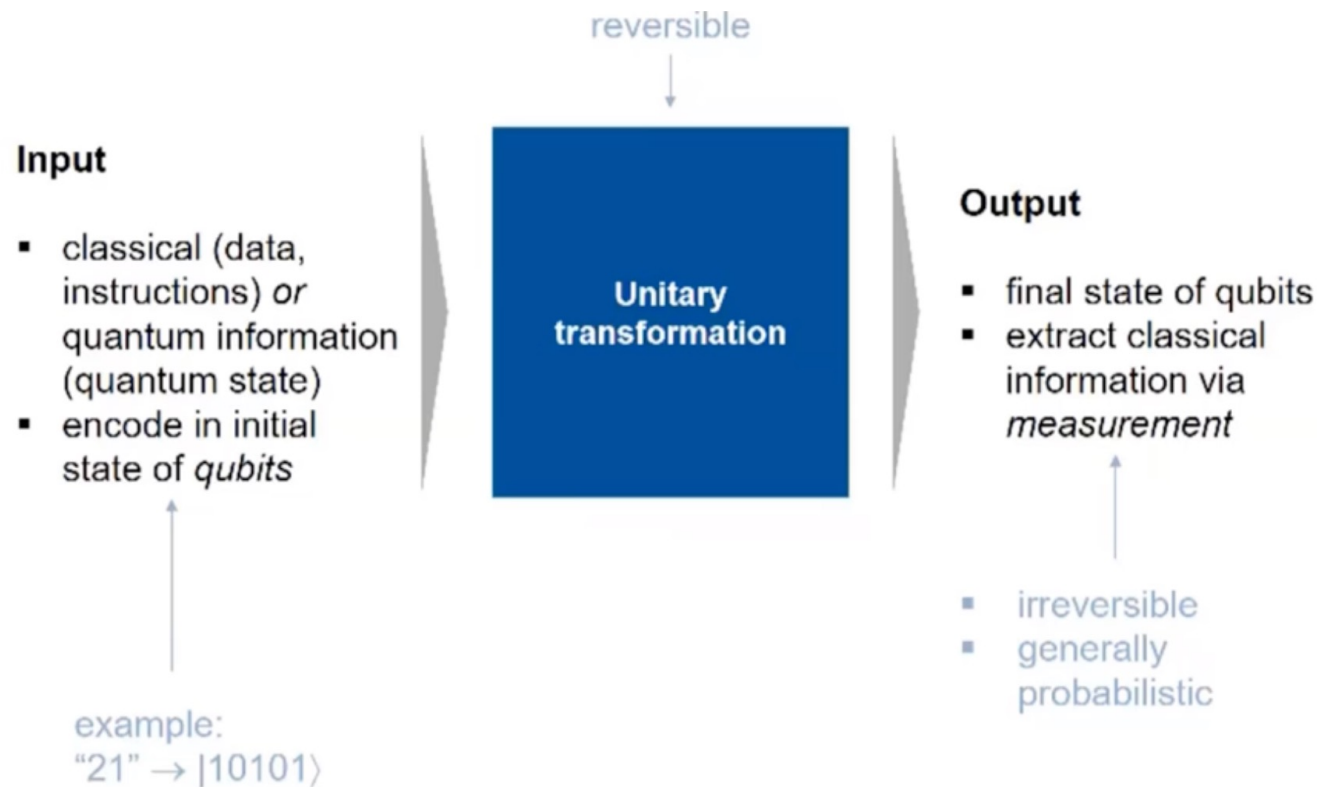
There are no restrictions in copying classical bits and putting them in multiple gate as inputs.

::... Quantum Computations



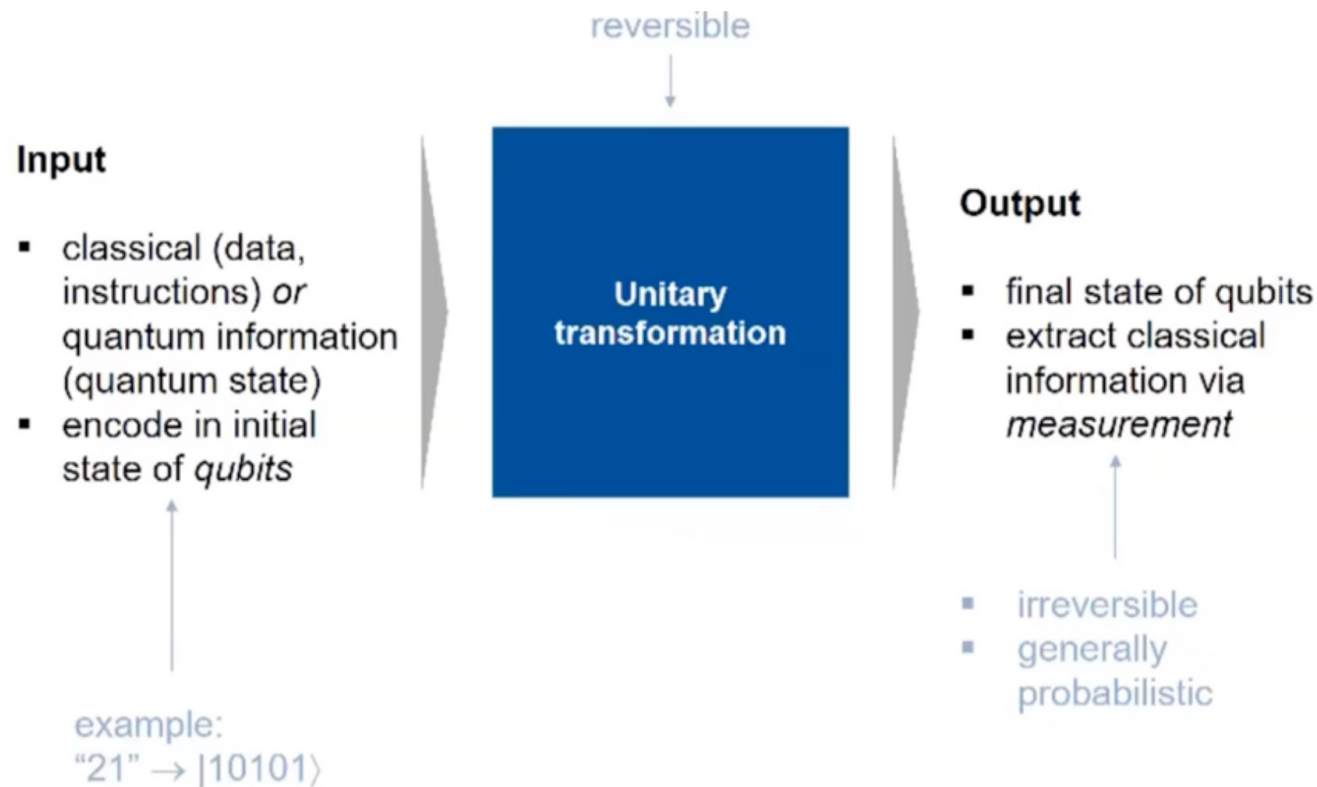
What is the schematic structure of a quantum computation?

::... Quantum Computations



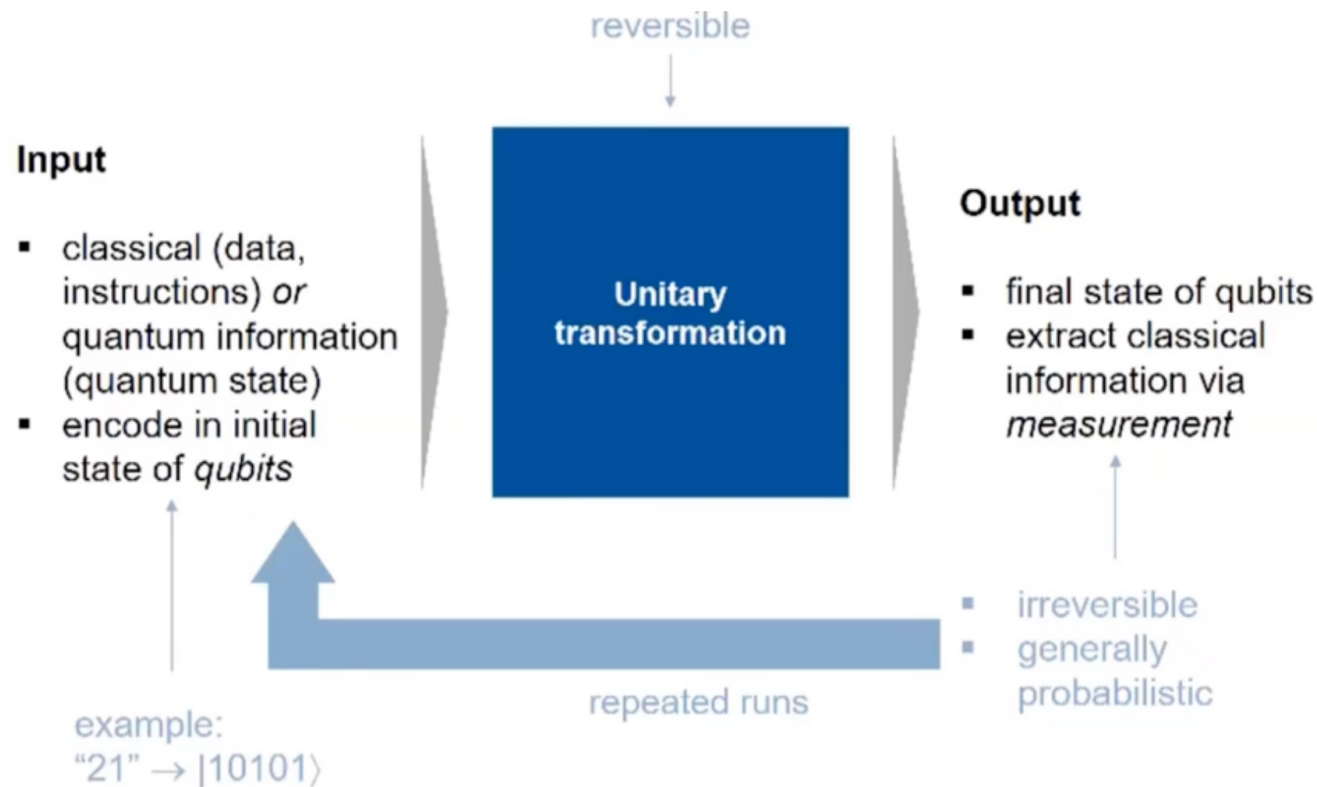
As in the classic computations, we start with inputs that can be classic bits being mapped onto quantum states (pure ones) or quantum information in mixed quantum states. Then, we have an output, which consists into the results of unitary transformations being applied on the input in a reversible manner. There is a final state where we extract classic information by means of a measurement, which is probabilistic and irreversible.

::... Quantum Computations



From the measurement output, we cannot infer the quantum state we had before. This also introduces a probabilistic state: if we have a superposition state, we have a probabilistic measurement into a given classical bit or not. So quantum computations are always probabilistic, in contrast to classical ones, which are always deterministic.

::... Quantum Computations

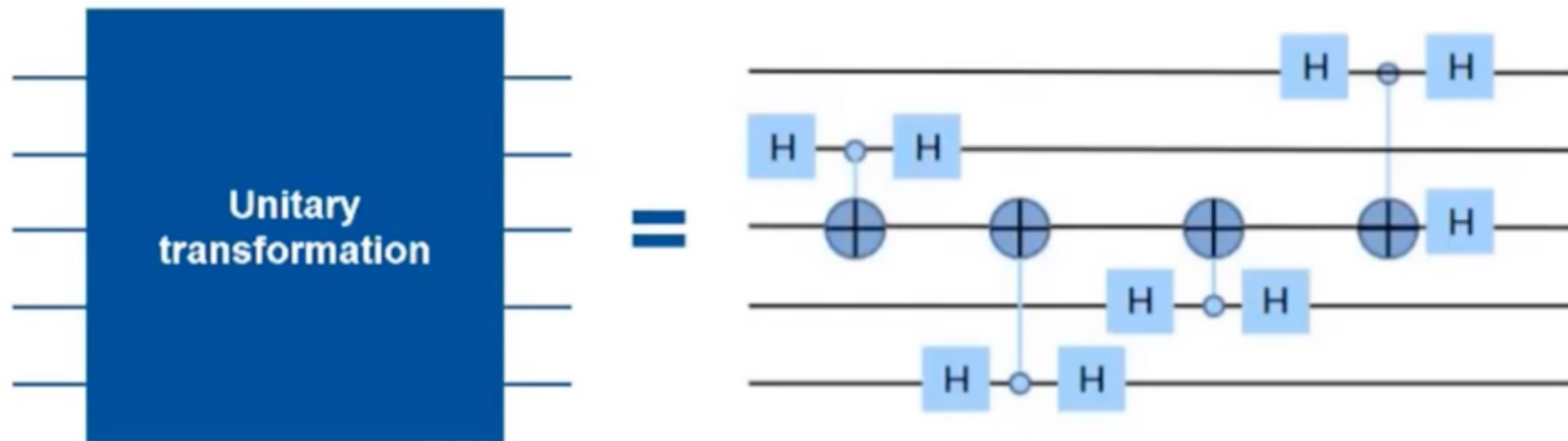


In a single run, we immediately have the output, but we need to collect a sufficient number of data points to identify the most likely solutions to a given problem.

Quantum algorithms can solve some problems more efficiently and quickly than classical algorithms, even though they must be run many times.

::... Quantum Computations

In analogy with the classic computations, where Boolean functions can be realised by using a set of elementary gates in the universal set, each involving one or two bits at a time. In the quantum case, any arbitrary unitary transformation in the n -qubit Hilbert space can be built up using elementary quantum gates involving one or two qubits at a time.



We have infinitely many unitary transformations of quantum states, but we typically use a finite set of elementary quantum gates, each reversible, and the resulting unitary transformation is reversible as well.

::... Single Qubit Gates

The simplest type of gate is the one that takes a single qubit and returns a single qubit as output.

$$|\psi\rangle \xrightarrow{U} \hat{U} |\psi\rangle$$

- Pauli gates apply the Pauli operators to the input in order to obtain the output by means of a unitary transformation.

$$|t\rangle \xrightarrow{X} |t \oplus 1\rangle, \quad \leftarrow \oplus: \text{addition mod 2}$$

$$|t\rangle \xrightarrow{Z} (-1)^t |t\rangle, \quad t \in \{0, 1\}$$

- Hadamard gate maps the $|0\rangle$ to the point $|+\rangle$ and $|1\rangle$ to $|-\rangle$ by making a rotation of the visual representation of the state.

$$|0\rangle \xrightarrow{H} |+\rangle, \quad |1\rangle \xrightarrow{H} |-\rangle$$

::... Single Qubit Gates

- Phase and $\pi/8$ gates are Z-axis rotations that respectively apply a 90° and 45° rotations.

$$\begin{array}{cc} \text{---} \boxed{S} \text{---} & S := \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \\ \text{---} \boxed{T} \text{---} & T := \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \end{array}$$

We have a continuous set of single-qubit transformations, in contrast to the classical case where there is only the NOT gate. However, within this infinitely many, only a few of them are used.

Among the gates there are a series of identities relationships when put in series:

$$\text{---} \boxed{H} \text{---} \boxed{H} \text{---} = \text{---} \quad \text{Two Hadamard gates nullify themselves if put in series.}$$

::... Single Qubit Gates

Two Hadamard gates having a Pauli X in the middle returns the Pauli Z:

$$\text{---} \boxed{H} \text{---} \boxed{X} \text{---} \boxed{H} \text{---} = \text{---} \boxed{Z} \text{---}$$

::... Single Qubit Gates

Two Hadamard gates having a Pauli X in the middle returns the Pauli Z:

$$\begin{array}{ccccccc}
 |1\rangle & & |-\rangle & & -|-\rangle & & -|1\rangle \\
 \text{---} & \boxed{H} & \text{---} & \boxed{X} & \text{---} & \boxed{H} & \text{---} = \text{---} \boxed{Z} \text{---} \\
 |0\rangle & & |+\rangle & & |+\rangle & & |0\rangle
 \end{array}$$

Similarly there are more simplifications:

$$\text{---} \boxed{H} \text{---} \boxed{Z} \text{---} \boxed{H} \text{---} = \text{---} \boxed{X} \text{---}$$

$$\text{---} \boxed{S} \text{---} \boxed{S} \text{---} = \text{---} \boxed{Z} \text{---}$$

$$\text{---} \boxed{T} \text{---} \boxed{T} \text{---} = \text{---} \boxed{S} \text{---}$$

::... Single Qubit Gates

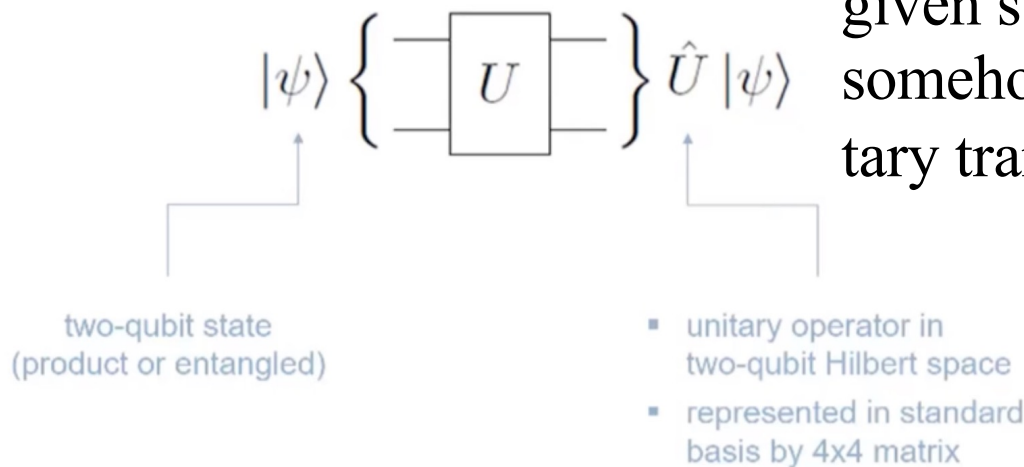
There is an extra gate for a single qubit, and this is the measurement gate:



It takes as input a qubit and, as output, returns a classical bit with one of the two possible outputs. This is the only gate that is probabilistic and irreversible, unless we deal with pure states.

::... Multi Qubit Gates

With two-qubit gates, things start to be more fun: the inputs are in a given P state (product where each qubit is in a given state or entangled where their states are somehow related), upon which a proper unitary transformation $\hat{U}$ is applied to obtain an output in a new state belonging to a two-qubit Hilbert space.

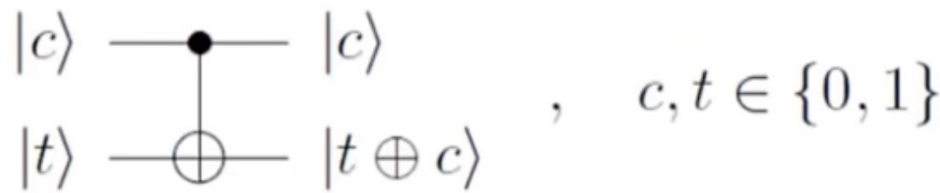


$$\begin{array}{c}
 |c\rangle \\
 |t\rangle
 \end{array}
 \begin{array}{c}
 \bullet \\
 \oplus
 \end{array}
 \begin{array}{c}
 |c\rangle \\
 |t \oplus c\rangle
 \end{array}
 , \quad c, t \in \{0, 1\}$$

Here're some examples of two-qubit gates.

- C-NOT (Controlled NOT) gate that flips the target qubit if the control qubit is in the state $|1\rangle$, and leaves it unchanged if the control qubit is $|0\rangle$. It is essential for entangle and disentangle Bell states and universal quantum computing, mapping input $|c, t\rangle$ to $|c, t \oplus c\rangle$.

::... Multi Qubit Gates



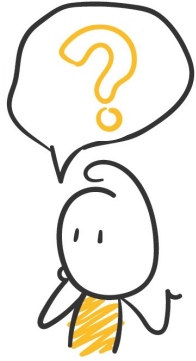
$|c\rangle$ —●— $|c\rangle$
 $|t\rangle$ —⊕— $|t \oplus c\rangle$, $c, t \in \{0, 1\}$

$U_{\text{CNOT}} = \begin{matrix} & \begin{matrix} 00 & 01 & 10 & 11 \end{matrix} \\ \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{X} \end{pmatrix}$

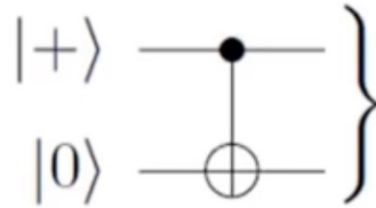
In fact, any quantum circuit can be simulated to an arbitrary degree of accuracy using a combination of CNOT gates and single qubit rotations.

- The combination of CNOT gates and single-qubit gates forms a universal set for quantum computation.
- Any-qubit quantum gate can be decomposed into elementary, single-qubit gates and CNOT gates.
- While CNOT + single-qubit gates is a standard set, other, more efficient sets exist, but they still rely on the same principle.

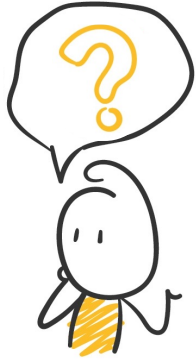
::... Multi Qubit Gates



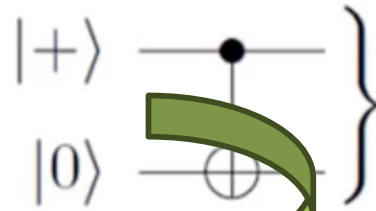
What happens in this situation?



::... Multi Qubit Gates

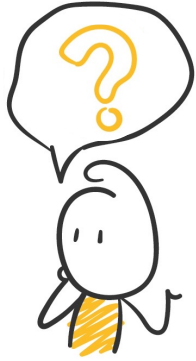


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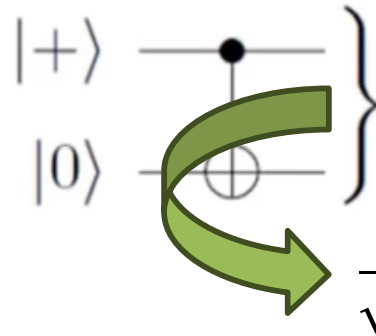


$$|+0\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

::... Multi Qubit Gates



What happens in this situation?



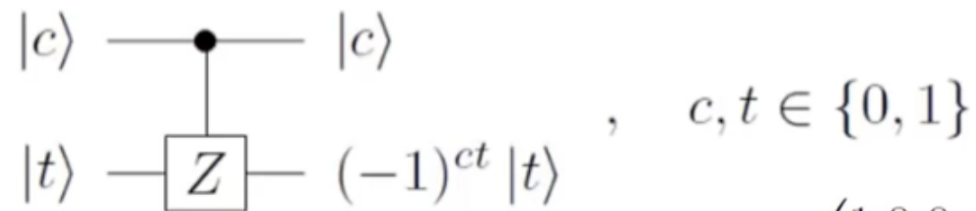
In output we get a Bell state.

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\psi^+\rangle$$

::... Multi Qubit Gates

Here're some examples of two-qubit gates.

- C-Z (Controlled Z) gate that applies the Z gate on the second qubit based on the state assumed by the first qubit.



C-NOT is equivalent to a C-X gate.

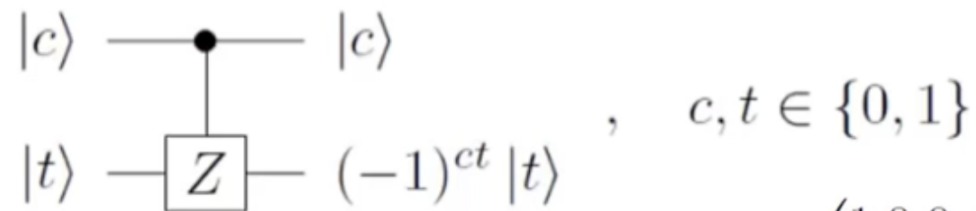


$$U_{C-Z} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{Z} \end{pmatrix}$$

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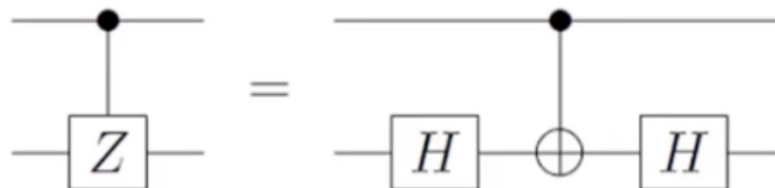
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It is possible to demonstrate that the C-Z gate is equivalent to a series of two Hadamard gates on the second qubit and a C-NOT in between:

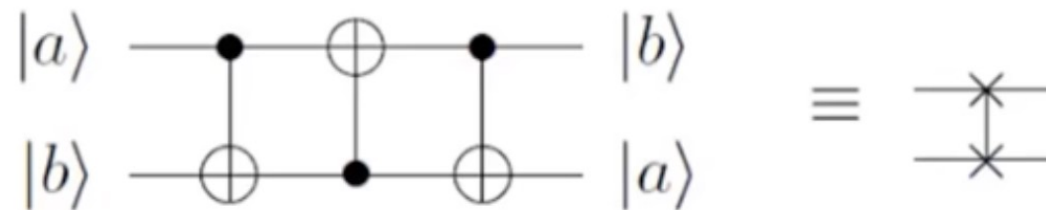
$$= \begin{pmatrix} \mathbf{H} & \mathbf{0} \\ \mathbf{0} & \mathbf{H} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{X} \end{pmatrix} \begin{pmatrix} \mathbf{H} & \mathbf{0} \\ \mathbf{0} & \mathbf{H} \end{pmatrix}$$



::... Multi Qubit Gates

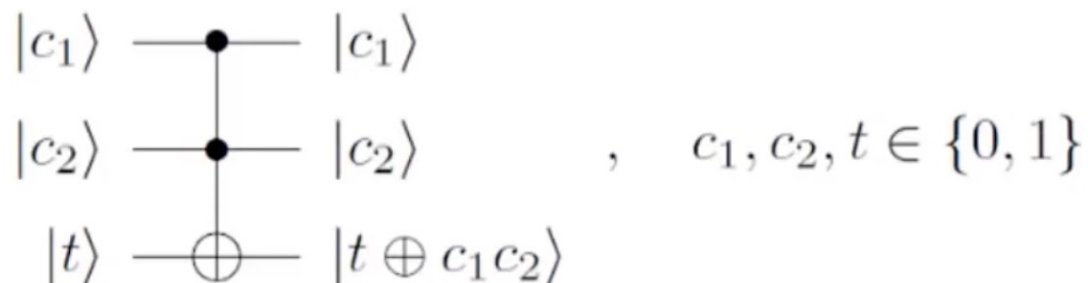
Here're some examples of two-qubit gates.

- Three C-NOT can be deployed in series where the first and the third is controlled by the first qubit, while the second is inverted:



The effect is to swap the state between the two qubits. This configuration is fundamental for transferring information between qubits, often used in quantum computing for rearranging state information.

There is an important three-qubit gate called Toffoli gate, consisting into a

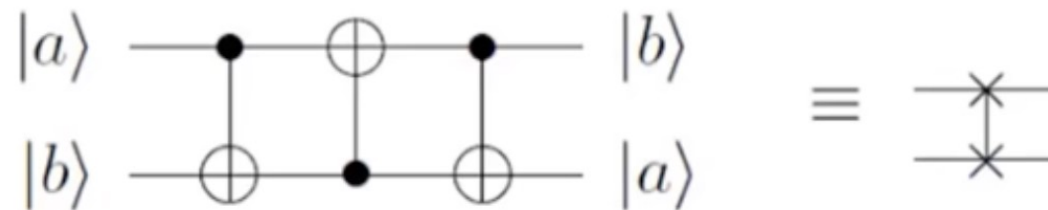


gate controlled by multiple qubits, where the target qubit is flipped if both the controlled gates are in the state $|1\rangle$.

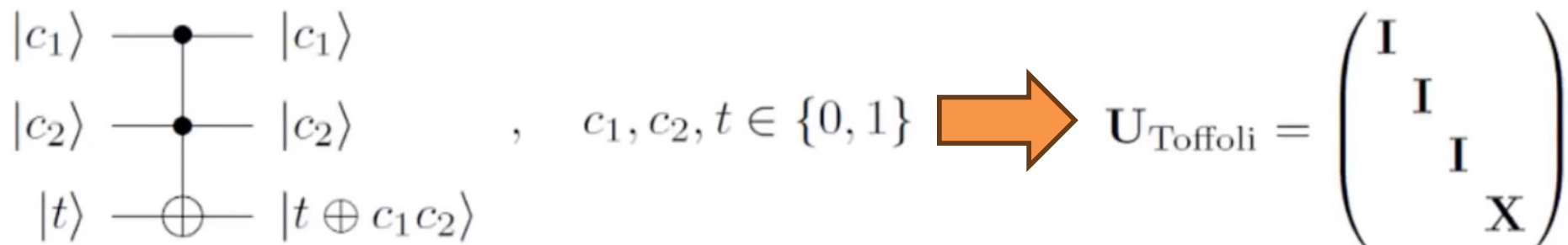
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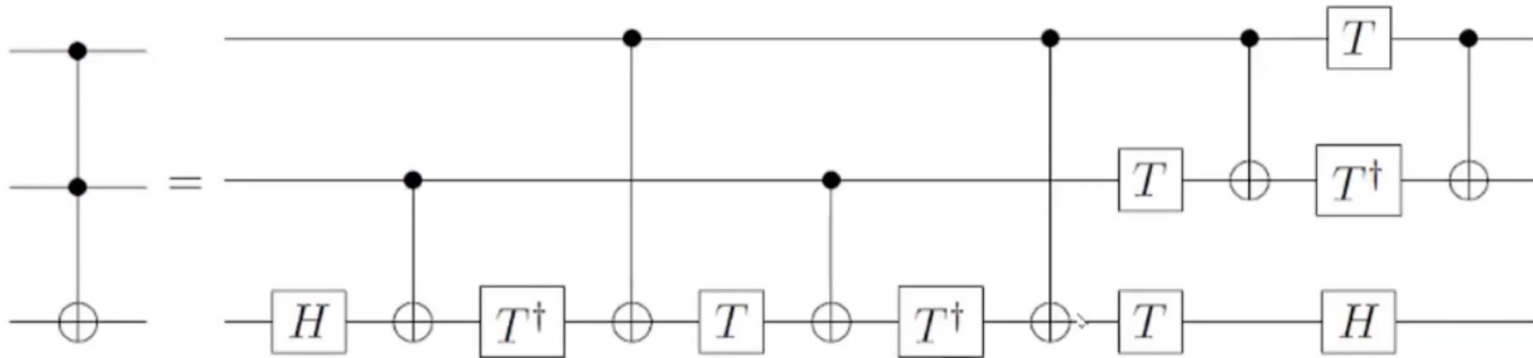


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::: Multi Qubit Gates

The Toffoli gate can be reduced as a proper circuit of C-NOT, Hadamard and $\pi/8$ gates:

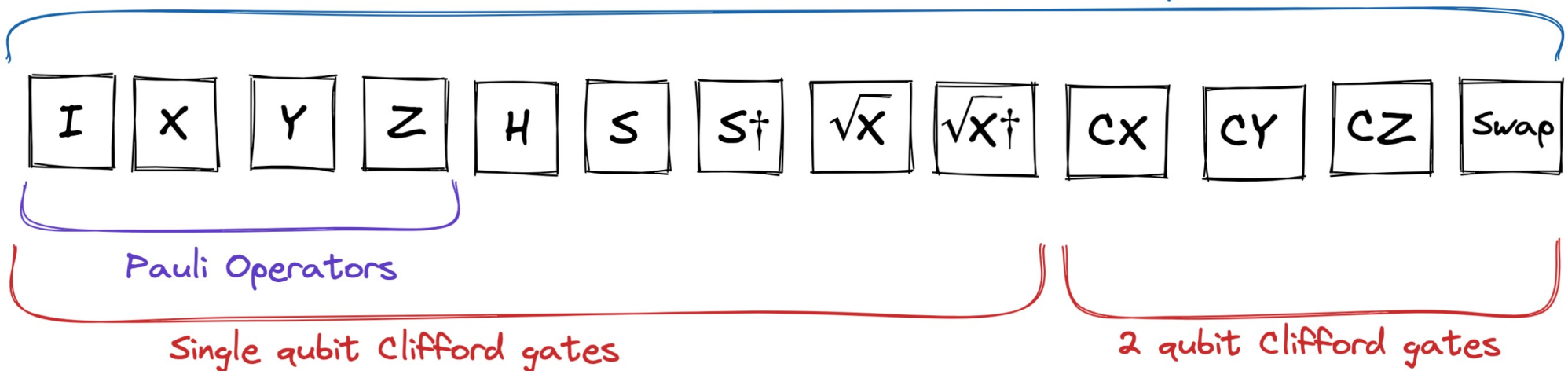


::... Clifford Gates

Clifford gates are the quantum gates that map Pauli operators (X, Y, Z, and their multi-qubit versions) to other Pauli operators when you surround the Pauli by the gate and its inverse. In other words, if C is a Clifford gate, then $CPC^\dagger$ is still a Pauli operator (up to a global phase).

- Let us consider Hadamard gate, that X into Z under conjugation:
 $HXH^\dagger = Z$.
- The T gate does not satisfy this property, as for example $TXT^\dagger$ does not result in a Pauli operator.

Qiskit Gates in the Clifford Group



::... Clifford Gates

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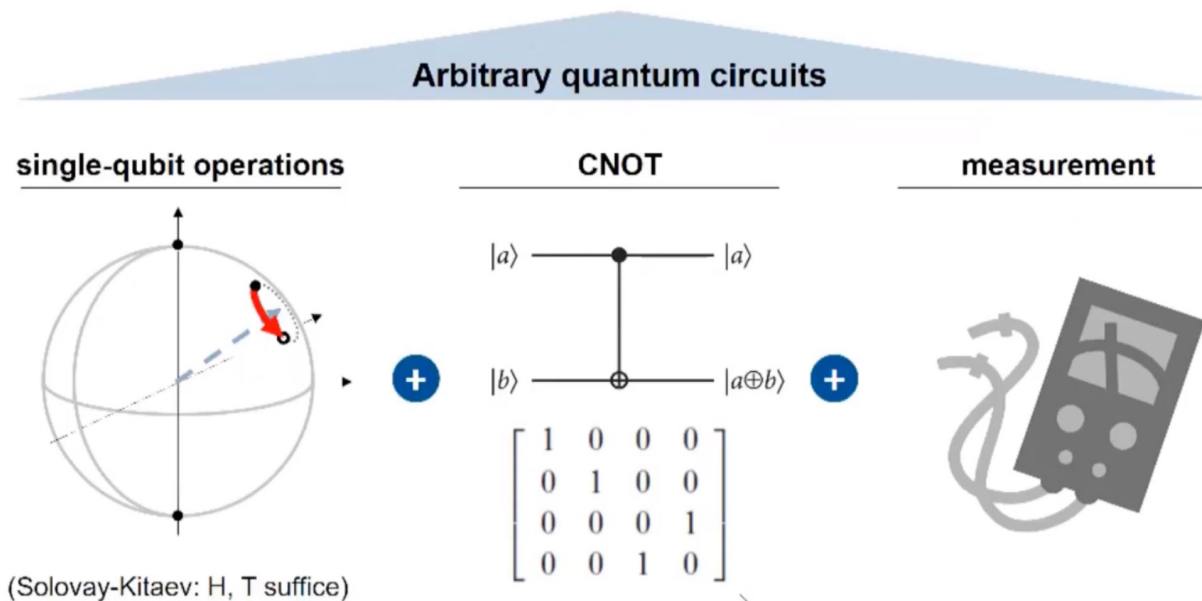
- Let us consider Hadamard gate, that X into Z under conjugation:
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That “Paulis stay Paulis” behavior is the hallmark of Clifford gates.

- These gates are crucial, as they are key building blocks for stabilizer circuits and quantum error correction.
- Circuits made only of Clifford gates can be simulated efficiently on a classical computer (the Gottesman–Knill idea).
- Clifford gates alone are not enough for universal quantum computing. To obtain a universal gate set, you need to include at least one non-Clifford gate, typically the T gate.

::... Universal Set

Any possible unitary transformation for inputs made of N qubits can be implemented by considering a universal set of gates:

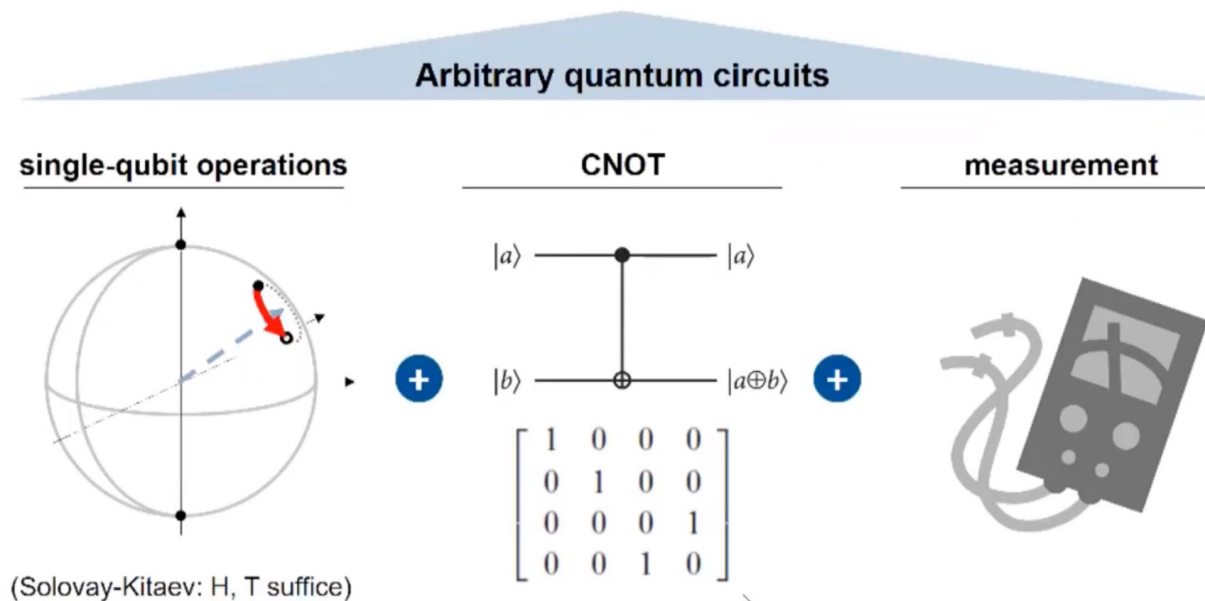


Because there are infinitely many possible quantum operations, but a finite gate set has only finitely many basic pieces. So in practice, a universal set means you can get as close as you want to any target operation by chaining enough gates together.

A universal gate set = enough basic quantum gates to construct any quantum algorithm.

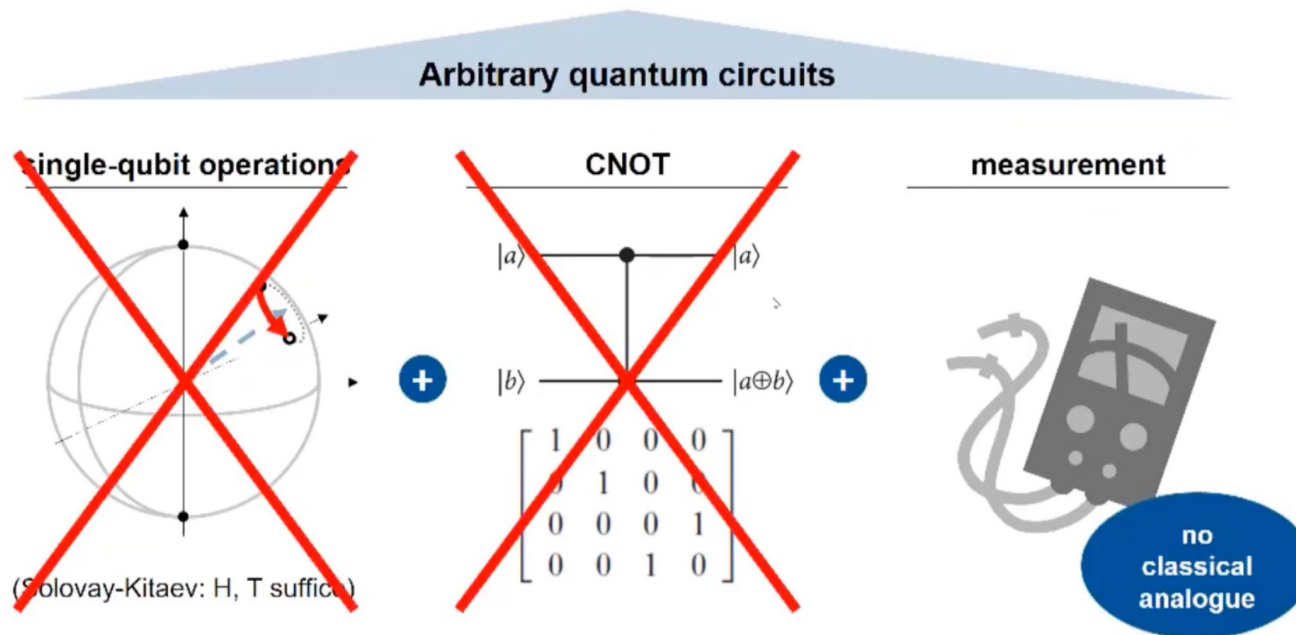
::... Universal Set

Any possible unitary transformation for inputs made of N qubits can be implemented by considering a universal set of gates:



We have only two single-qubit gates: the Hadamard gate, representing the set of Clifford gates, and the T gate (to increase the expressivity, as not all functions can be implemented only with Clifford gates). The universal set is completed by the C-NOT gate, the only two-qubit gate. At the end of the computation, in order to extract the result, a measurement gate is needed.

::... Universal Set

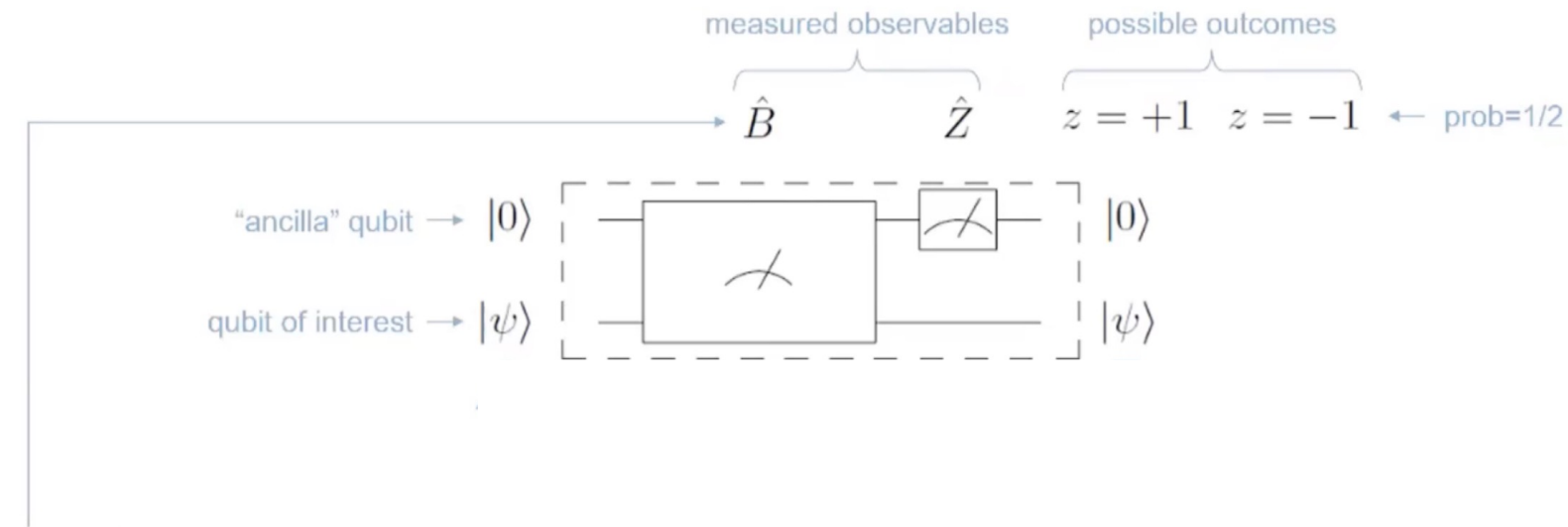


In theory, it is possible to perform computation using only measurements, without any gates, which is something completely unseen in the classical world.

- Imagine having a single wubit in the arbitrary state $|\psi\rangle$, then we have a secondary qubit as an auxiliary or ancilla with state $|0\rangle$. We want to execute a unitary transformation $\hat{U}$ on $|\psi\rangle$ with only measurements.

::... Universal Set

We will have two measurements in a row: the first one on a two-qubit observable $\hat{B}$ as a sum of two terms, and the second is performed on the ancilla qubit only and determines which of the states the first qubit is.



$$\hat{B} := \frac{1}{2}(\hat{X} - i\hat{Y}) \otimes \hat{U} + \frac{1}{2}(\hat{X} + i\hat{Y}) \otimes \hat{U}^\dagger$$

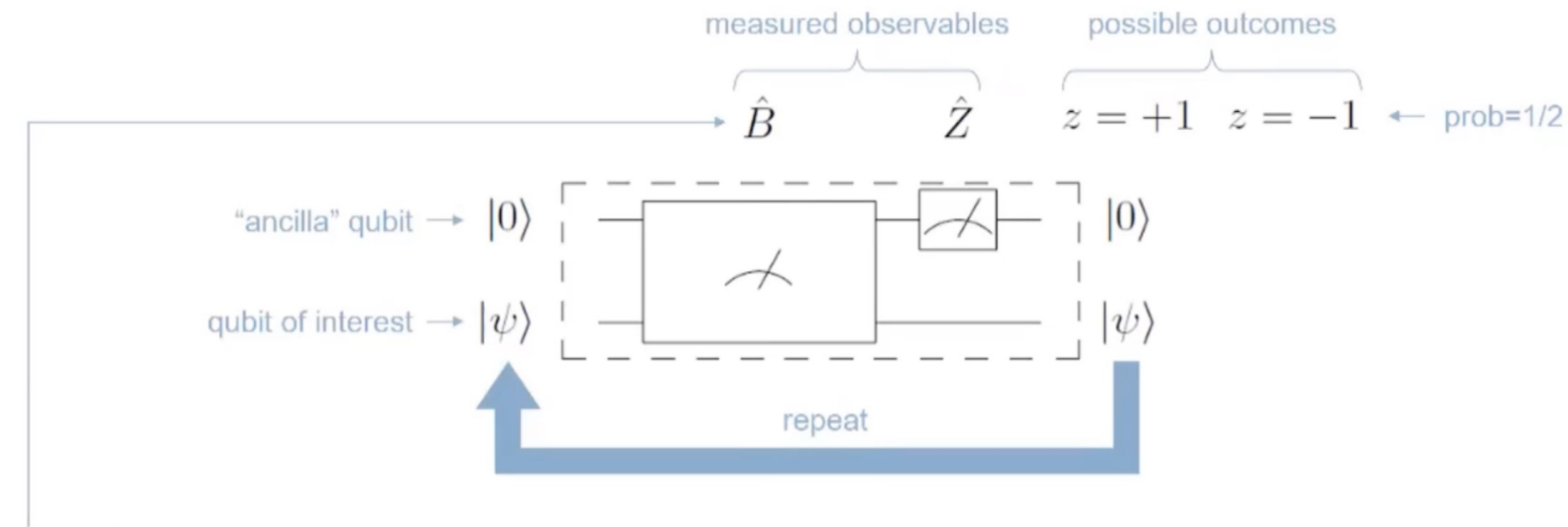
properties:

(1) $\hat{B}^\dagger = \hat{B}$ (2) $\hat{B}^2 = \hat{I}$ (3) $\hat{B}(|0\rangle \otimes |\psi\rangle) = |1\rangle \otimes \hat{U}|\psi\rangle$



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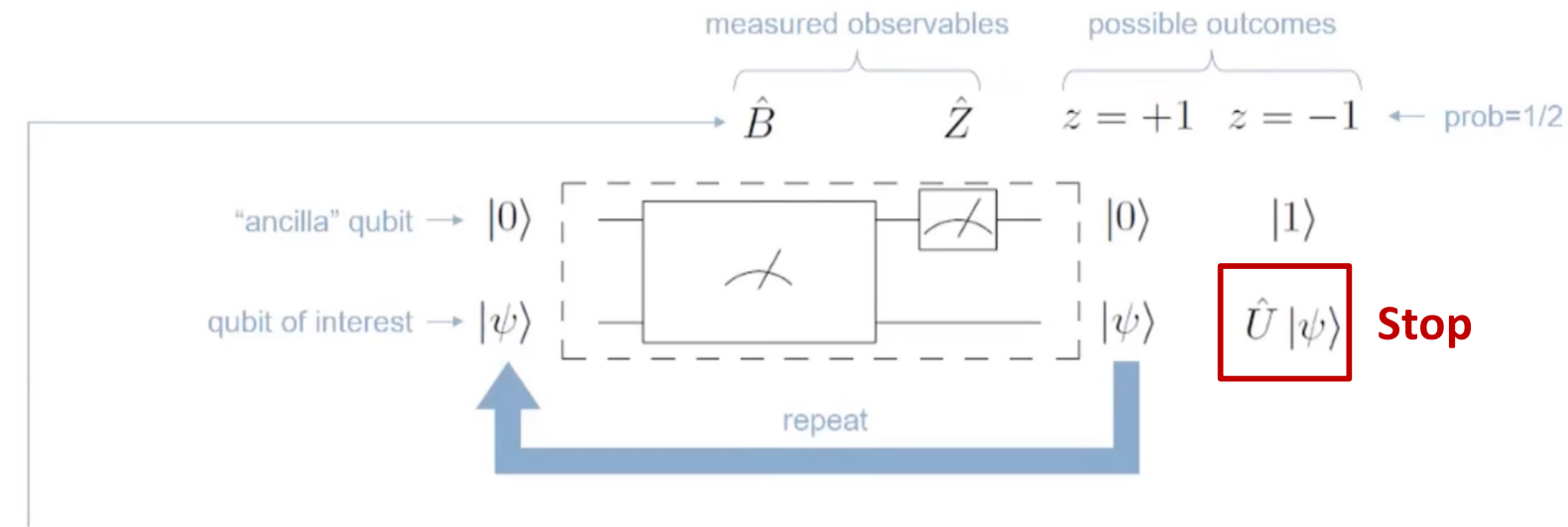
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The computation is repeated until the ancilla is not in state $|1\rangle$, then we stop and in the second qubit we have the results we want. This is probabilistic as we run as many times as needed to have the ancilla $|1\rangle$.

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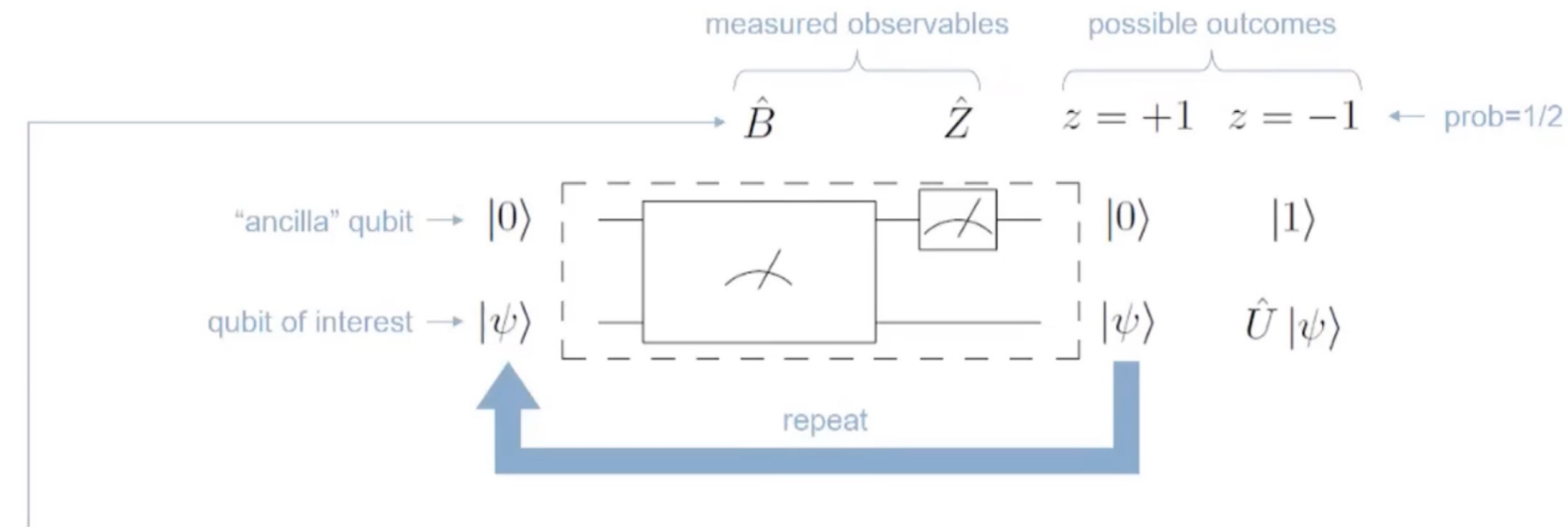
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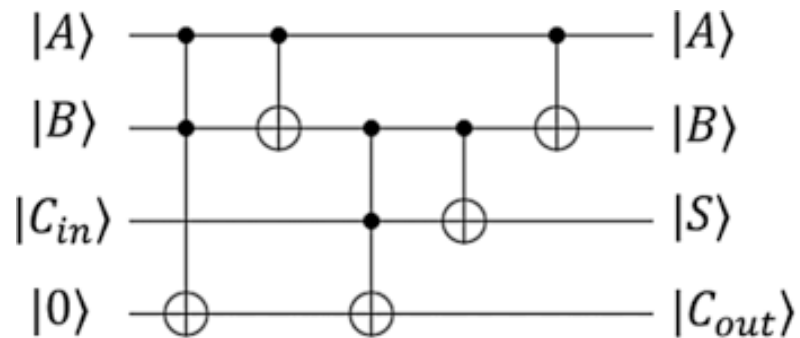
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The expected number of runs is $\langle n \rangle = \sum_{i=1}^{\infty} \text{prod}(n)n = \sum_{i=1}^{\infty} \frac{n}{2^n} = 2$.

::... Example of Circuit

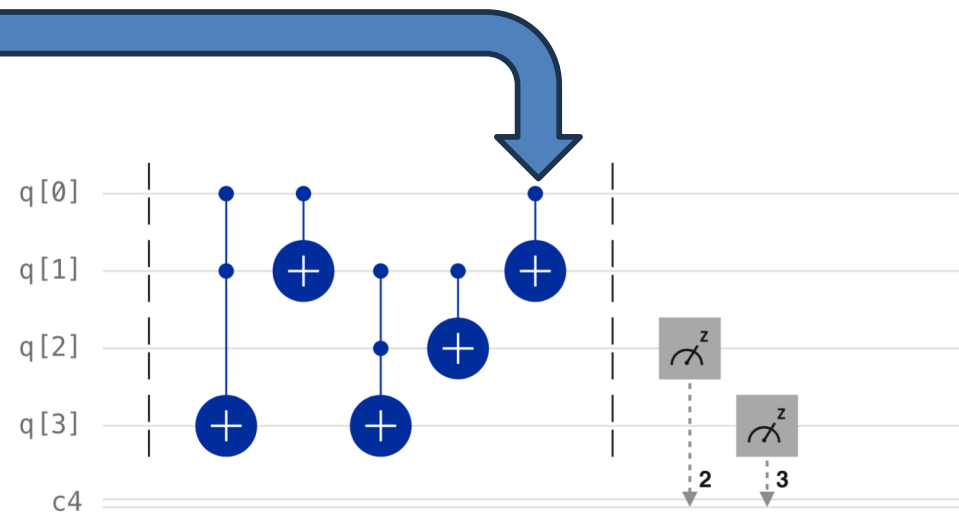
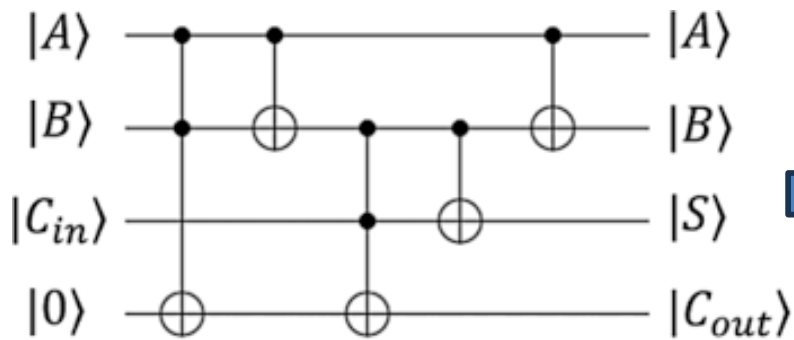
Let us experiment with the realization of a full-adder by means of a quantum circuit.



Example of Circuit

Let us experiment with the realization of a full-adder by means of a quan-

tum circuit by using IBM Composer at <https://quantum.cloud.ibm.com/composer>.

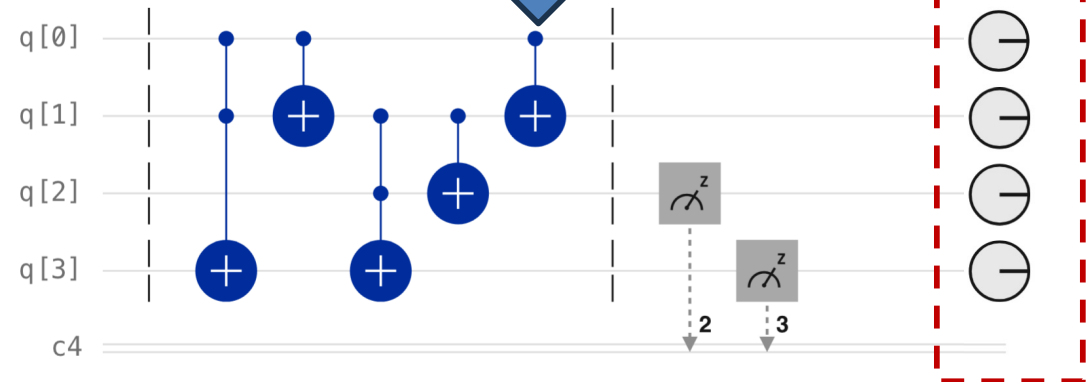
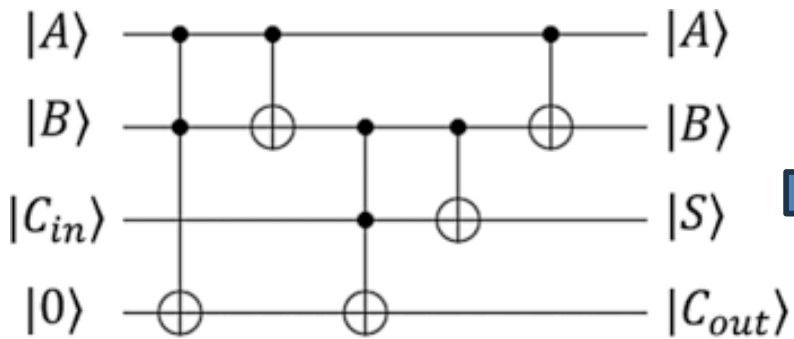


::: Example of Circuit

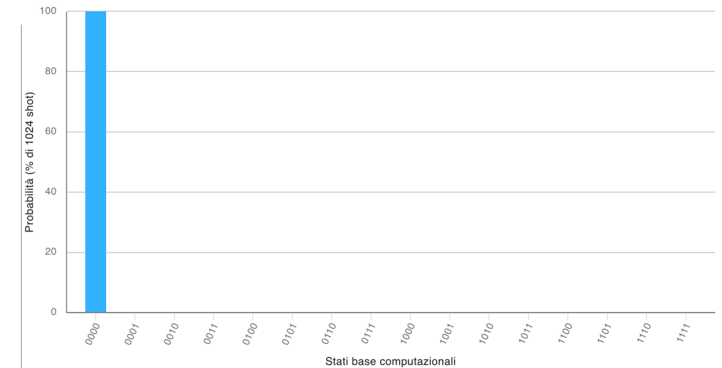
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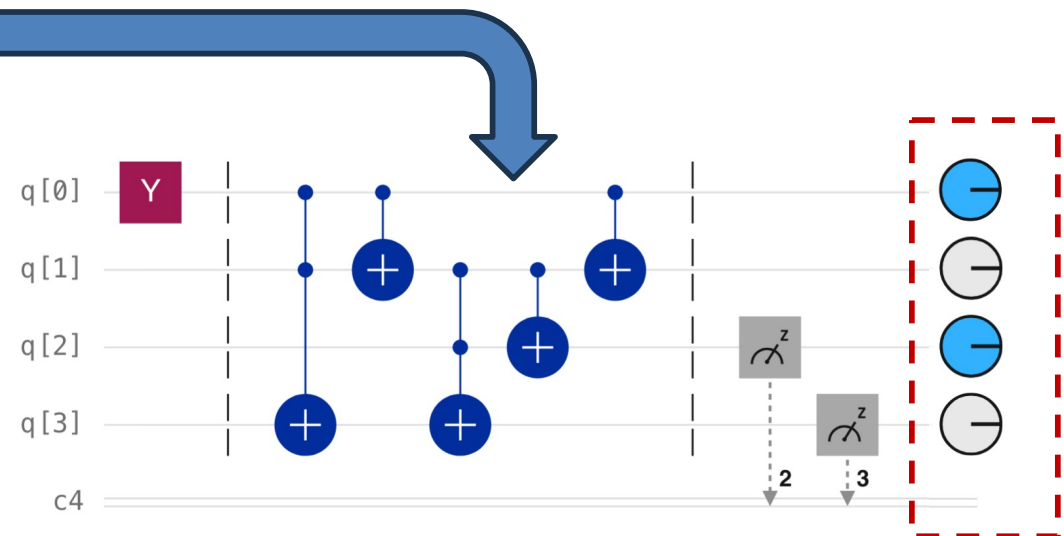
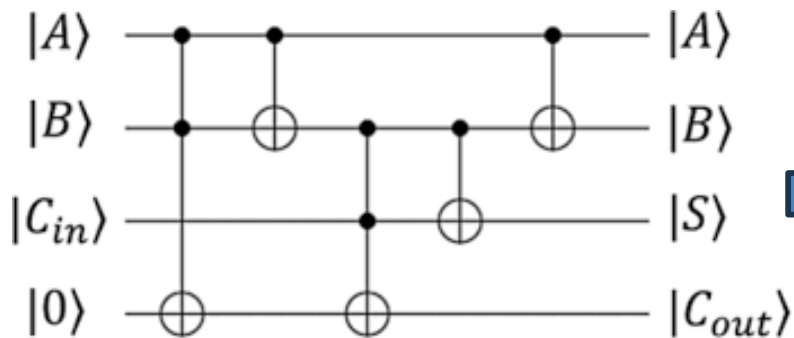
When the involved input qubits are in the $|0\rangle$, then the output are the expected ones.



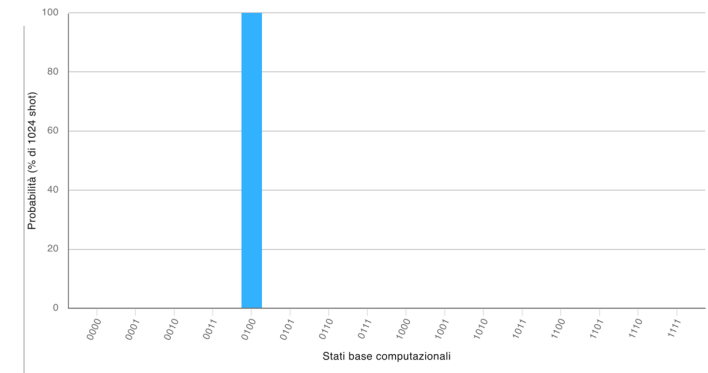
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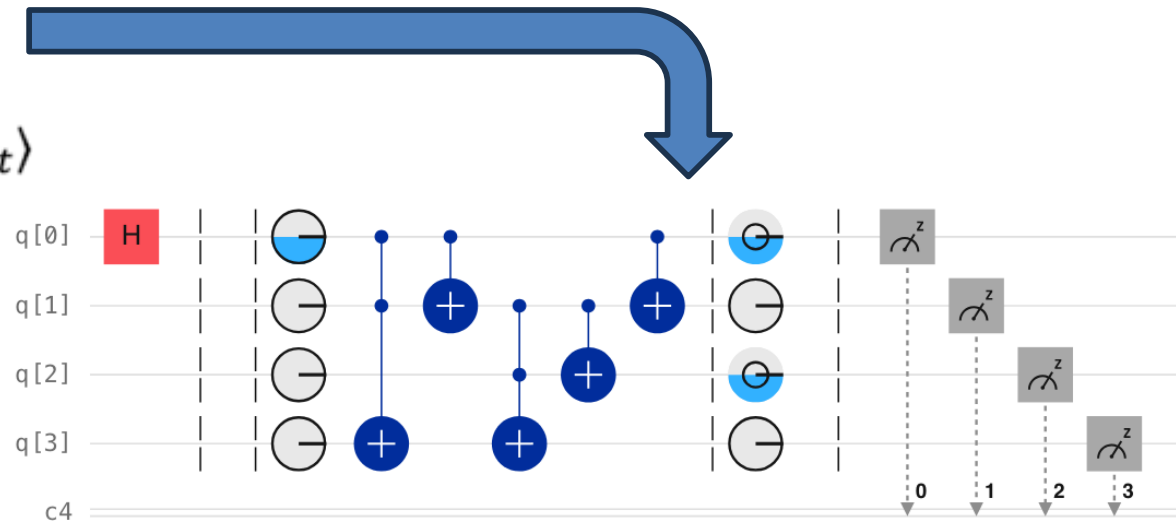
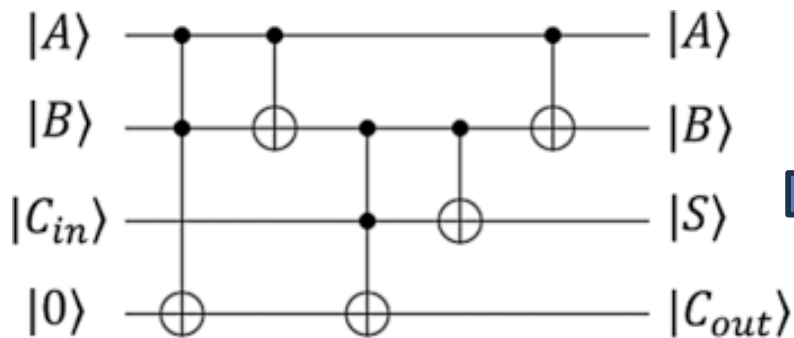
When the involved input qubits are in the $|0\rangle$, then the output are the expected ones. If the first input is set to $|1\rangle$ by using a Pauli Y gate, then the sum is done.



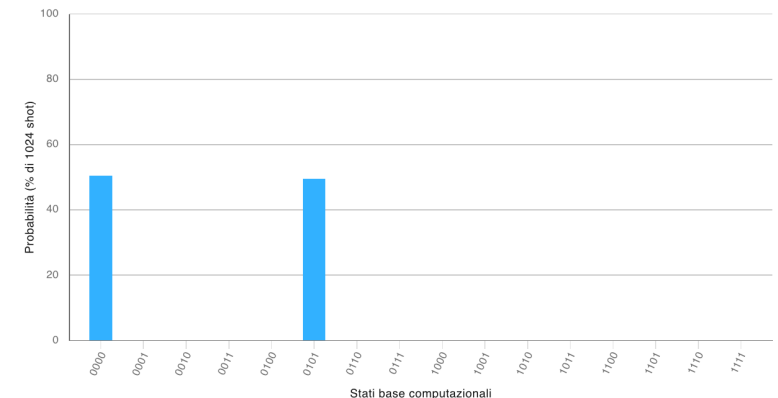
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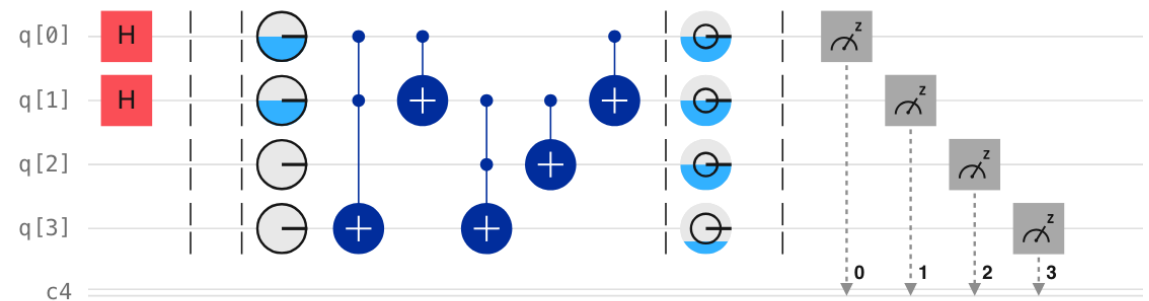
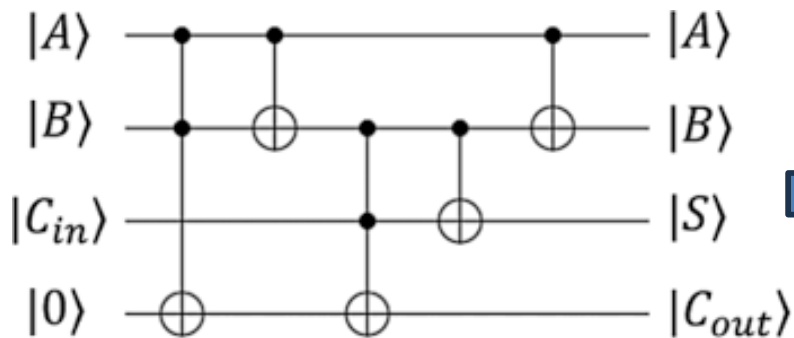
When the involved input qubits are in superposition thanks to the Hadamard gate, then also the output is in the superposition of the two possible output state.



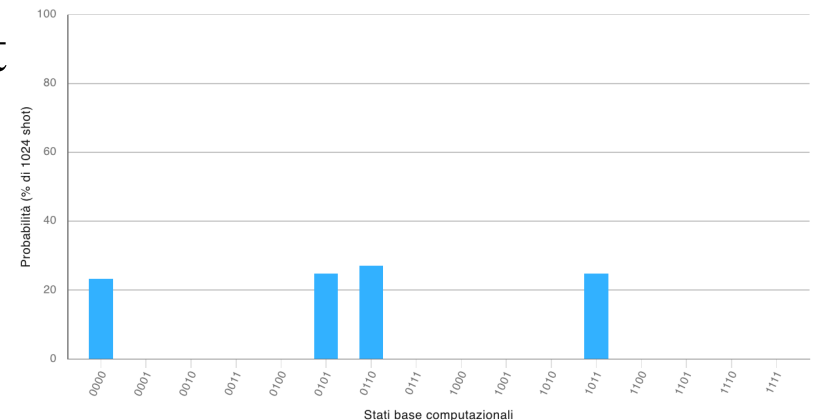
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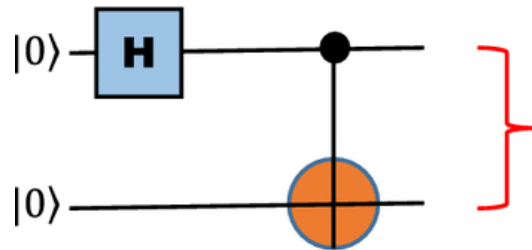


If both inputs are in superposition, the output is even more interesting, where one configuration is dominant over the other three.

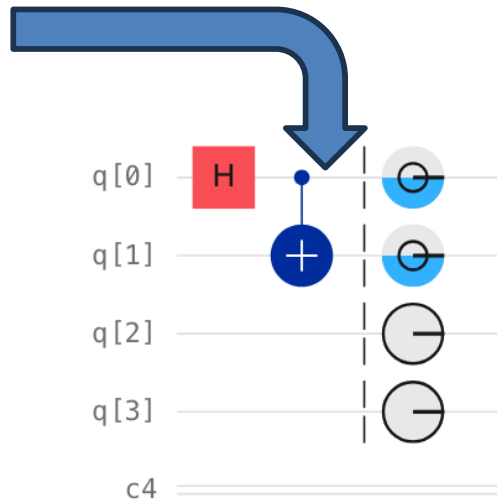


::... Example of Circuit

Let us experiment with the realization of a two entangled qubits with the application of an Hadamard gate and C-NOT by using IBM Composer at <https://quantum.cloud.ibm.com/composer>.



$$|\psi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



We obtain the qubits in the state $|\psi^+\rangle$.

